Independent Evaluations of Seasonal Antarctic Sea Ice Extent Reconstructions During the 20th Century

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Abstract

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Independent Evaluations of Seasonal Antarctic Sea Ice Extent Reconstructions During the 20th Century

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Changes in Earth's cryosphere, including sea ice in both polar regions, often result in amplified climate-related impacts through positive feedback loops, making it crucial to understand how and why the cryosphere is evolving now and into the future. Observations of Antarctic sea ice are comparably short as the records have only been reliably observed since the satellite era began in 1979. Prior to the satellite era, limited data sources on Antarctic sea ice are available, including various reconstructions and discrete information in ship logbooks from expeditions and commercial vessels written in the remarks sections. This research aims to further understand how independent early sources of information can be used to evaluate the skill of seasonal Antarctic sea ice extent reconstructions (Fogt et al. 2022).

Ship logbook observations of sea ice are extremely valuable because they are able to give insight into where the ice edge was prior to satellite measurement in 1979. As this is a new area of research, little is yet known on what potential data have been recovered. Not only is it important to know what data are available, but also if the ship logbooks are reliable enough to be used to evaluate Antarctic sea ice extent reconstructions. This research compiled a list of available ship logbooks at Ohio University, only a handful of which were able to be digitized in time for this project. The latitudes from the ship

logbooks of the ice edge show to be within an acceptable range of the latitudes of the sea ice from reconstructions (on average within 3°), indicating the skill of the sea ice extent reconstructions, spatially and temporally, is good prior to 1979.

The Antarctic sea ice extent reconstructions were also compared with estimates of early twentieth century sea ice extent from coupled climate models in terms of the ranges of variability and linear trends. Overall, the variability in the sea ice, both temporally and spatially, are similar between the coupled climate models and the reconstructions, suggesting the statistically generated reconstructions are able to produce physically plausible variations contained in the climate models. The linear trends, especially in the early twentieth century, are similar between the coupled climate models and the reconstructions, even though the end of the twentieth century in the models is showing a decreasing trend in sea ice and the reconstructions have a positive trend.

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Chapter 1: Introduction

The changes seen in the last century in Earth's climate, especially near surface warming, are creating positive feedback loops which have strong impacts on the cryosphere through changes in surface albedo. These changes are seen through observations over extended periods of time. The rapid changes in sea ice loss in the Arctic and the primary increasing trend in sea ice around Antarctica make these two regions a leading research concern. Surrounding Antarctica, the Southern Ocean plays a vital role on a global scale with influences through thermal expansion along with a uniform global sea level rise (Hobbs et al. 2016), even more than in the relatively land-locked Arctic Ocean. Given the potential for more global impacts through the ocean, there is a strong justification for more research in the climate-cryosphere connections in and around Antarctica.

Sea ice plays a crucial role in the Earth's climate system. It has many functions by regulating Earth's energy balance, hindering ocean-atmosphere heat, momentum and gas exchanges, and affecting the circulation of the atmosphere and ocean (Shu et al. 2020). Antarctic sea ice provides insulation, influences the freshwater budget, and the ice-albedo feedback to the Southern Ocean (Liu et al. 2004). The atmosphere around Antarctica creates a feedback with sea ice, for example, via changes in surface temperature. An increase in surface temperature leads to a loss in sea ice. This loss in sea ice would expose more open ocean and lower the albedo, thus, raising the temperatures more as it absorbs in incoming solar radiation. With these interactions, decreases in sea ice can often be attributed to warmer temperatures both within the atmosphere and the oceans. In the regions of Antarctic sea ice, the Antarctic Peninsula has seen near-surface warming temperatures that can be associated with the changes seen in the regional sea ice (Steig et al. 2009); however other regions of Antarctic sea ice have been expanding at the same time. It is imperative to understand the forces that influence sea ice and why there is no even distribution in the changes in the sea ice surrounding Antarctica.

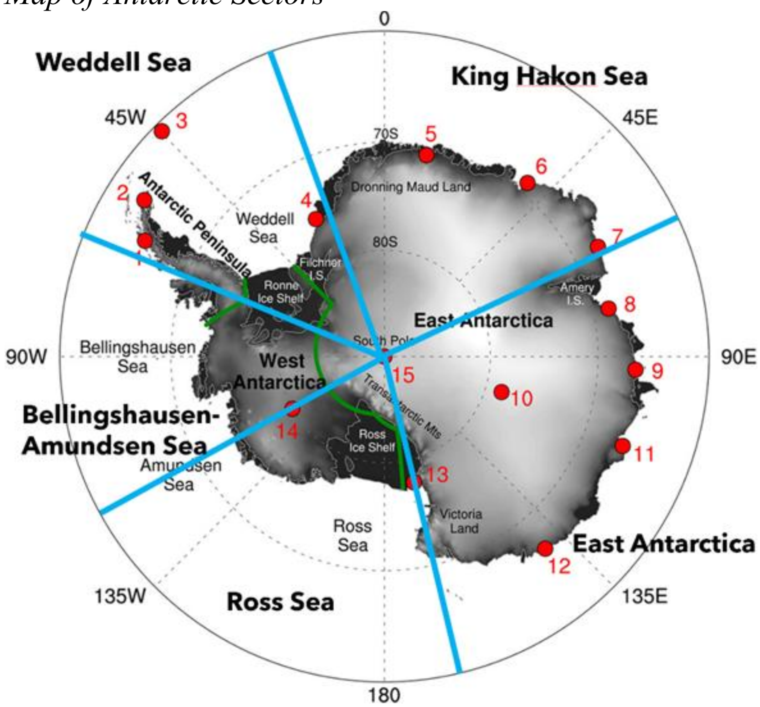
Typically, maximum Antarctic sea ice extent (SIE) occurs mid-to-late September, followed by the minimum SIE in late February. Unlike the Arctic, where sea ice rapidly decreases, satellite data show that the total SIE has a positive trend since 1979 up until around 2016 (Parkinson 2019). The only exception to the positive trend is the Amundsen and Bellingshausen Seas, where SIE is decreasing (Shu et al. 2020). Determining the cause of this phenomenon is an active area of climate research.

Antarctica is divided into different sectors to better analyze spatially-varying sea ice and climate patterns. These sectors are based on close geographical proximity defined through different studies. The broader sectors of Antarctica (Antarctic Peninsula, West Antarctica, and East Antarctica) were outlined by Nicolas and Bromwich (2014) in relation to near surface temperature trends. Then, it was delineated even more by Raphael and Hobbs (2014) who defined the sectors pertaining to sea ice extent variance and decorrelation length scale, where regions show different changes each year from neighboring regions. Figure 1.1 shows the sectors from Nicolas and Bromwich (2014) with the more defined sectors (in blue) outlined by Raphael and Hobbs (2014) on top. Specifically, the delineations of Raphael and Hobbs (2014), split Eastern Antarctica into the King Hakon Sea sector and a separate East Antarctica sector. Furthermore, the Ross and Weddell Seas became their own sectors in portions of West Antarctica and the Antarctic Peninsula. Finally, the Bellingshausen and Amundsen Seas were combined into one sector.

Up until 2016, the Antarctic was experiencing a significant increase in the sea ice. However, in 2016 there was a sudden large decrease in Antarctic sea ice extent (Turner et al. 2017). A new record high since 1979 for total Antarctic sea ice extent (SIE) was set in 2014 but shortly after a new record low occurred in 2017. Just five years later, this low was broken on 25 February 2022 (Turner et al. 2022), with again a new record low set in February 2023. These decreases in sea ice occurring across Antarctica recently (and in contrast to a previous long-term positive trend) highlight the importance of understanding what influences regional Antarctic sea ice, especially why some regions are seeing more changes in sea ice extent than others.

Figure 1.1

Map of Antarctic Sectors



Note: Map of Antarctic sectors from Nicolas and Bromwich (2014) with lines in blue representing the more defined sectors described from Raphael and Hobbs (2014).

Given the connections to the atmosphere mentioned previously, understanding the different climate modes and how they influence sea ice in the Antarctic region is crucial. If the relationships between the different climate modes and Antarctic sea ice changes are strong, it is possible for them to play a role in understanding both current and past changes in Antarctic sea ice.

However, when looking at the SIE in Antarctica, one significant limiting factor to better understanding these connections with climate is the short length of records. Sea ice observations are sparse and, when compared globally, have a short historical period due to Antarctica having harsh conditions that are typically not habitable for people. The easiest and most spatially and temporally complete method for observations of sea ice is from satellites, which detect the sea ice from space across the entire Antarctic continent. Unfortunately, the satellite record is so short compared to other global measure of climate, beginning around 1979. As such, satellite-based observations of Antarctic SIE show a limited range of natural climate variability covering roughly four decades (Parkinson 2019). Apart from satellite measurements, there are other means in capturing Antarctic sea ice observations before the satellite era through ship logbooks and whaling records. Observations from ship logbooks have biases since the sea ice information was recorded more for the summer months (December-February) because of the harsh conditions in other seasons, and the relative ease in reduced ice of catching seals and whales. Ship logbooks and whaling records, even with certain biases, do give potential insight into what the SIE was like before the satellite era. These daily reports of sea ice at specific latitudes and longitudes are able to show how far the sea ice may have extended on a certain day as sea ice is extremely variable day-to-day; however much of these records remain in archives, and only a few have been scanned, let along digitized.

To place the short satellite observations in a longer historical context, seasonal Antarctic SIE reconstructions were performed through robust, mid latitude pressure and temperature observation-based reconstruction ensembles (Fogt et al. 2022). The reconstructions are well correlated with the satellite observations, clearly showing the positive trend of Antarctic sea ice until 2016, when there was a sudden decline. By using reconstructions, it is concluded the sudden changes seen from one year to another have occurred in the past. A negative trend in sea ice occurred in the early to mid-twentieth century, followed by a strong increase in the late twentieth century (Fogt et al. 2022). Given their importance in extending a short and critically important observational record, the purpose of this research is to further evaluate these seasonal SIE reconstructions (Fogt et al. 2022), which will help scientists better understand their strengths and weaknesses and ultimately guide them in their use and application. Independent data sources not used in the reconstructions will be helpful in further evaluating the skill. Ship logbooks, paleo-based sea ice reconstructions, and climate models are all independent sources that will be used in this research. Since Ohio University has recently obtained a vast quantity of ship logbook scans, the focus of this work will be on comparisons of Antarctic sea ice with earlier ship-based estimates of sea ice, as well as sea ice from coupled climate models. Specifically, the research questions are as follows:

1. What are the available early estimates of sea ice extent prior to the modern satellite era from the logbooks provided to Ohio University, and are these reliable enough to be used to evaluate Antarctic sea ice extent reconstructions?
2. Using ship logbook data, how does the skill of seasonal Antarctic sea ice extent reconstructions vary spatially and temporally prior to 1979?
3. How do estimates of early twentieth century sea ice extent from historical simulations of Antarctic SIE in coupled climate models compare to the seasonal sea ice reconstructions in terms of ranges of variability and linear trends?

This thesis is structured as follows: Chapter 2 will explain the literature that is relevant to Antarctic sea ice extent and the phenomena that influence sea ice extent along with explaining the historical observations prior to the satellite era and representations of Antarctic sea ice extent from current climate models. Chapter 3 describes the data collection, processing, and analysis techniques to statistically analyze the skill of seasonal Antarctic sea ice extent reconstructions and estimating the early twentieth century sea ice extent from coupled climate models. Chapter 4 discusses the statistical skill of sea ice extent reconstructions prior to 1979, the historical data that are currently available and their reliability on being able to be used to evaluate the Fogt et al. (2022) reconstructions, and an analysis on how the coupled climate models compare to seasonal sea ice reconstructions. Finally, Chapter 5 will answer each research question in detail found through the results in this work.

Chapter 2: Literature Review

2.1 Fundamentals of Sea Ice Extent

A critical component of the global climate system is sea ice. Sea ice is frozen water that floats over ocean water. It forms, grows, and melts entirely in the ocean. As sea ice melts, it does not contribute to sea-level rise as its displaced volume is already accounted for within the ocean. Sea ice develops around Antarctica as far north as 55°S (NSIDC 2020a). The seasonal cycle of sea ice affects global climate dynamics in various ways. Interaction with albedo, atmospheric circulation, ocean productivity, and thermohaline circulation are all affected by sea ice (Eayrs et al. 2019).

Areas with considerable sea ice coverage will have a higher albedo as these surfaces are more likely to reflect incoming solar radiation. Regions with high albedos will have colder surface temperatures as less solar radiation is absorbed. Warming temperatures over time will cause the sea ice to melt. As the sea ice melts, the ocean absorbs incoming solar radiation due to a lower albedo. The temperatures will warm even more in the region. In autumn, a delay in sea ice formation will occur due to a more extended ice-free season with sea ice loss in early spring (Eayrs et al. 2021). Atmospheric circulation is affected by sea ice also. In the summer, the sea ice retreats, allowing the ocean to increase its temperature by absorbing solar radiation. The air-sea temperature gradient increases in the fall due to the cooling of the atmosphere. The ocean surface cools from the increasing air-sea gradient, and the atmosphere warms and moistens (Feldl et al. 2020).

Sea ice affects ocean productivity as warmer ocean temperatures occur due to the lack of sea ice. The thermohaline circulation is also affected by sea ice. Transportation of heat from the equator to the poles occurs through ocean currents, known as the thermohaline circulation (NSIDC 2020a). The sea ice melt freshens the ocean and impedes the thermohaline circulation. When sea ice forms, it expels salt. The thermohaline circulation may slow due to a reduction in sea ice growth (NSIDC 2020a). As noted above, sea ice affects many processes of the climate and is measured remotely from satellites. When talking about sea ice growth and reduction, SIE is typically the topic of discussion. SIE is the total area covered by some amount of ice, usually above 15% of the standard 25km x 25km area, including open water between ice floes (large packs of floating ice) (NSIDC 2020a). The 25km x 25km is the standard grid box size of sea ice measurements from satellite data, which have a much higher spatial resolution than 25km (but vary through time as satellite sensors change). A region usually has two categories in terms of SIE: ice-covered (>15% sea ice concentration) or not ice-covered (<15% sea ice concentration). Another term discussed when looking at sea ice is sea ice area (SIA), which is calculated as the sum of the product of the area and the ice concentration of each grid cell. SIE is often used over SIA because the uncertainties in passive microwave retrievals have less of an effect on SIE than SIA (Eayrs et al. 2019). Since 1979, the beginning of the satellite period, the total (circum-) Antarctic SIE has shown a statistically significant increase. This expansion is evident in all months of the year. The increase in SIE may result from the near cancelation of ice loss in the Bellingshausen and Amundsen Seas and ice gain in the Ross and Weddell Seas (Fan et al.

2014). Several studies have shown a relationship between SIE and sea surface temperatures (SST). For example, ice gain is associated with lower SST, while ice loss is associated with higher SST. However, a debate remains on the overall increase of Antarctic SIE. Several mechanisms are being proposed with contributions from anthropogenic and natural forcings (Fan et al. 2014).

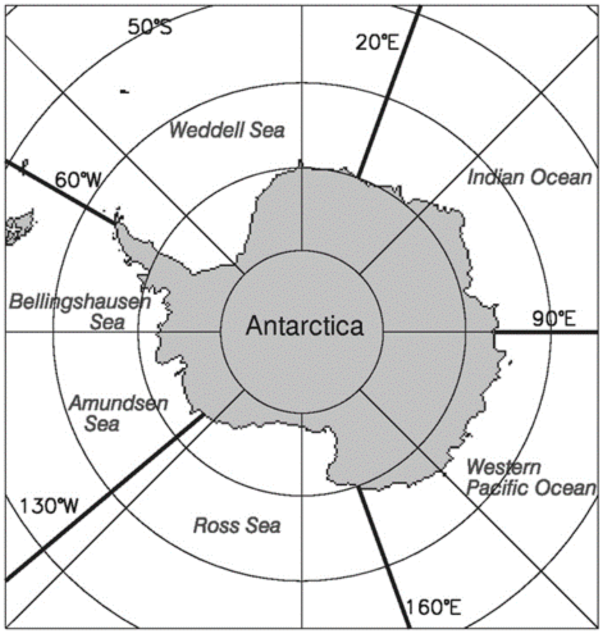
2.2 Regional Sea Ice Extent Variations

Regional variations and trends in Antarctic SIE tell a story of Antarctica's climate. The latitude of sea ice maximum is roughly zonally symmetrical as it forms a circle around Antarctica (NSIDC 2020b). A weak positive trend of 0.9 ± 0.2% per decade of SIE is occurring. Whaling records and ice cores around Antarctica helped in deriving the ice edge before the satellite era (Goosse et al. 2009). The regional trends in SIE have been significant. Changes in the sea ice cover have contributed from large low-pressure anomalies in the eastern Ross Sea and the high-pressure anomalies extending from the Antarctic continent (Lefebvre and Goosse 2008).

Each of Antarctica's five sectors (Fig. 2.1) has a pattern regarding sea ice minimums and maximums. In the Weddell Sea, the monthly minimum ice extent occurs in February. The maximum extent, however, varies from September to October. The highest monthly average SIE occurred in September 1980, followed by the lowest February and March SIE. Negative trends have been noticed in winter and spring within the Weddell Sea, while positive trends have been shown in summer and autumn (Parkinson 2019).

Figure 2.1

Five Sea Ice Sectors of Antarctica



Note: Extracted from Parkinson 2019.

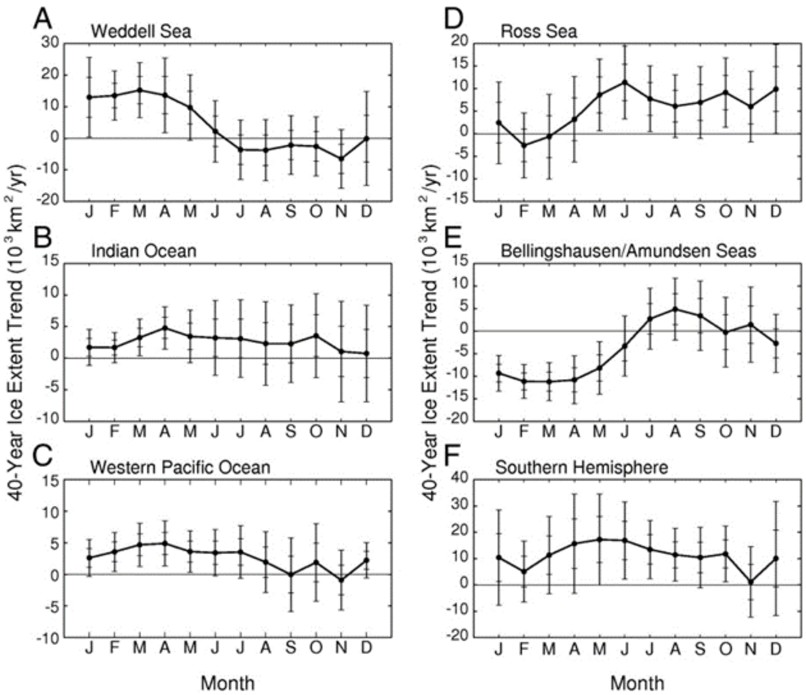
While the Weddell Sea has a maximum SIE in September, the Indian Ocean typically sees the SIE maximum occurring in October, and the SIE minimum occurs in February. The Indian Ocean saw a record-high monthly ice extent in October 2010 (Parkinson 2019). The Western Pacific Ocean has a February minimum and a September maximum in the SIE. A record-high SIE was documented in 2013, and a notable prominent downward trend in SIE from 2013-2018 occurred in this sector (Parkinson 2019). The Ross Sea has consistent monthly minimums occurring in February. A large variability occurs in its month of maximum SIE, with maximums occurring in July, August, September, October, and November. A reduction of sea ice since 2007 has led to an almost total disappearance of sea ice cover followed by a record low of SIE in February 2017 (Parkinson 2019).

The Bellingshausen and Amundsen Seas sector is characterized with much different variability and seasonal cycle than the sectors mentioned above. The month of minimum ice coverage for this sector was March rather than February. A significant contrast between this sector and the other sectors is that it had an overall downward trend in SIE for most of the record after 1979, followed by an upward trend. This contrast relates well with the warming recorded on the Antarctic Peninsula adjacent to the Bellingshausen Sea (Parkinson 2019).

Overall, the Southern Ocean SIE trends remain positive for each of the 12 months. The Indian Ocean, Western Pacific Ocean, and the Ross Sea had positive trends through 2014. Nevertheless, only the Indian Ocean remains consistent with the commonality with the entire Southern Ocean as having positive trends for each of the 12 months. The Western Pacific and the Ross Sea have positive trends occurring in 10 months and negative or no trends occurring in the remaining two months. As mentioned before, the Bellingshausen and Amundsen Seas sector is quite different from the rest of the sectors as all 12 monthly trends were negative through 2014. However, looking at the 40-year record, its summer and autumn values remain negative, and the SIE trends in winter are positive while autumn trends are mixed (Parkinson 2019). Figure 2.2 summarizes the trends in each sector for monthly SIE trends for the 40-year period 1979-2018.

Figure 2.2

Monthly Antarctic SIE Trends from 1979-2018 in Each Sector



Note: Extracted from Parkinson 2019.

Overall, Antarctic sea ice extent dropped rapidly from a record high in 2014 to a record low in 2017. This dramatic loss in sea ice equaled the 30-year observed record decrease in the Arctic Sea Ice (Eayrs et al. 2021). The positive trend in sea ice cover strongly correlates with SST. The average SIE ranges from a summer minimum in February to a winter maximum in September. A better representation of surface temperatures in climate models will better agree with observed trends (Eayrs et al. 2019).

2.2.1 The 2017 and 2022 Record Sea Ice Minimums

A long-term increase in the annual mean Antarctic SIE was observed since 1978. The SIE hit a record high in 2014 before falling to a record low in 2017 (Wang et al. 2022).

Various mechanisms have been proposed as to why the SIE saw a record low in 2017 but the reasons behind the variability in the Antarctic sea ice are complicated. Not only did the annual mean SIE see a record low, but the seasonal minimum SIE also hit a record low on 1 March 2017. Prior to the 2017 record low SIE, Antarctic sea ice saw an unprecedented springtime retreat in 2016. The sea ice retreat varies in rate and time in all of the sectors in Antarctica. In the Ross Sea, Weddell Sea, and Amundsen/Bellingshausen Seas, the variability of the SIE is strongly influenced by change in the depth and location of the ASL while the SIE in the west Pacific Ocean sector is strongly related to a SAM like pattern with more SIE associated with a positive SAM (Turner et al. 2017). In the spring months (September to November) of 2016, the Antarctic SIE decreased about 46% greater than the mean loss and 18% more than any other spring season since the beginning of the satellite era. This early melt of the ice coincided with numerous record atmospheric circulation anomalies over three months (Turner et al. 2017). There was a record deep ASL in September 2016 that aided in a greater sea ice retreat than usual in the Amundsen/Bellingshausen Seas and Weddell Sea sectors. Amplified planetary waves and strong meridional flow in October 2016 led to strong poleward heat transport, which further contributed to the sea ice retreat (Turner et al. 2017). After the record low SIE in 2017, there was a moderate recovery in the annual mean SIE (Turner et al. 2022).

The record-low SIE in 2017 was broken five years later, on 25 February 2022 (Wang et al. 2022; Turner et al. 2022). This new record was mostly a result from large negative SIE anomalies in the Amundsen/Bellingshausen Seas, the Weddell Sea, and the western Indian Ocean Sector (Wang et al. 2022). Starting in early September, the sea ice retreated and the negative anomalies became larger until mid-November, eventually leading to a quick decrease in SIE that exceeded two standard deviations of the climatology on 8 February 2022. Furthermore, on 25 February, the sea ice in the western Amundsen Sea and the eastern Ross Sea completely disappeared. At this time, all five sectors in Antarctica had negative SIE anomalies. The Ross Sea had the largest contribution with 46.3% and the Weddell Sea having the second biggest contribution with 21.6% (Turner et al. 2022). It is important to note that even though the total Southern Ocean SIE had a record minimum in February in 2022, none of the five sectors had a record low SIE. One reason for the record low level in 2022 was the presence of negative sea ice anomalies in all the sectors, especially in the Ross Sea. Also, the ASL had a record low depth in October to November 2021. This led to southerly winds of record strength off of Antarctica. The role of the ASL that led to this record minimum is consistent with previous work as it showed that a deeper ASL leads to a greater transport of ice away from the Ross Ice Shelf and coastal areas to the east and new, thinner ice over the southern Ross Sea. In spring 2021, the sea ice was much thinner along the coast of the Amundsen Sea and it was also accompanied with strong northward dispersion of sea ice. This dispersion of sea ice lead to more open water which aided in the ice-albedo feedback. Thus, leading to more favorable conditions for summer sea ice melting (Wang et al. 2022). A positive phase of the SAM in the previous spring was another contributing factor that led to the record SIE in February. A positive SAM combined with La Niña conditions favors a deep ASL (Turner et al. 2022). Interestingly, at the time of writing this thesis, Antarctic SIE is poised to set new record minimums in 2023.

2.3 Large-Scale Patterns that Affect Antarctic Sea Ice Extent

2.3.1 Amundsen Sea Low

As suggested in the records for 2017 and 2022, large-scale patterns of atmospheric variability can explain substantial portions of Antarctic SIE trends. The Amundsen Sea Low (ASL) plays a significant role in Antarctic SIE as it changes in depth and location, and it highly influences the SIE variability in the Amundsen, Bellingshausen, Ross, and Weddell Seas. The ASL is the climatological area of low pressure located in the South Pacific sector of the Southern Ocean. The ASL is present due to many depressions in the South Pacific sector of the Southern Ocean. These depressions move from the midlatitudes to the south. They can also develop in the highly baroclinic zone of the circumpolar trough around Antarctica (Raphael et al. 2016). The ASL occurs because of the zonal flow around the continental topography and is closely related to the Southern Annular Mode. The cyclonic flow around the ASL brings warm northerly winds into the Antarctic Peninsula/Bellingshausen Sea region and cold southerly winds over the Ross Sea. Observations of sea ice extent trends are consistent with the deepening of the ASL (Eayrs et al. 2019). Kausahara et al. (2019) found that the activity of the ASL has a connection to the variability in the sea ice in West Antarctica.

When the ASL is in its average location, southerly airflow on the western side pulls cold air from the continent over the Ross Sea. This expands the sea ice cover to the north and moves ice away from the coast, exposing the open water at the coastline, which refreezes and drifts equatorward. On the eastern side of the ASL, the opposite occurs. SIE in the Amundsen and Bellingshausen Seas is limited due to the warm northerly flow associated with the ASL (Eayrs et al. 2021). Strong southerly flow promotes ice growth when the low is shifted toward the Antarctic Peninsula and over the West Antarctic ice sheet. When the low shifts toward the northwest, northerly flow restricts sea ice growth and expansion (Raphael and Hobbs 2014). Strong winds associated with the ASL impact sea ice by forcing it to drift into areas that influence other large-scale patterns such as El Niño/Southern Oscillation and Southern Annular Mode.

2.3.2 Southern Annular Mode

Another large-scale pattern that affects the Antarctic SIE is the Southern Annular Mode (SAM). In the Southern Hemisphere, the SAM is the primary pattern of atmospheric variability. The SAM is the zonal mean atmospheric pressure difference between the Southern Hemisphere mid-latitudes (~40ºS) and Antarctica (~65ºS) (Abram et al. 2014). A positive SAM is associated with low-pressure anomalies over Antarctica and high-pressure anomalies over the mid-latitudes. A strengthening of the Southern Hemisphere westerly jet results from the enhanced atmospheric pressure gradient due to a positive SAM. The Antarctic Peninsula temperature variability is sensitive to the strength of the westerly winds because of the topography (Abram et al. 2014). A positive SAM also has a significant impact in the summer and can help to explain some of the decrease in SIE in the Bellingshausen Sea (Eayrs et al. 2019).

Strengthened westerlies are responsible for increased cooling and sea ice expansion by delivering cold, surface mixed layer water northward by the Ekman effect (Eayrs et al. 2021). The Ekman effect is when surface water molecules move by the force of the winds and drag deep layers of water molecules below with them. The water molecules are moved from shallower layers to deeper layers by friction, and deeper layers move more slowly than shallower layers. As a result, each successively deeper layer of water moves more slowly to the left (in the Southern Hemisphere), creating a spiral effect (NOAA). The northward transport of surface mixed-layer water suggests the upwelling of warm, deep waters from below the mixed layer, leading to sea-ice reduction (Eayrs et al. 2021). The rising trend in the SAM in the last 25 years could have induced a cooling effect over the continent. An increase in the ice cover may counterbalance the warming occurring due to the greenhouse effect (Lefebvre and Goosse 2008).

2.3.3 El Niño/Southern Oscillation

The El Niño/Southern Oscillation (ENSO) is another large-scale pattern that affects SIE. ENSO is a recurring climate pattern involving changes in the temperature of waters in the central and eastern tropical Pacific Ocean. ENSO has a strong response in tropical SST anomalies, which plays a vital role in the sea ice in Antarctica (Eayrs et al. 2019), especially in the South Pacific and Atlantic Ocean sectors, where ENSO induces a dipole response in the sea ice. During a year with a high Southern Oscillation index (i.e., La Niña), there is more ice in the Amundsen sector and less ice around the Antarctic Peninsula (Lefebvre and Goosse 2008). During warm ENSO events (El Niño), there are warm temperature anomalies resulting in reduced sea ice anomalies in the Pacific center of the Antarctic Dipole while having an opposite response in the Atlantic center of the Antarctic Dipole. The Antarctic Dipole is a high-latitude climate variability in the air-sea-ice system that responds strongly to the ENSO forcing. During cold ENSO events, colder temperatures result in an expansion of sea ice in the Pacific center of the Antarctic

Dipole. In the Atlantic center of the Antarctic Dipole, a reduction of sea ice occurs resulting from warmer temperatures (Yuan 2004).

2.3.4 Interdecadal Pacific Oscillation and Pacific Decadal Oscillation

The Interdecadal Pacific Oscillation (IPO) is another large-scale climate pattern that influences the sea ice in Antarctica. The IPO is the difference between the sea surface temperature anomaly (SSTA) averaged over the central equatorial Pacific and the average of the SSTA in the Northwest and Southwest Pacific. It measures the interdecadal variability of surface ocean conditions throughout the Pacific. The IPO is different from the Pacific Decadal Oscillation (PDO) because it spans both hemispheres, whereas the PDO only occurs in the mid-latitudes in the Northern Hemisphere. The IPO influences the sea ice in the eastern Ross, Amundsen, and Bellingshausen Seas (Purich et al. 2016).

A positive phase of the IPO allows for a warmer tropical Pacific. A warmer tropical Pacific alters the location of the ASL through changes in the Rossby Waves, which affect the sea ice through changes in the atmospheric circulation. The ASL position and magnitude are strongly tied to the Southern Hemisphere Rossby Waves. Rossby Waves are driven by tropical convective heating and precipitation anomalies during IPO phases (Clem and Fogt 2015). A negative phase of the IPO deepens the ASL. A negative phase allows for warm air advection, resulting in lower sea ice concentration in the Bellingshausen Sea (Meehl et al. 2016). Before 1999, the Antarctic sea ice experienced increases during the positive phase of the IPO. The sea ice expansion was due to cold air being advected from the south (Meehl et al. 2016). The IPO, along with other large-scale climate processes, impacts the SIE, depending on whether the IPO is in a positive or negative phase.

The PDO also has a statistical connection to changes in Antarctic sea ice. As mentioned above, the PDO primarily occurs in the mid-latitudes in the Northern Hemisphere, with a smaller area of opposite SST anomalies in the tropical Pacific. In the warm phase of the PDO, SSTs are warm in the tropical Pacific, and SSTs are cold in the North Pacific (Wilby 2017). El Niño events are likely to occur during the positive (warm) phase of the PDO. A deepening of the ASL and west Antarctic warming after 1979 has a connection to the PDO (Clem and Fogt 2015).

2.3.5 Local Wind

Regional variations in wind, even distinct from large-scale patterns of climate variability, also play a significant role in influencing SIE. Meridional winds can impact sea ice production and melt. Cold, dry air can be advected from southerly meridional winds extending from the continent. Warm, moist air is advected from meridional winds originating at lower latitudes (Hobbs et al. 2016). The primary influence in SIE trends in the West Antarctic region was wind-driven ice advection changes. Northerly winds (bringing in warmer air) caused the SIE to decrease in the Bellingshausen Sea, while southerly winds (bringing in colder air) caused the SIE to increase in the Ross Sea (Goose and Zunz 2014).

2.4 Strategies to Overcome the Short Observational Record

One of the problems when looking at the Antarctic climate is the short time period in records compared to much of the rest of the world, especially across the Northern

Hemisphere. Historical records and paleo data can help in understanding past variability of Antarctic SIE prior to direct observation. Routine satellite monitoring of the sea ice began in the 1970s. Models generally do a poor job in capturing sea ice trends. There is a lack of information on climate models with pre-satellite sea ice coverage (Edinburgh and Day 2016). One limiting factor of using early satellite data prior to the advent of the modern satellite era in 1979 was poor resolution. Another limiting factor was the influence of cloud cover, which looks similar to sea ice in static (non-looping) satellite images (both visible and infrared). The early satellite images were taken over a short period, making it challenging to observe the variability in the SIE (Hobbs et al. 2016).

Studies aiming to understand the historical significance of Antarctic climate variations lack the evidence needed from long-term observations (Curran 2003). There are some places within Antarctica where long-term observations spanning nearly a century exist; however, almost all Antarctic climate records deal with substantial interannual variability. Interannual variability makes it difficult to evaluate the impacts of the climate trends (King and Harangozo 1998). Sea ice is no different than other Antarctic climate variables, marked with large interannual variability. This is further complicated by the even shorter nature of satellite observations compared to conventional surface observations of Antarctic climate, most of which begin in 1957. In comparison, SIE records are only reliable from 1979, the beginning of the continuous satellite observations for sea ice (Turner et al. 2007).

2.4.1 Ship Logbooks

One way to help bridge the gap in determining the history of SIE is to use evidence from proxies and discrete historical records from ship logbooks. Proxy records can give an insight into multidecadal to multimillennial time scales (Eayrs et al. 2019). Proxies and historical records suggest that SIE has decreased over longer time scales. Due to the small size of SIE records, scientists are making efforts to extend the record by analyzing marine sediments, ice cores, and coastal records for biogenic or geochemical proxies. It is essential to be careful when comparing proxy and historical records with satellite observations. Since proxy and historical records are the only available information on longer time scales, these records are valuable as their data put recent changes in SIE into a longer-term context (Eayrs et al. 2019).

Meteorological data kept in logbooks from whaling and sealing commercial vessels is one of the forms of historical direct observations to help determine SIE before the satellite era. Before the satellite era, direct information about Antarctic sea ice is highly sparse. Since information about Antarctic sea ice is sparse, this has led to an attempt to produce reconstructions of 20th-century sea ice change using historical information from discrete observations of the ice edge and conditions (cover and size of icebergs) from whaling ships. Additionally, records of the location of whale catches help estimate the summer SIE as whale catches were assumed to have occurred near the ice edge (Brace et al. 2020).

Although spatially and temporally discrete, the first-ever weather observations for this region come from logbooks. Commercial, fishing, and whaling vessels in the

Southern Ocean tend to be overlooked as a tool for reconstructing sea ice (Teleti et al. 2019). One advantage of using whaling logbooks is that the vessels usually would visit the same area year after year, thus providing sustained temporal coverage. The Teleti et al. 2019 study shows the meteorological observations from whaling vessel logs are consistent with existing observational datasets. A deeper look into the historical records of the Antarctic whale catches show the catches shifted southward on average by about 2.8^{o} latitude in the beginning of the 1950s (de la Mare 2009; Ackley et al. 2003). To come to this conclusion, an analysis of the estimated mean catch position for each year was calculated using a generalized linear model that corrected for systematic changes in the coverage. The average latitude of catches remained mostly stable from 1931 to 1959 and again from 1972 to 1987, the end of the commercial whaling in Antarctica (de la Mare 2009). A series of cruises from the 1920s to the 1930s occurred to the Southern Ocean by the UK Discovery Committee. The purpose of the cruises was to investigate the oceanographic properties and plankton distribution with relation to whale conservation (Ackley et al. 2003). It was during this period that direct observations of SIE were made as it was written down when the ship had encountered pack ice.

Whaling fleets operated in the proximity of the ice edge and followed it southward as it retreated in the summer whaling season. Blue whales and minke whales often overlap in a geographic area. Both species are known to concentrate near the ice edge. The possibility of whaling fleets being constrained by ice conditions is great. If this is the case, then the whaling positions are a direct and consistent measure of the ice edge change over time (de la Mare 2009). Mackintosh and Herman (1940) compiled a circumpolar map of the monthly variation of the average ice edge from the observations from whaling vessels (Ackley et al. 2003). A comparison is made between the historical maps to the modern record through digitizing and comparing with satellite data derived from passive microwave satellite imagery from 1979 to 1998.

However, potential biases may occur by using whaling data as a sea ice indicator. Some of these biases include the species of whales caught, spatial and temporal heterogeneities in the coverage of whaling data, and offsets between satellite and chart records of SIE (Abram et al. 2010). Some argue further that the species composition of the whale catch is different between the early period and late period of whaling (blue-fin-humpback in the early period and minke in the later period). There is also a potential bias with the modern satellite edge being too far south and when it is corrected it reduces the difference in whaling positions to within the bounds of observed variability (de la Mare 2009). Abram et al. (2010) pointed out that regional changes produced using direct chart observations differ from those derived from whaling records. With these limitations, data from whaling vessels will still provide a glimpse of what the SIE was like in the past.

2.4.2 Reconstructions From Paleoclimatological Data

Another proxy used to determine SIE before the satellite era is ice cores. Antarctic ice cores provide an alternative way to reconstruct past sea ice changes. Ice cores contain general conditions of the ice in a particular sector of the ocean (Abram et al. 2010). One way ice core based reconstructions work is that poleward advection of air masses transport aerosols from the open ocean and the sea ice zone to Antarctica. The aerosols are then deposited onto the ice sheet and are trapped in annual snow layers. The concentration of aerosols in the atmosphere is affected by the amount of ice each year, thus, allowing for the Antarctic ice cores to be robustly used to reconstruct past sea ice conditions over centennial to millennial timescales (Thomas et al. 2019).

Conditions at the sea ice source, prevailing wind direction, deposition on route, distance from source, post dispositional processes in the snow, and species migration in the ice are all important factors that influence the concentration of all proxies in an ice core (Thomas et al. 2019). Ice core based sea ice reconstructions are best when they originate from high accumulation coastal sites. This is where the dominant wind direction is onshore, transporting species associated with sea ice directly to the ice sheet. However, Antarctica's wind direction and meteorological conditions are highly variable. In this region, the majority of the ice core based sea ice reconstructions represent regional sea ice changes (Thomas et al. 2019).

Curran and Jones (2000) first proposed that methanesulphonic (MSA) concentration may be interpreted as a sea-ice proxy under the following conditions: (1) Antarctic winters with a more significant sea-ice decay during the following spring melt, (2) the more extensive sea-ice break-up zone enhances phytoplankton blooming and production of MSA, and (3) atmospheric transport deposits MSA onto ice caps, incorporating the varying yearly concentrations into ice-core records. MSA was one of the only first organic aerosol components detected in polar ice cores (Thomas et al. 2019).

In Antarctic ice cores, the only source of MSA comes from the atmospheric oxidation of dimethylsulphide (DMS) (derived from marine phytoplankton that lives in the sea ice zone). The blooms of these sea ice algae come from the seasonal melting of sea ice. This leads to an increase in emissions of DMS into the atmosphere (Abram et al. 2010). Because of this, enhanced winter SIE and sea ice algae blooms during seasonal ice melting can lead to an increased MSA being transported to ice cores the following summer. MSA and SIE have a number of statistically significant correlations between them observed at coastal sites around Antarctica. With a greater SIE in the winter months, a larger area of sea ice breakup occurs the following spring and summer. This enhances the biological activity and DMS production; thus, a high MSA represents winter SIE (Thomas et al. 2019). There is a similar positive relationship between ice core MSA records and Antarctic sea ice variability at various other sites in the Ross Sea/Indian Ocean sector (Abram et al. 2010). In the Amundsen Sea, a high MSA (increased SIE) years are associated with a deepening of the ASL and a regional expansion and intensification of the westerly jet. A dual effect occurs from this in the Amundsen/Ross Sea with an increase in SIC and an enhancement in the transport of MSA from the ice edge to the ice core site (Thomas and Abram 2016).

Sea salts have also been proposed as a proxy. Sea salt aerosol originates from breaking waves over the sea surface. The large particles are quickly deposited over the open water, but the smaller particles are able to travel farther distances and eventually deposit on the ice sheets. Once deposited on ice sheets, sea salts are normally measured in ice cores as sodium. It is suggested that the positive correlations with sea ice results from the strengthened atmospheric circulation at the sea ice margin (Thomas et al. 2019).

Ice cores have a range of advantages associated with them. They have a high seasonal to annual resolution. They have the potential to provide centennial to millennial reconstructions. A variety of proxies are available from ice cores, and the records can provide local to regional information. However, there are many disadvantages associated with ice core based climate reconstructions, including sea ice. They have limited spatial coverage and are limited to mainly coastal regions. The proxies are dependent on meteorological conditions, which are assumed to be consistent through time. There is a potential for post-depositional changes. Finally, there have yet to be any records that cover a full 2000-year period (Thomas et al. 2019).

2.4.3 Observation-Based Reconstructions

Although these reconstructions provide some information on historical Antarctic sea ice prior to the satellite observations, Fogt et al. (2022) look into the challenges of the short and highly variable Antarctic SIE limits by creating seasonal reconstruction ensembles of Antarctic SIE since 1905. Through their research, a strong connection is noted between Antarctic sea ice and regional and large-scale climate variability. A positive trend is captured through the reconstructions from 1979-2016, when there was a sudden decline in Antarctic sea ice, similar to satellite observations. The seasonal total of Antarctic SIE reconstructions is well correlated with the observed, exceeding 0.71 in all seasons. One exception found in this is for September-November where the reconstructions were either equally or more highly correlated with the observations (Fogt et al. 2022).

A strong positive trend for Antarctic SIE is shown in the shorter observational period through 2014 (the year of a record maximum based on satellite observations). 2016 was the year of a marked decline in Antarctic SIE. These positive and negative trends are based on just over 40 years of observations from satellite data. Reconstructions from various periods throughout the twentieth century help to highlight the historical perspective of the positive trends since 1979 (Fogt et al. 2022). The positive trends, in most seasons and sectors, in the observational period are different from the negative trends in the early to mid-twentieth century. Fogt et al. (2022) sea ice extent reconstructions show the negative trends in the early to mid-twentieth century in all seasons except for December-February. Their reconstructions also suggest a regime shift in Antarctic SIE in the middle of the twentieth century. Before 1960, Antarctic SIE were declining but by the late twentieth century there were strong increases as seen through satellite observations. The observed period may have also started at a time when the SIE was relatively low in the last century compared to the rising positive trend that is part of multidecadal variability (Fogt et al. 2022).

A small, weaker reversal of sea ice changes in the early to late twentieth century can be hinted through proxy-based reconstructions of Antarctic SIE, especially in the Ross and Weddell Seas. The regime shift seen is not equal in magnitude or timing among all the sectors. Since it is not equal, this helps to highlight the known independent nature of Antarctic sea ice around the continent. King Hakon VII and Ross-Amundsen Seas show the most pronounced shift from early to mid-twentieth century ice loss to late twentieth century ice gain (Fogt et al. 2022). The only sector in Antarctica to have a statistically significant increase in winter SIE during 1979-2020 was the Ross-Amundsen Seas. The sector reconstructions from Fogt et al. (2022) the Ross-Amundsen and Amundsen-Bellingshausen Seas counteract each other as the winter ensemble mean reconstructions show the ice was increasing during 1905-1978 in the Amundsen-Bellingshausen Seas, while the Ross-Amundsen Seas experienced a decrease during the same time period.

Beginning in 2016, a new regime since approximately 1960 characterizes persistent years of below average SIE. Before 2016 the total Antarctic SIE had been consistently increasing. After 2016, Antarctic SIE was below average for several years. The changes in the average sea ice between two consecutive years emerge as a unique characteristic of the recent Antarctic total sea ice regime. Changes from one year to the next are common in seasonal total Antarctic sea ice variability. However, the switch from above average to below average Antarctic total SIE as seen in the 2016 event has occurred infrequently since the beginning of the twentieth century (Fogt et al. 2022). These reconstructions by Fogt et al. (2022) thus provide critical information on Antarctic sea ice history, yet their reliability and performance are based on a direct comparison with the short satellite record beginning in 1979. It is therefore necessary to provide an additional assessment of the performance of these reconstructions in the earlier 20th century using independent sources of data. Ultimately, this will help other researchers to know the strengths, weaknesses, and reliability of these new data so that scientists can appropriately analyze them for future research.

2.4.4 Sea Ice Estimates From Coupled Climate Models

In absence of continuous measurements, the primary tool to look at spatiotemporal centennial-scale projections of the climate system are coupled climate models (Holmes et al. 2022). Given their completeness (both spatially and temporally) and dynamic and physical constraints, it is possible to further the understanding of the climate system variability and provide projections of future climate change through global coupled climate models. Coupled climate models are also used to study the evolution of sea ice and the potential environmental impact of the decline of sea ice. The Couple Model Intercomparison Project (CMIP) provides a common model experiment protocol that allows for an evaluation of climate models developed by at least 40 modeling groups worldwide (Roach et al. 2020). CMIP also provides the modeling community a way to assess the performance of sea ice simulations in state-of-the-art climate models to help improve fidelity (Shu et al. 2020).

Since the satellite era began, observations indicate there has been a slightly positive trend in Antarctic SIE, as discussed earlier. Coupled climate models, however, struggle to show this trend. Even though there have been advances in climate modeling, a simulation of Antarctic sea ice has remained a problem for the models. In fact, most climate models show a decrease in the Antarctic SIE, opposite of observations. Climate models that were part of the Coupled Model Intercomparison Project Phase 5 (CMIP5) simulate a strong decline in the Antarctic SIE for September in the recent decades. In contrast, the satellite data shows a slight positive trend in September for the same time period of 1979-2018 (Rackow et al. 2022). The CMIP6 (the most current generation coupled climate models) models are similar to the CMIP5 as they show the seasonal cycles of Antarctic SIE are well produced; however, the negative biases are slightly larger in CMIP6 models (Shu et al. 2020). It is clear that both CMIP5 and CMIP6 models simulate negative SIE trends instead of the positive trends the observations show.

During the period 1979-2005, the annual mean Antarctic SIE observed trends were 0.13 (±0.10) and 0.14 (±0.10) million km^2 per decade. The trends from both CMIP5 and CMIP6 simulate negative values of -0.34 (±0.03) and -0.35 (±0.03) million km^2 per decade. Nonetheless, Shu et al. (2020) found in their study that several models and some realizations do indeed have positive Antarctic SIE trends. They found that 11 percent (33 out of 307) of CMIP6 realizations have a positive trend during the period 1979 to 2005. Some CMIP6 models show a consistency with observations at the winter maximum in September. Various models also retain high concentration of summer sea ice cover. These consistencies with observations are a positive result for CMIP6 models, however, they still broadly underestimate the summer minimum sea ice area (Roach et al. 2020).

Looking at previous multi-model ensembles, a relationship exists between the historical state and future response of Antarctic SIE. In terms of SIA (sea ice area), CMIP6 projections show a loss in all forcing scenarios in both summer and winter by the end of the 21st century (Holmes et al. 2022). There is at least one model under all the scenarios, with the exception of the strongest forcing, that simulates an increase in SIA by the end of the century. A linear relationship between SIA historical and SIA 8.5 scenario exists, indicating that models lose most of their ice by the end of the century for the February SIA. For September SIA, the models with more sea ice in the historical period lose more ice (Holmes et al. 2022).

Roach et al. (2020) research suggests there is a small discrepancy between models and observations than previously identified. The satellite data have extended the observational record an additional four years showing the sea ice area has weakened, and the regional signals are less clear over 1979-2018 than 1979-2014. Models that have large trends in global mean surface temperature show the most negative trends in sea ice area over 1979-2018. The interannual variability of sea ice is larger than that observed over the historical period. Many of the models simulate an unlikely mean sea ice area, which results in a larger intermodel spread that is vastly larger than that in the Arctic (Roach et al. 2020). Even with these issues, there should be a moderately higher confidence in the simulation of Antarctic climate in CMIP6 than previous generations. In this study, due to the concerns of CMIP6 trends, sea ice extent reconstructions will be cautiously compared to CMIP6 – mostly in terms of validating physically possible changes present in the reconstruction if they are commonly seen in the CMIP6 ensemble.

2.5 Synthesis

This research aims to use independent data sources to evaluate further the skill of the sea ice reconstructions that were recently performed (Fogt et al. 2022). Available sources of early estimates of SIE prior to the satellite era will be examined, and it will be determined if these are reliable enough to be used to evaluate Antarctica SIE reconstructions. The skill of seasonal Antarctic SIE reconstructions and the use of independent data will be examined to determine how it varies spatially and temporally prior to 1979 (beginning of the satellite era). Once these two aspects have been researched, they can be of significant use for future research. Further evaluation of the reconstructions will help scientists better understand their strengths and weaknesses, guiding them in their research.

Climate change is a big topic being discussed today and has a significant impact on the Earth. This is seen through sea-level rise from melting ice sheets and glaciers in both the Antarctic and Arctic. This research will help provide context over the changes seen in the Antarctic climate. The research will help determine if the region's changes can be linked to human-indued climate change or if the changes are primarily due to natural variations.

Sea ice plays a vital role in the climate variability of Earth as the positive feedback loop between the cryosphere, atmosphere, and ocean are important in global climate change. By gathering more information on different Antarctic processes, an understanding of the extent of sea ice can be made and how it will impact areas around Antarctica in the future. With the research that will be done in this project, other research can be continued from these data and information.

Chapter 3: Data and Methods

3.1 Area of Study

The area of study for sea ice extent reconstruction will be anywhere there is sea ice around the Antarctic region. This area can extend out to the furthest extent the sea ice will grow, extending as far north as 50°S. Following the natural boundary set by the sea ice extent will allow for a complete reconstruction in the Antarctic region to give a full spatial picture of what the SIE was like before the satellite era.

3.2 Antarctic Sea Ice Observations

Fogt et al. (2022) computed sea ice extent (SIE) using satellite-observed sea ice data from Nimbus-7 Scanning Multichannel Microwave Radiometer (SMMR) and Defense Meteorological Satellite Program (DMSP) Special Sensor Microwave Imager - Special Sensor Microwave Imager/Sounder (SSM/I-SSMIS). Climate Data Record (CDR) daily concentration fields from the National Oceanic and Atmospheric Administration / National Snow and Ice Data Center (NOAA/NSIDC) Climate Data Record of Passive Microwave Sea Ice Concentration, Version 4 (https://nsidc.org/data/g02202) were used (Meier et al. 2021). The CDR algorithm output combines ice concentration estimates from the National Aeronautics and Space Administration (NASA) Team algorithm (Cavalieri et al. 1984) and the NASA Bootstrap algorithm (Comiso 1986). The data span the period 25 October 1978 to 31 December 2020 and are daily. There is an exception prior to July 1987 when they are given every other day. Data are gridded on the SSM/I polar stereographic grid (25 km × 25 km). The SIE used in their analysis is calculated using the equatorward limit of the 15% sea ice concentration (SIC) isoline. The SIC is the sum of the area of every grid cell that is 15% or more covered with sea ice. The monthly SIE was calculated as the average of the SIE for days in that month. In addition to the alternate day observations from 1978 to 1987, there are a number of days and segments of days with no observations. In particular, there are no data between 3 December 1987 and 12 January 1988. Since these days had no observations, Fogt et al. (2022) fit a stochastically imputed daily SIE using the following procedure: they constructed anomalies by subtracting off the invariant annual cycle, estimated using the methods of recent work56. They then fit a Bayesian AutoRegressive Integrated Moving Average (ARIMA) model to the observed daily SIE. Then they stochastically imputed the missing days by drawing for their posterior distribution. All the days in the month were averaged to compute the monthly SIE values. It is noted that, except for the period from 3 December 1987 and 12 January 1988, the impact of this stochastic multiple imputation scheme is small since the daily data are nearly complete. The sectoral SIE was computed as the sum of the area within the sector of every grid cell that is above 15% or more covered with sea ice (most cells are completely within a single sector and do not cross into the adjacent sector). Although contiguous, the sectors are defined based on their spatial decorrelation scale as discussed in earlier work21. The longitude bounds for the sectors are as follows: Amundsen-Bellingshausen Seas (250°-290°E), Weddell Sea (290°-346°E), King Hakon VII (346°-71°E), East Antarctica (71°-162°E) and Ross-Amundsen Sea (162°-250°E); the total Antarctic sea ice extent represents the area of all sea ice surrounding Antarctica, and is precisely equal to the sum of the areas in the five sectors.

3.3 Data

3.3.1 Sea Ice Extent Historical Data

To obtain a sense of what the SIE was like before the satellite era, historical records are used to gain a greater understanding. Ship logbooks were obtained to gather sea ice information. The historical logbook data came from collaborators in the United Kingdom, specifically Julie Jones and Clive Wilkinson. Student researchers from Ohio University digitized the ship logbooks from scans. Many expeditions to Antarctica were logged. Each page was carefully examined to determine where ice packs and ice edges were located. The location of the edges and packs are used to help approximate where the SIE was during that time period. These data are valuable to assist in answering the question of the available sources of early estimates of SIE prior to the modern satellite era. These data will also help in answering the question of if the historical data set is reliable enough to use to evaluate Antarctic SIE reconstructions.

To construct the data gathered from the ship logbooks, the approximate latitude and longitude are mapped of the SIE. It is important to stress these are approximate locations, although they are manually observed. First, when the observations of sea ice were written down in the logbooks, the location was based on a specific time and day – and the ice edge changes quickly, especially in active winter. Second, these locations are typically biased north of where the sea ice actually was. The sea ice could have been miles off from the location recorded in the logbooks as the vessels were not going to sail straight into the ice in order to minimize their risk. Because of this bias, there needs to be a correction for the latitude and longitude of the sea ice. Finally, the ice that is observed from the deck on the ship and entered into the logbook is likely the farthest edge of the ice, not the 15% concentration that is used to map the ice edge from modern satellite records. This disparity creates even more of a northern bias in sea ice edges determined from logbooks.

Since these logbooks are for a specific time, location, latitude, and longitude, these observations differ quite substantially from the sea ice reconstructions to which they are compared in this thesis. First, the Fogt et al. (2022) sea ice extent reconstructions deal with seasonal averages taken over several months, not the ice conditions at a specific time. Even a few day gaps in manual observations from a ship deck in a specific location can lead to quite a different average of the ice edge over the course of a season. Second, Fogt et al. (2022) reconstructions not only represent a seasonal mean, but the total area of sea ice >15% concentration within a longitudinal band, not the ice conditions at a specific latitude and longitude. Typically, logbook observations only see about 1% to 5% of the ice cover in a 25km x 25km (the area of a satellite measurement of sea ice). As such, in order to make comparisons with point measurements at a specific day, relationships will need to be derived between the daily variation in ice edge as a function of seasonal and sector mean values used in the reconstructions. Finally, another issue that is less common but does occur, is mistaking icebergs (ice that originated on land, from a glacier or ice sheet, and is quite tall in comparison) for sea ice (ice that grows and melts entirely on the ocean, and as such is typically only a few meters thick in the Antarctic). This would sometimes occur in whaling ship records. When making observations, this could be a common mistake as the people making the observation are not well versed in the difference of an iceberg or sea ice, especially if they see the iceberg from a few miles away and cannot adequately discern its shape or height. There is an important distinction between the two and this error can be huge when trying to use logbooks to aid in reconstructing the sea ice extent before the satellite era; indeed most ice reports in ship logbooks are that of icebergs instead of the ’pack ice’ (a common logbook term for sea ice).

Although a great source of data, there are many challenges when trying to compare a daily sea ice report to a seasonal, sector mean sea ice extent reconstruction. Since this is a challenge, there needs to be a correction for the visual observation, one which accounts for the large spatiotemporal variability of the daily Antarctic marginal ice zone / ice edge.

3.3.2 Sea Ice Extent Model Data

Climate model data using CMIP6 was used for sea ice extent reconstructions for each sector of Antarctica. These data include monthly SIE from 1850 to 2014 using multiple ensembles (Table 3.1). Using multiple ensemble members is a great advantage. By using ensembles, this allows for estimations of the certainty of the results. When using multiple ensembles, researchers are able to investigate interannual variations better as they have not been smoothed out like they would be in the ensemble mean. The ensemble members depict possibilities more similar to an observed climate variable on Earth; the ensemble or model means are best suited to depict consistencies, generally tied to applied forcing mechanisms, between the models.

The CMIP6 data were derived using a series of code in order to calculate the SIE. To ensure this data is consistent with existing data, the models and numbers were checked using Roach et al. (2020) models and data. Seasonal means were created using these data along with monthly means. A total of the SIE was also calculated for the Antarctic region.

The CMIP6 model data were obtained from World Climate Research Programme. When downloading the data from WRCP, specific areas were needed to help ensure the right data were chosen. For this research, historical forcings were used. The frequency that was needed was monthly data, and the realm that was focused on was sea ice. Finally, the variable that was important when downloading the climate models was making sure it was the Southern Hemisphere sea ice concentration only. Integrating the sea of sea ice concentration >15% in this variable gave the total extent; however, by altering code, each sector's extent was able to be determined using the longitude bounds of each sector and calculating this area separately.

When these CMIP6 sea ice concentration data were downloaded, they were converted into netCDF (network Common Data Format) files. This allowed for the file to be read correctly when ran through the processing code (NCL). The first step when running the code was to gather the monthly sea ice netCDF concentration files for the models historical simulations for the years 1850-2014. In CMIP6, the sea ice concentration is on the ocean model's grid. Because of this, the use of the area files of the ocean models (areacello netCDF files, when available) was used to calculate the accumulated area of the sea ice greater than 15% concentration, within the longitude bounds of each sector. This is achieved by multiplying the sea ice concentration as a percentage with the area (in square km), and adding these all up (integrating) over the sector to get the sector total sea ice extent, expressed as an area, just like the reconstructions. An output of monthly data was also achieved for further analysis. Since the ocean model's cell areas shrink as you move poleward closer to the continent, no weighing is needed as this is already accounted for in the cell area files. It is important to note that some of the area cell files could not be found for all the models used by Roach et al. (2022). Even if the historical CMIP6 sea ice concentration data were available, not all the SIE as in Roach et al. (2022) was able to be calculated for all the models without these area cell files.

Using these climate model data can aid in the process of comparing the outputs of the SIE to the SIE gathered from using the historical data from the ship logbooks. This helps to provide a better idea of understanding of the SIE prior to the satellite era while also answering the question of whether or not the historical data is reliable enough to reconstruct the SIE in the early 20th century. Using the historical simulations of the CMIP6 data will aid in determining physically plausible changes in Antarctic sea ice – such as the magnitude and variation in short-term trends, interannual changes, and the relationships between various sea ice sectors. The output of these simulations can then be compared to the historical observations from the historical data gathered (such as the ship logbook data). From here, it is possible to see how well the reconstruction estimates of past sea ice match up the range of physically possible variability within the dynamically constrained climate models.

Table 3.1

CMIP6 Historical Models and Number of Ensembles Used

Model	Number of Ensembles
ACESS	45
BCC	6
CAMS	3
CanESM5	65
CESM	20
CNRM	21
FGOALS	3
GFDL	2
HadGEM3	9
IPSL	35
MIROC	83
MRI	12
UKESM	18

3.4 Methodology

3.4.1 Collection of Historical Data

One major aspect of this research is to determine what historical data exist for sea ice data. The collection of available historical sea ice data allows for the data to be analyzed for reliability and completeness. After the collection of the ship logbooks for the historical sea ice data, digitization was necessary to be able to determine if they were reliable enough to reconstruct the sea ice extent. While reading through the ship logbooks, anytime one of the phrases ‘ice pack’, ‘pack of ice’, or ‘ice edge’ was written in the observation, it was logged into an excel document. The latitude and longitude were also recorded in an excel document. The latitude and longitude allow for the rough estimate of the sea ice edge to be mapped.

When going through the ship logbooks, it was important to keep in mind that the written observations of the sea ice have the potential to be biased, as discussed previously. When digitizing the records, it is important to note that not all of the sea ice observations had the latitude and longitude at the exact time of the observation. This led to an estimate of the location of the sea ice. Gaps also existed between the sea ice. The latitude and longitude give points of where the sea ice edge was.

3.4.2 Determining the Skill of Antarctic SIE Reconstructions

The validation of the seasonal Antarctic SIE reconstructions is completed using a variety of statistical approaches. These approaches compare the differences between the sea ice reconstructions and other estimates at approximately the same time and location. The statistical approaches used in this research are similar to historical data comparisons made for pressure reconstructions in Fogt et al. (2020).

Bias will help to depict if the reconstructions are on average or below historical independent estimates of Antarctic SIE. Mean absolute error will also be calculated to help determine the skill of the Antarctic SIE reconstructions. The mean absolute error takes the absolute value of the difference prior to averaging to avoid any cancellation between differences. Calculating this will help to understand how far the reconstructions deviate from historical estimates overall. Along with calculating the bias and mean absolute error, the root mean square error (RMSE) is also calculated. To calculate this, the differences are squared prior to averaging. The RMSE will help to further understand the reconstruction and SIE. The RMSE is calculated using the equation:

$$RMSE = \sqrt{\frac{(recon_i - observed_i)^2}{n}}$$

Equation 3.1

Where higher values of RMSE will indicate larger deviations from the observations. Lower values of RMSE will indicate the reconstructions and observations are closer together. If the RMSE is zero then this means there is a perfect reconstruction (no error).

3.4.3 Correlation

With multiple sources available through time, correlation analysis is also used to help determine the skill of the Antarctic SIE reconstructions. Correlation gives a statistical measure that expresses the extent to which two variables are linearly related, such as the reconstructed sea ice and other historical estimates of sea ice from ship logbooks or paleo reconstructions through time. Correlation makes it easy to see how two variables change together at a constant rate. It quantifies the strength of the linear relationship, and hypothesis test is employed to understand the statistical significance of various correlation values. Correlations are calculated using the equation:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Equation 3.2

where r is between a range of negative 1.0 to positive 1.0 and is the correlation coefficient, x_i and y_i are the variables at each year, $\bar{x}$ and $\bar{y}$ are the means of the two datasets. If the value of r reaches positive 1.0 then this indicates there is a perfect linear relationship. If the value of r reaches negative 1.0 then this indicates a perfect negative relationship between the datasets. For this research, it would be ideal for the value of r to be closer to positive 1.0 as this would indicate the observations and reconstructions are the same. The statistical significance of the correlation is found through hypothesis testing by using a null hypothesis of no linear relationship (r=0) and an alternative hypothesis that the correlation has a linear relationship that is either positive or negative (r≠0). The test statistic will be found using the following formula:

$$t_{n-2} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Equation 3.3

Where *n* is the sample size in years, and *r* is the correlation value from eqn. 3.2. This t-value is important as it is used to calculate the probability that the null hypothesis is true (p-value) using *n*-2 degrees of freedom.

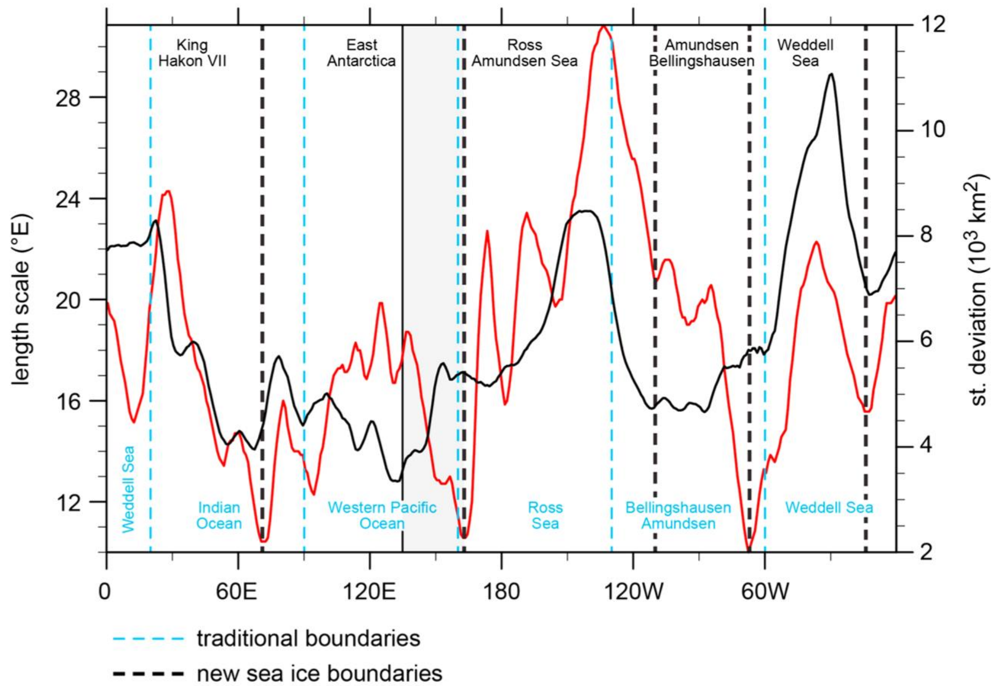
A limit to using correlation is that it does not look at the effect of other outside variables. The cause and effect are unknown when using correlation, and it does not accurately describe all relationships between two variables as it only looks at how two variables are linearly related.

3.4.4 Collection of CMIP6 Control and Historical Coupled Climate Models

Gathering of the climate models used specific parameters to make sure the correct models were used in this research. When selecting the appropriate climate models, the parameters needed were monthly sea ice concentration in the Southern Hemisphere. Downloading these data from WRCP gave the total SIE for the Southern Hemisphere. For this research, the monthly SIE for each sector (Amundsen/Bellingshausen, Weddell, Ross/Amundsen, East Antarctic, and King Hakon VII) was needed. Figure 3.1 shows the standard deviation of the sea ice extent anomalies by longitude for each of the sectors (Raphael and Hobbs 2014).

Figure 3.1

Standard Deviation of the Sea Ice Extent Anomalies for Each of the Sectors



Note: Standard deviation of the sea ice extent anomalies showing the tradition boundaries (blue dashed lines) and the new sea ice boundaries (dashed black lines). Extracted from Raphael and Hobbs 2014.

Once each sector had the SIE, an average was calculated to create seasonal means (DJF, MAM, JJA, and SON) for each ensemble member for each of the five sectors. Model averages are also calculated per year across each ensemble member for specific model configurations (such as the resolution, or variants in forcing). Once the seasonal means were created, standard deviations across the ensemble members (Table 3.1) were calculated for each year using the following equation:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{N}}$$

Equation 3.4

where x_i is the value of each population, μ is the population mean (or sample mean), and *N* is the size of the population. The standard deviation shows how the data are dispersed from the mean. If there is a low standard deviation then the data are more clustered around the mean. A high standard deviation would indicate the data are more spread out from the mean. Once the standard deviation was calculated for each year, a plot of the model averages of +/- 2 standard deviations was created along with the reconstructions. This gives a visual representation to see where the models lie within the mean to indicate how the data are spread. The closer the model standard deviations are to the mean, the more consistent each ensemble member in its depiction of the sea ice conditions in that year.

Interannual changes in the CMIP6 data are also calculated. This is achieved by calculating the difference for each consecutive year for each ensemble member. A plot is then created for the average differences along with +/- 2 standard deviations of the differences by model. This is done to be able to compare the models to the reconstructions and test for physically plausible changes in sea ice conditions from year to year in the reconstructions.

Calculating the 30-year trend for each ensemble member is important because this allows for an identification of the direction of the trend. Once this is calculated, plots of the distribution of all possible 30-year trends are created from these data. Creating these plots allows for a comparison to be made with the observations and the reconstructions to see how the trends have changed over a 30 year period, and to again determine if they trends in the reconstructions are physically valid and consistent with possible changes in sea ice extent.

3.4.5 Comparison of Seasonal Sea Ice Reconstructions to Control and Historical Coupled Climate Models

Comparisons with climate model simulations from the Coupled Climate Model Intercomparison Project 6 (CMIP6) historical and control simulations were used. Even though the CMIP6 simulations are known to have errors in Antarctic sea ice compared to observations, understanding their ranges of Antarctic sea ice variability and trends can help to understand physically valid changes in Antarctic ice on longer timescales than afforded by observations. The comparisons are used to assess if the trends and variability present in the SIE reconstructions are physically valid and therefore possible, and not just a statistical artifact of the reconstruction procedure.

Linear regression analysis is used to calculate the statistical significance of temporal trends. The trends examine how good of a fit of the least squares ordinary linear regression line through the data. For this research, the linear relationship in question is how SIE changes through time. Regression analysis focuses on dependent and independent variables. Regression analysis lets the linear trends be compared between historic sea ice information from climate models and the reconstructions. This is another means of validating the reliability of the reconstructions. The regression trends use the following equation for a best fit line:

$$y = mx + b$$

Equation 3.5

where *y* is the dependent variable, *m* is the slope of the line, *x* is the independent variable, and *b* is the intercept of the y-axis. The independent variable in this research is time (in years) as this is a factor in seasonal SIE and the dependent variable is the seasonal SIE as this is the main factor trying to be predicted or understood. The slope is calculated using the equation:

$$m = \frac{n\sum_{i=1}^{n} x_i y_i - (\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n\sum_{i=1}^{n}(x_i)^2 - \left(\sum_{i=1}^{n} x_i\right)^2}$$

Equation 3.6

which can be simplified into the following:

$$m = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Equation 3.7

where x_i is the independent variable, y_i is the dependent variable, and $\bar{x}$ and $\bar{y}$ are the means of the dataset. In order to test the statistical significance of the slope, the standard error of the slope is calculated using the following equation:

$$s_m = \frac{\sqrt{\frac{1}{n(n-2)}\left(n\sum_{i=1}^{n} y_i^2 - (\sum_{i=1}^{n} y_i)^2 - \frac{(n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i)^2}{n\sum_{i=1}^{n}(x_i)^2 - (\sum_{i=1}^{n} x_i)^2}\right)}}{\sqrt{n\sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2}}$$

Equation 3.8

The null hypothesis with no slope ($m_0 = 0$) is used as a test statistic in order to determine the probabilities of the null hypothesis using a t-distribution with the following equation:

$$t_{n-2} \sim \frac{m - m_o}{s_m}$$

Equation 3.9

where s_m is from Eqn. 3.8 , t_{n-2} is the t-distribution with *n*-2 degrees of freedom, m is the actual slope, m_o is the hypothesized slop (which is zero), and *n* is the sample size of the data.

The final calculation for the trend analysis is the 95% confidence interval (CI). This will observe the range of the values where the slope will have a high probability of occurring. The CI is calculated using the following equation:

$$95\%\ CI = m \pm\ t_{n-2} * s_m$$

Equation 3.10

where *m* is the slope, t_{n-2} is the two-tailed t-value corresponding to $p = 0.05$ with *n*-2 degrees of freedom, and s_m is the standard error of the slope from Eqn. 3.7.

Comparison of means is also used to compare the seasonal sea ice reconstructions to control and historical coupled climate models. This allows for two or more variables in a group of data to be compared statistically to assess their differences. The goal is to estimate the mean, assess the variation based on sample estimates, and use the information to provide evidence of the central tendency. There are several ways to assess the comparisons such as t-tests and the analysis of variance.

3.4.6 Determining Decadal Scale Changes Using a T-Test and F-Test

Decadal scale changes will be analyzed using t-tests. The t-test is the method used for comparing one group to a value or the mean of one group to another. These perform well when there are minor or moderate departures from normal. Calculating the t-test is performed by using the following equation:

$$t_{m+n+2} = \frac{(\bar{X}-\bar{Y})-(\mu_x-\mu_y)}{S_p\sqrt{\frac{1}{n}+\frac{1}{m}}}$$

Equation 3.11

where $\bar{x}$ and $\bar{y}$ are the means of the two data sets, μ_x and μ_y are the population mean (sample mean) of the two separate datasets, *n* and *m* are the population size of each of the datasets, and S_p is the pooled sample standard deviation. S_p is calculated using the following equation:

$$S_p = \sqrt{{S_p}^2}$$

Equation 3.12

where S_p^2 is the pooled sample variance. The pooled sample variance is calculated using the following equation:

$${S_p}^2 = \frac{(n-1){S_x}^2 + (m-1){S_y}^2}{m+n-2}$$

Equation 3.13

where *n* and *m* are the population size of the two data sets and S_x^2 and S_y^2 are the variances for each of the data sets, which are assumed to be equal for each data set. The variances for each of the two data sets were calculated using the following equation:

$$S_x{}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - x)^2$$

Equation 3.14

When calculating the t-test, the null and alternative hypotheses need to be determined. The null hypothesis, that is hoping to be disproved, is that means or equal ($\bar{x} = \bar{y}$), or that there is no change in average sea ice extent on the decadal time scale. The alternative hypothesis, what is hoping to be accepted, is that there is a change on the decadal time scale. The acceptance or rejection of the null hypothesis is based on the critical acceptance value of $p < 0.05$. The equation used to calculate the t-test statistic for two mean comparison was:

$$\bar{x} - \bar{y} - (\mu_x - \mu_y) \pm t_{m+n-2(\frac{\alpha}{2})} S_p \sqrt{\frac{1}{n} + \frac{1}{m}}$$

Equation 3.15

The t-tests are calculated on comparison of means for 10-year periods. Performing the t-test will allow a comparison in the decadal changes between the two groups. A determination can then be made on whether or not the process has an effect on the decadal scale changes. This same process is also used for the model ensemble means.

To look at the variability in the CMIP6 ensemble members in one model, and between various CMIP6 models, the standard deviation, using Eqn. 3.4, was calculated for 10-year periods. Once these standard deviations are calculated, f-tests are performed on all possible pairs of standard deviations. The equation used to calculate the f-test was:

$$f = \frac{\sigma_1{}^2}{\sigma_2{}^2}$$

Equation 3.16

where σ_1 and σ_2 are the variance of two separate datasets. The F-test is used to determine if there is a potential model overfitting.

Performing the f-test will help determine if there are realistic changes in variability in the reconstructions and models. The f-test statistic will help determine whether or not there is a probability to reject the null hypothesis (they are equal). This gives a chance to look at the different time periods to see the variability changes through time.

Chapter 4: Analysis

4.1 Ship Logbook Data

4.1.1 Scanned Logbook Archives Previously Unkeyed

Antarctic sea ice plays a vital role in Earth's climate, as the literature review describes. Images of the Antarctic sea ice extent are short, as the satellite era of remote sea ice observations began in 1979. To be able to determine what the sea ice extent was like prior to the satellite era, ship logbooks are utilized. The ship logbooks help put together a greater picture of what the SIE was like by being able to synthesize pieces from day-to-day explorations and commercial ventures that tell where the ice edge may lie. The ship logbooks contain a wide variety of information ranging from meteorological data such as temperature or pressure to the position of whale catches or if they have encountered anything unusual on their trip. Many of the logbooks will mention sightings of sea ice. This is usually written in the logbook in a separate remarks area, qualitatively described as 'pack ice' or 'ice pack' and is usually written when they either see it in the distance or if they have come into contact with the sea ice itself.

Ohio University was able to gather digital scans from multiple ship logbooks that ranged from various expeditions to commercial logbooks from whaling or sealing industries. This list of available ship logbook data provided to Ohio University is shown in table 4.1, detailing the archive the logbooks came from, the relative starting and ending years the ships traveled, the general location where the ships sailed, and how many expeditions are within the archives. While the information in Table 4.1 seems exhaustive, it is in reality only a small portion of available logbook data; there are an unknown additional number of voyages and information that have not yet been recovered. Thus, additional work is needed for historical data recovery to aid in other research projects like this one aiming to understand climate variations in regions where direct observations are scarce or altogether absent.

The first step in getting the ship logbook ready for examination for this project was to get them digitized. Thousands upon thousands of pages were carefully examined for mention of sea ice. In addition to my own work, several undergraduate student researchers at Ohio University worked to digitize the data from the ship logbook to excel files. Originally, the undergraduate students were key entering all meteorological data, to provide a new and robust dataset of not only sea ice, but also of many other climate parameters from these important sources. However, this was time consuming, taking over 30 minutes to read the difficult writing, and in some cases, translate the information into English, a single page (roughly 1 month). Because most of these observations did not contain information on sea ice, the considerable time to key enter/digitize all the meteorological observations was deemed not efficient, and the Ohio University team started scanning pages and only keying in dates when sea ice information was included. To expedite the process even further, only the ship's date, latitude, longitude, and relevant sea ice information were digitized. This dramatically reduced the number of observations from the available sources in Table 4.1.

Table 4.1

Archive of Available Ship Logbooks at Ohio University

Archive	Start Date	End Date	Locations Traveled	Number of Voyages
British Antarctic Survey	1948	1970	North Atlantic, South Atlantic	184
Sveta	1929	1940	Southern Ocean	23
National Meteorological Archive	1873	1960	Caribbean, South Pacific, North Atlantic, Southern Ocean	1,044
Christian Salvesen Archive – University of Edinburgh	1932	1959	Southern Ocean, Weddell Sea	65
Sea Mammal Research Unit St. Andrews University	1853	1963	Indian Ocean, Pacific Ocean, Weddell Sea, Ross Sea, Antarctic whaling grounds	284
National Archives of Australia	1947	1961	South Pacific	4
Museo Maritimo Nacional, Valparaiso	1916	1970	South Pacific, North Atlantic	79
Archives Nationales, Paris	1738	1840	North Atlantic	4
National Archive of New Zealand, Wellington	1956	1969	South Pacific	15
Hvalfangstmuseet (Whaling Museum), Sandefjord	1893	1942	North Atlantic	120
Norwegian Maritime Museum Oslo	1910	1931	Southern Ocean	32
Vestfold Archivet, Sandefjord	1916	1962	Southern Ocean, Antarctic whaling grounds	108
British Antarctic Survey	1819	2011	North Atlantic, Southern Ocean	387
British Geological Survey	1844	1845	Southern Ocean	1

Table 4.1 continued:

Archive	Start Date	End Date	Locations Traveled	Number of Voyages
Climatic Research Unit, University of East Anglia	1897	1947	North Atlantic, Southern Ocean	9
Edinburgh University Special Collections	1932	1930	North Atlantic, South Atlantic	64
Met Office Hadley Centre	1898	1930	Southern Ocean	5
National Oceanographic Centre, Southampton	1925	1985	South Atlantic, Southern Ocean	16
National Archives, Kew	1793	1969	South Atlantic, South Pacific, Southern Ocean	65
National Maritime Museum, Greenwich	1855	1922	South Atlantic	2
National Meteorological Archive, Exeter	1845	1957	Southern Ocean, South Atlantic	151
Royal Geographical Society	1819	1904	North Atlantic, Southern Ocean	10
Scott Polar Research Institute	1971	1970	South Pacific, Southern Ocean, South Atlantic	185
University St. Andrews SMRU	1929	1967		170
G.W. Blunt White Library, Mystic Seaport	1836	1887	South Atlantic	11
ICOADS	1932	1970	Southern Ocean	35
National Archives (NARA), Washington	1838	1973	Southern Ocean, Ross Sea	160
New Bedford Whaling Museum	1853	1951	South Atlantic, Southern Ocean	24
Gallica	1838	1840	Southern Ocean	2
National Meteorological Archive, Exeter	1839	1895	South Pacific high latitude	403
Maury	1830	1858	South Pacific high latitude	25
Norwegian Logbooks	1930	1957	South Atlantic, Southern Ocean	24

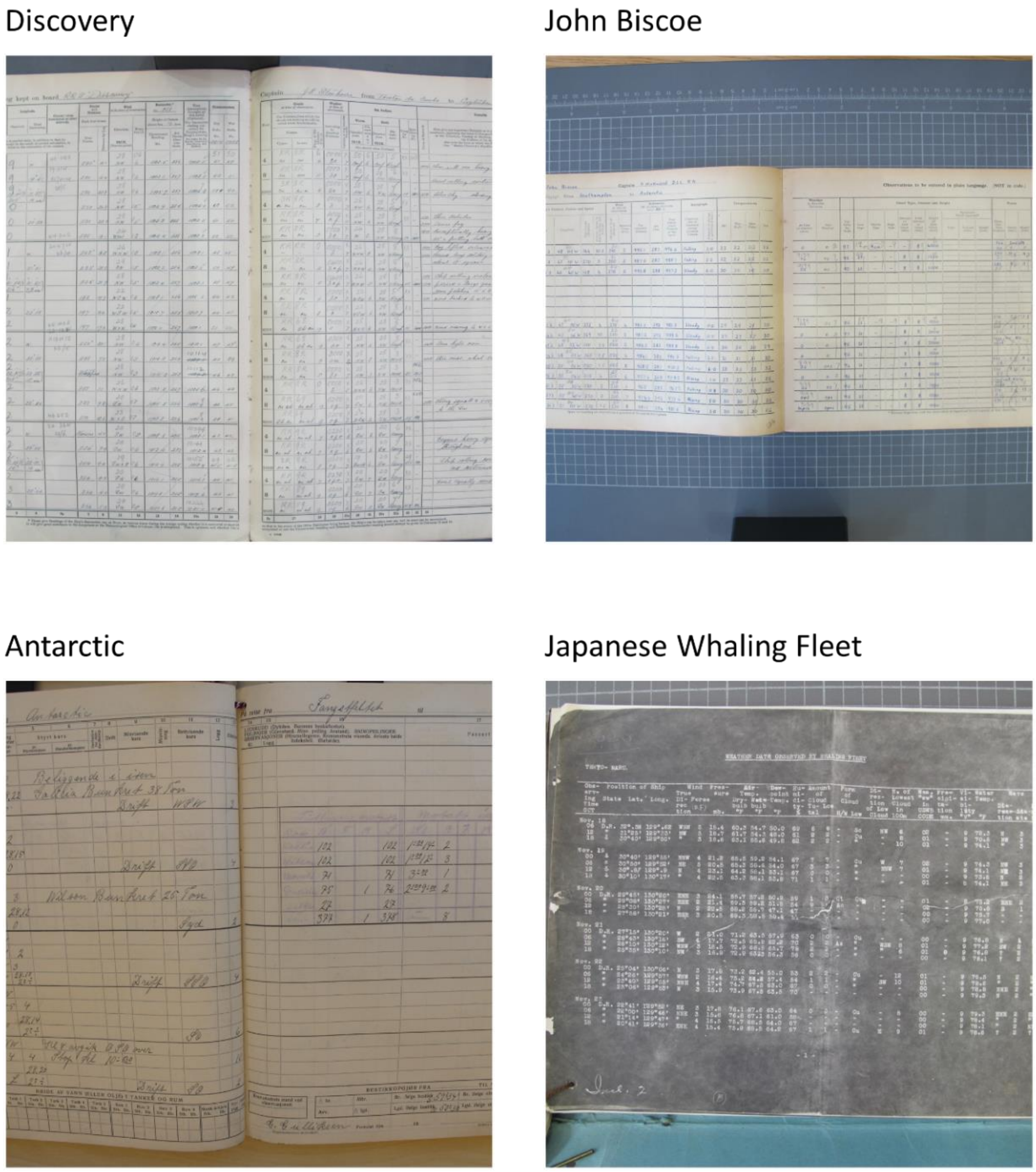
Note: This table gives the general data associated with the available archives of logbooks at Ohio University.

As mentioned before, it is important to recall that within these archives in Table 4.1, not all of the expeditions had sea ice written or logged within the logbooks, and it isn't always clear if this meant there was not ice encountered, or if that company/group chose not to make remarks about the sea ice information. When digitizing the logbooks, the contents written in were all copied over into Excel files. If sea ice was mentioned in the logbooks, it was important to find the associated latitude and longitude of where the sea ice sighting was. The latitude and longitude is then used in this research to help determine where the sea ice edge was to be able to complete the full picture of what the SIE looked like. While the majority of entries in the logbook did not include sea ice information, there were times within the logbooks when sea ice was mentioned, but there was no latitude and longitude listed. Because of the harsh conditions in Antarctica during the winter months, a majority of the expeditions occurred in the months of November through March. The logbooks start as early as the 1700s and go as late as the 1980s. The setup of the logbooks can be different from one to the next, as seen in Fig. 4.1, but they are able to all still give the same information. While a majority of the ship logbooks at Ohio University are in English, there were some logbooks in Norwegian that needed to be translated. In this case, specific Norwegian words for ice were searched for, such as 'beliggende i isen', 'pakkis', or 'isen' since these words translated to ice pack, pack ice, or ice edge.

During the digitization process of the logbooks, an Excel file was created for each of the major group/company/operation from the logbooks at Ohio University in Table 4.1. When there was mention of sea ice in the remarks column, the year, month, and day were recorded for that sighting. The latitude and longitude, if written in the logbooks, were recorded in the excel file, as this information is extremely helpful in determining where the SIE was. The latitude and longitude are used to take the single data point and convert it into an area to determine how reliable these logbooks can be in evaluating Antarctic sea ice extent reconstructions. Along with the dates and coordinates, any remarks about the sea ice were logged. The remarks ranged anywhere from being at the ice edge to spotting the sea ice or sea ice being a couple of miles from a location.

Figure 4.1

Examples of Pages from Ship logbooks



Note. The top left photo is from the Discovery expedition, the top right photo is from the John Biscoe expedition, the bottom left photo is from the Antarctic expedition, and the bottom right is from the Japanese whaling fleet.

4.1.2 Other Exiting Logbook Data, Previously Keyed

There is more information beyond what was keyed from the Ohio University undergraduate student researchers. Table 4.2 outlines the additional logbook data beyond the archives that have been previously keyed by other researchers outside of Ohio University, which may prove beneficial. Most of the listings in Table 4.2 are from specific early expeditions to Antarctica. The table details the name of the expedition (or external existing data source, where applicable), the start date, the end date, the locations traveled during the expedition, and remarks of the expedition.

Table 4.2

Additional Logbook Data Beyond the Archives Keyed from Ohio University Undergraduate Student Researchers

Expedition / Data Source	Start Date	End Date	Locations Traveled	Remarks
Aurora	1911	1947	Southern Ocean	Australian expedition with objective of exploring Antarctica directly below Australia
Imperial	1914	1917	Transatlantic	Objective of Antarctic exploration
Nimrod	1907	1909	North Atlantic, Southern Ocean	British expedition with objective of reaching South Pole

Table 4.2 continued:

Expedition / Data Source	Start Date	End Date	Locations Traveled	Remarks
Porquoi Pas	1908	1910	North Atlantic, Southern Ocean	French expedition with objective of sailing to Antarctica
Discovery	1929	1931	North Atlantic, Southern Ocean	British expedition with objective of working in Ross Sea sector
Terra Nova	1910	1913	Southern Ocean	British expedition with objective of reaching the South Pole
USS Bear	1939	1940	Southern Ocean	Sealing vessel
Teleti	1933	1960	Southern Ocean	Historical Southern ocean climate dataset from whaling records (Teleti et al. 2019)
Edinburgh and Day	1897	1917	Southern Ocean	Estimating the extent of Antarctic summer sea ice during Heroic Age (Edinburgh and Day, 2016)
de la Mare	1931	1995	Southern Ocean	Changes in Antarctic SIE from whaling records (de la Mare 2009)
SOFI	1903	2008	South Orkney Islands	Timing of formation and breakout of fast-ice

Note: Expeditions not keyed by Ohio University undergraduate student researchers.

The objective of a majority of these expeditions in Table 4.2 was to explore Antarctica, reach the South Pole to conduct further research. During these expeditions, just like the ship logbooks from Table 4.1, mention of sea ice was often noted in the expeditions logbooks listed in Table 4.2. Teleti et al. (2019), Edinburgh and Day (2016), and de la Mare (2009) are all papers that aim at using various data sources to estimate the pre-satellite observation SIE in Antarctica. Teleti et al. (2019) used historical logbooks, specifically the Christian Salvesen Whaling Company logbooks from whaling ships, to look at the meteorological observations in the Southern Ocean. The goal of Teleti et al. (2019) is to be able to use the meteorological observations found in logbooks of whaling vessels to help fill a data void in the Southern Ocean and potentially be useful in generating more accurate premodern climate datasets. Edinburgh and Day (2016) analyzed observations from ship logbooks during the Heroic Age of Antarctica Exploration (1897-1917) during the summertime to look at the sea ice edge. Their research was used to estimate the summer SIE in Antarctic in this period. Finally, de la Mare (2009) looked at the changes in Antarctic SIE using historical observations and whaling records. Through de la Mare (2009) data archive , the sea ice edge, ice edge mean, southern most catch of whales, and whale catch estimated ice edge were calculated using the historical observations and whaling catch records to analyze possible values of early 20th century Antarctic SIE.

4.1.3 Actual Logbook Data Used in this Research

As mentioned before, there is an overwhelming number of ship logbooks to be digitized, and since the process requires reading every single entry from digital scans of hand-written logbooks, it is a tedious process to do manually. Because each page of the logbook was scanned individually to preserve the book form of the data, each image is saved as a separate file, requiring an individual to open up to 100 files just to read every possible entry within a logbook. The process can therefore take several hours or longer for just one entry of ice information to store for this research. Even with additional undergraduate students aiding in the data keying aspects, it is not possible within a single year of these research time to key enter all the data available in Table 4.1 Since this is very time-consuming, and much more research can be conducted further from the ship logbooks, only certain logbooks were used for this research to estimate the SIE in Antarctica. Table 4.3 outlines the ship logbooks used specifically in this research, including some of the existing data in Table 4.2. The latitude, longitude, date, and remarks were utilized to make the estimates to be able to compare them to the reconstructions from Fogt et al. (2022). The observations from the logbooks listed in Table 4.2 were compiled into an Excel file. Since these observations are daily observations of the ice edge, a procedure was created in order to compare the daily ice edge seasonal, sectoral means. This procedure is detailed in the following section.

Table 4.3

Ship Logbooks Used in this Research

Archive	Start Date	End Date	Locations Traveled
British Antarctic Survey	1948	1970	North Atlantic, South Atlantic, Southern Ocean
National Meteorological Archive	1936	1960	Caribbean, South Pacific, North Atlantic, Southern Ocean
Teleti	1933	1960	Southern Ocean
Nimrod	1907	1909	North Atlantic, Southern Ocean
Discovery	1929	1931	North Atlantic, Ross Sea sector
Terra Nova	1910	1960	North Atlantic, Southern Ocean
Aurora	1911	1947	South Pacific, Southern Ocean
USS Bear	1939	1940	Southern Ocean
Norwegian Logbooks	1930	1957	South Atlantic, Southern Ocean

Note: The latitude, longitude, and date were used from these logbooks to compare the daily ice edge with seasonal, sectoral means.

4.2 Comparing the Daily Ice Edge with Seasonal, Sectoral means

Since the ship logbooks gave the observations of sea ice in a single point, on a single day, they have to be adjusted to relate to the seasonal and sectoral means of the Fogt et al. (2022) sea ice reconstructions. In order to achieve this, the daily ice edge recordings needed to be converted to an area that would be able to compare to the seasonal, sectoral means. A procedure to make this comparison was created as follows.

The first step in the procedure was to create a program to get the ice edge for each day in the satellite observational record. The program extracted daily values of the ice edge (as a latitude of the 15% sea ice concentration) for each longitude, for each year (1979-2020). This program created netcdf files that were used in later steps. The first dimension in the netcdf file was the date (yyymmdd), the second dimension was the longitude (ranges from 0-359, at 1° increments), and the third dimension was the statistic (maximum equatorward latitude of at least 15% SIC, and two values to check the longitude).

Once this netcdf data file was created, the next step was to create another program to create a regression relationship, by longitude, between the daily values of the maximum latitude 15% SIC (what the logbooks give us) and the seasonal mean, sectoral mean SIE expressed as an area (what the reconstruction gives us). This program read from the first program (the ice edge statistics from 1979 to 2020) netcdf file, and these daily values of longitude for each day within the sector were used for the dependent variable (y) along with the seasonal, sectoral mean SIE as the independent variable (x) for all years. Once this was created, a regression line was then fit for each longitude over all the possible years. The sector means shifted depending on the longitude, using the longitude boundaries for each sector (Weddell Sea 290°E-364°E, King Hakon Sea 346°E-71°E, East Antarctica 71°E-162°E, Ross-Amundsen Seas 162°E-250°E, and Amundsen-Bellingshausen Seas 250°E-290°E). The regression statistics were then saved into a text file for later use. In the text file, the first column represented the season, the next column is the longitude (0-359), the regression coefficient (slope of the regression line), the y-intercept, the regression standard error (2 sigma range of the regression coefficient), r-squared value (a measure of how good the regression is, at each longitude for a given season), the average area for the season and sector (this value remains the same until moved into a different sector, or a different season), and the average maximum latitude from 1979-2020 based on daily data for that longitude, within that season.

The regression statistics text file was used to make the comparisons with the ship logbook data. A specific day (yyyymmdd) was used where the logbook specified it saw sea ice. To make an estimate of where the reconstruction says the 15% latitude of the sea ice edge is, the reconstruction seasonal, sectoral mean value of SIE for the sector that falls within the longitude bounds of the ship logbook data is used (the reconstruction value). This reconstruction value is then multiplied by the regression coefficient, and then the y-intercept is added to convert from seasonal, sectoral area mean, to a daily estimate of the ice edge at a specific longitude. This is the reconstruction's best estimate at the ice edge. However, the ice edge varies a lot daily. Since this is the case, the 95% confidence interval needed to be looked at as well. This was achieved by multiplying the reconstruction value by the regression standard error value and adding the y-intercept value. This value is then added and subtracted to the reconstruction best estimate value of the ice edge to get the 95% range of possible daily values; it is anticipated that approximately 95% of the daily ice edge ranges would occur within this latitude range specified by the 95% confidence interval.

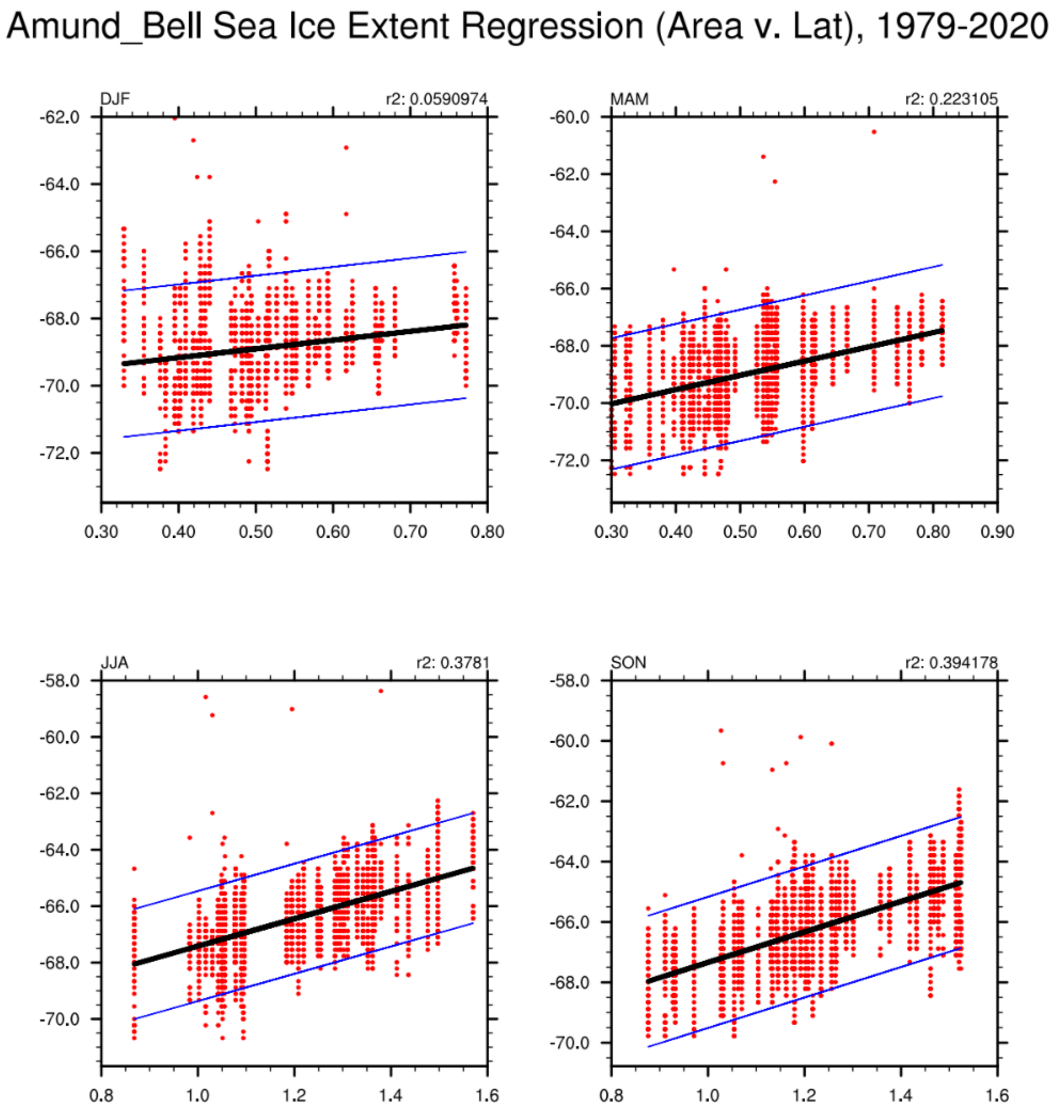
To help visualize the relationship between the maximum latitudes of the 15% SIC with the seasonal mean, sector average sea ice extent expressed as an area, the program that plotted regression for all years was used. This program extracted the daily values of the maximum latitude of the 15% sea ice concentration for each longitude from the ice edge statistics from 1979 to 2020 netcdf file. The latitudes from this were compared to the SIE of the sector mean, seasonal mean sea ice extent. From here, figures (Figs. 4.2-4.6) were created to plot the seasonal mean, sector mean sea ice area (a measure of the sea ice extent used in the reconstructions, based on satellite data) with all the possible daily values of latitude for that year within the season, at a specified longitude as an example for each sector. The specified longitudes (in degrees East) are: 330°E (Weddell Sea), 45°E (King Hakon Sea), 120°E (East Antarctica), 200°E (Ross-Amundsen Seas), and 270°E (Amundsen-Bellingshausen Seas). Least-squares linear regression was calculated to measure the fit of the data.

The regression figures (Figs. 4.2-4.6) detail how complex the Antarctic sea ice is from day to day and year to year, and how challenging it is to relate a single point observation in the logbooks of the ice edge latitude to the seasonal, sectoral mean SIE expressed as an area from the reconstructions. Figures 4.2-4.6 tell the story of how the sea ice edge latitude can change dramatically within a given season each season and each sector. Within these figures, the red dots represent the daily values of where the ice observations were for each year between 1979-2020. The y-axis represents the latitudes and how far north or south the daily ice values are. The x-axis represents the seasonal mean area of the sea ice within that sector, from the reconstruction. The two blue lines on each figure represent the 95% confidence interval, where a majority of the daily latitude observations should fall between. Any red dot which falls outside the 95% confidence interval line is an outlier in the data not captured by the regression relationship. The black line in each of the figures represents the regression line relating the ice edge based on the daily observations to the seasonal, sectoral mean sea ice extent as an area. When looking at Figures 4.2-4.6, a better relationship between these two measures would be marked with a positive regression line; further, the greater the slope, the greater the change in daily sea ice edge to changes in seasonal mean, sectoral mean sea ice extent. For this research, it is ideal for the slope to be greater as this would make reconstructions better for predicting the daily SIE as given in historical logbooks, at any given longitude.

Figures 4.2-4.6 show the variability in the sea ice. The ice edge varies, quite a bit, depending on the season. The is prevalent especially during the summer months (DJF). In the Ross Amundsen (Fig. 4.5), the range of the latitude for the daily SIE within any given sector spans over 12° latitude. As the seasons progress, the range of the SIE decreases, and is less in the winter months. The range is higher in the summer season compared to the other seasons because the daily sea ice edge is much more variable, influenced by the greatly reduced amount of sea ice in summer. Because the SIC is lower in the summer months, the pack ice is able to move more freely from one day to the next. For example, if a storm comes through on a summer day, it can easily push the sea ice around, whereas if a storm passes during a winter day, it is harder to move as the sea ice is more dense. In East Antarctica (Fig. 4.3) and King Hakon (Fig. 4.4) sectors, the relationship between the seasonal mean area of the sea ice (reconstruction) versus the latitude of the daily observation is not great. Apart from the width of the 95% confidence intervals (blue lines in Figs. 4.2-4.6), the ‘goodness of fit / r^2’ value can (given in the upper right of each regression plot), can approximate how much of the daily sea ice variations are linearly explained by seasonal sectoral means of sea ice extent. Unfortunately, most of the relationships in each season between the two fall below 10%. In fact, Fig. 4.3 and Fig. 4.4 barely have a slope, indicating the predictions of the daily latitude of the sea ice edge from a seasonal sector mean reconstruction value would not be great. Unsurprisingly, the relationship is weak in East Antarctica (Fig. 4.3) and King Hakon (Fig. 4.4) sectors, as these regions experience the lowest amount of summer Antarctic sea ice (nearly all the ice melts in these regions most summers).

Figure 4.2

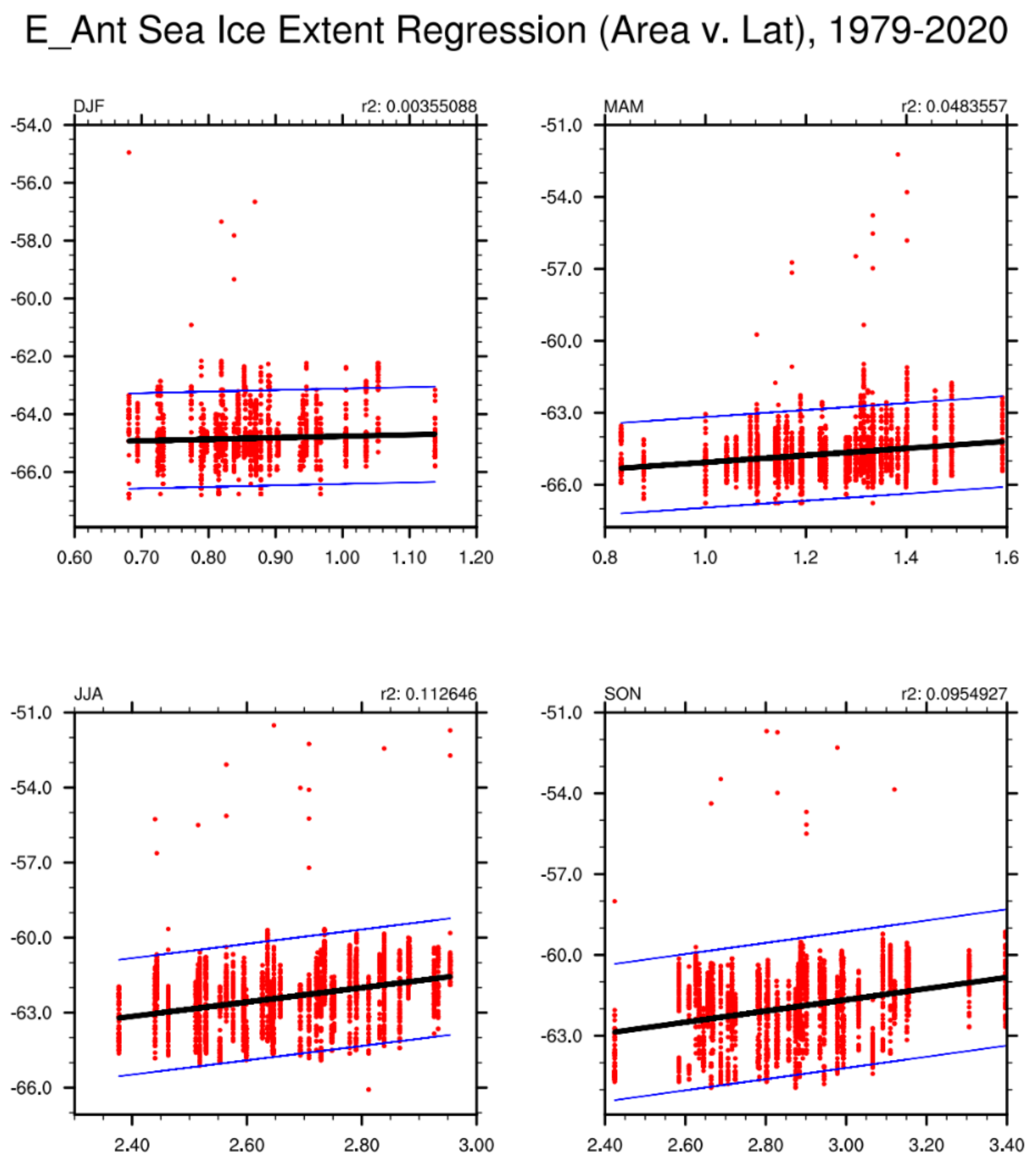
Relationship Between the Daily Maximum (equatorward) Latitudes of the 15% SIC with the Seasonal Means for Amundsen-Bellingshausen Seas



Note: Seasonal mean SIE along the x-axis is expressed as an area, 10^6 km^2.

Figure 4.3

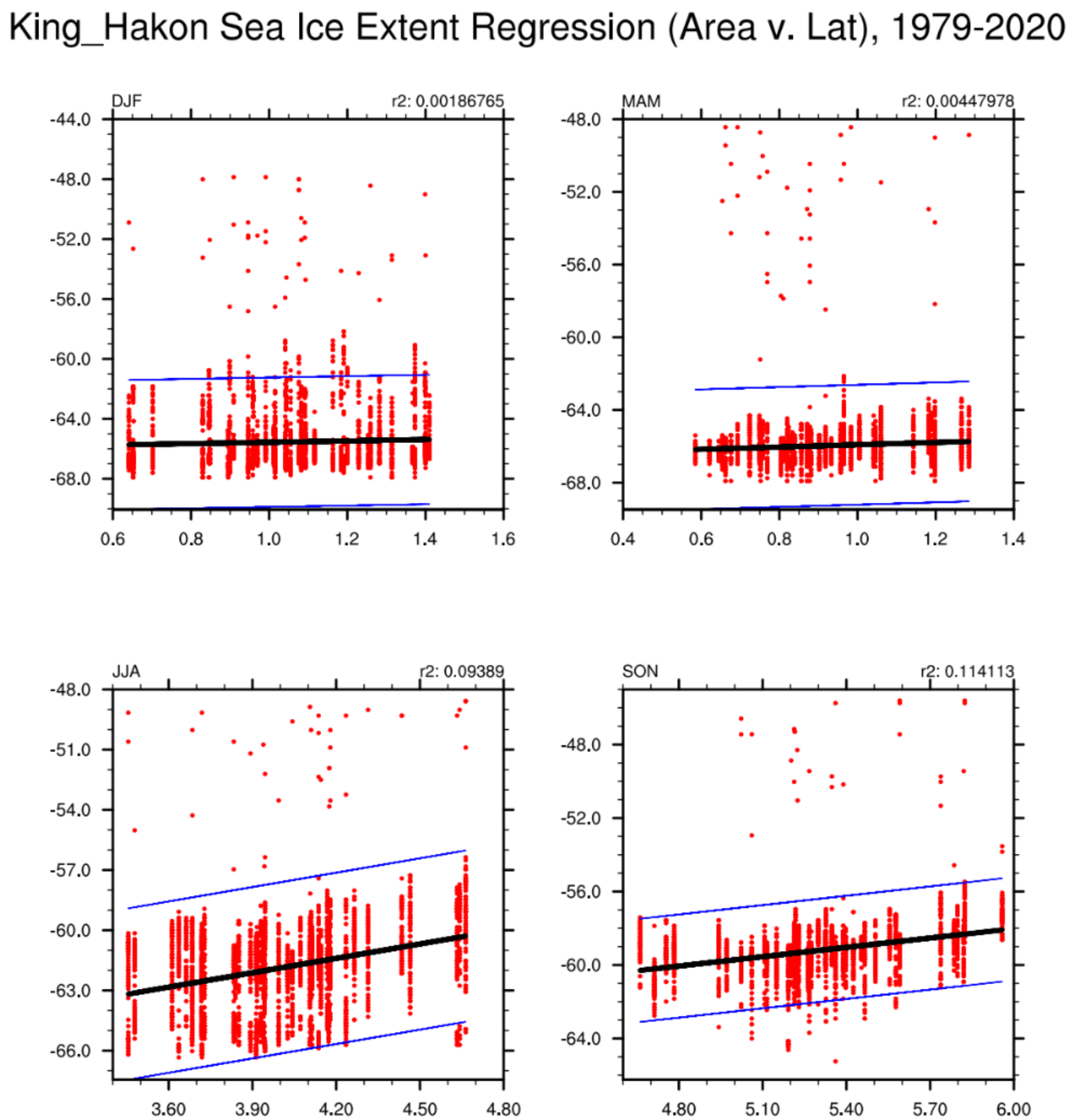
Relationship Between the Daily Maximum (equatorward) Latitudes of the 15% SIC with the Seasonal Means for East Antarctica



Note: Seasonal mean SIE along the x-axis is expressed as an area, 10^6 km^2.

Figure 4.4

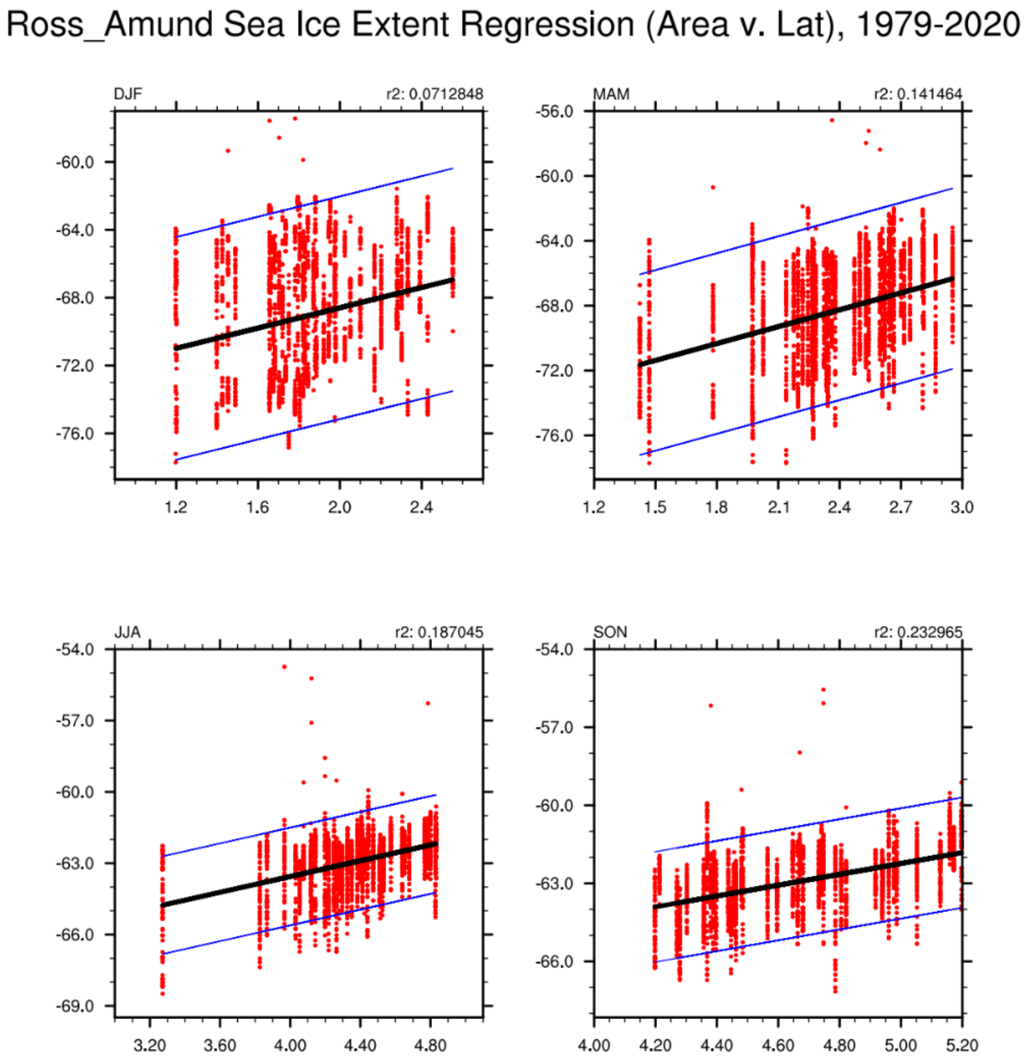
Relationship Between the Daily Maximum (equatorward) Latitudes of the 15% SIC with the Seasonal Means for King Hakon Sea



Note: Seasonal mean SIE along the x-axis is expressed as an area, 10^6 km^2.

Figure 4.5

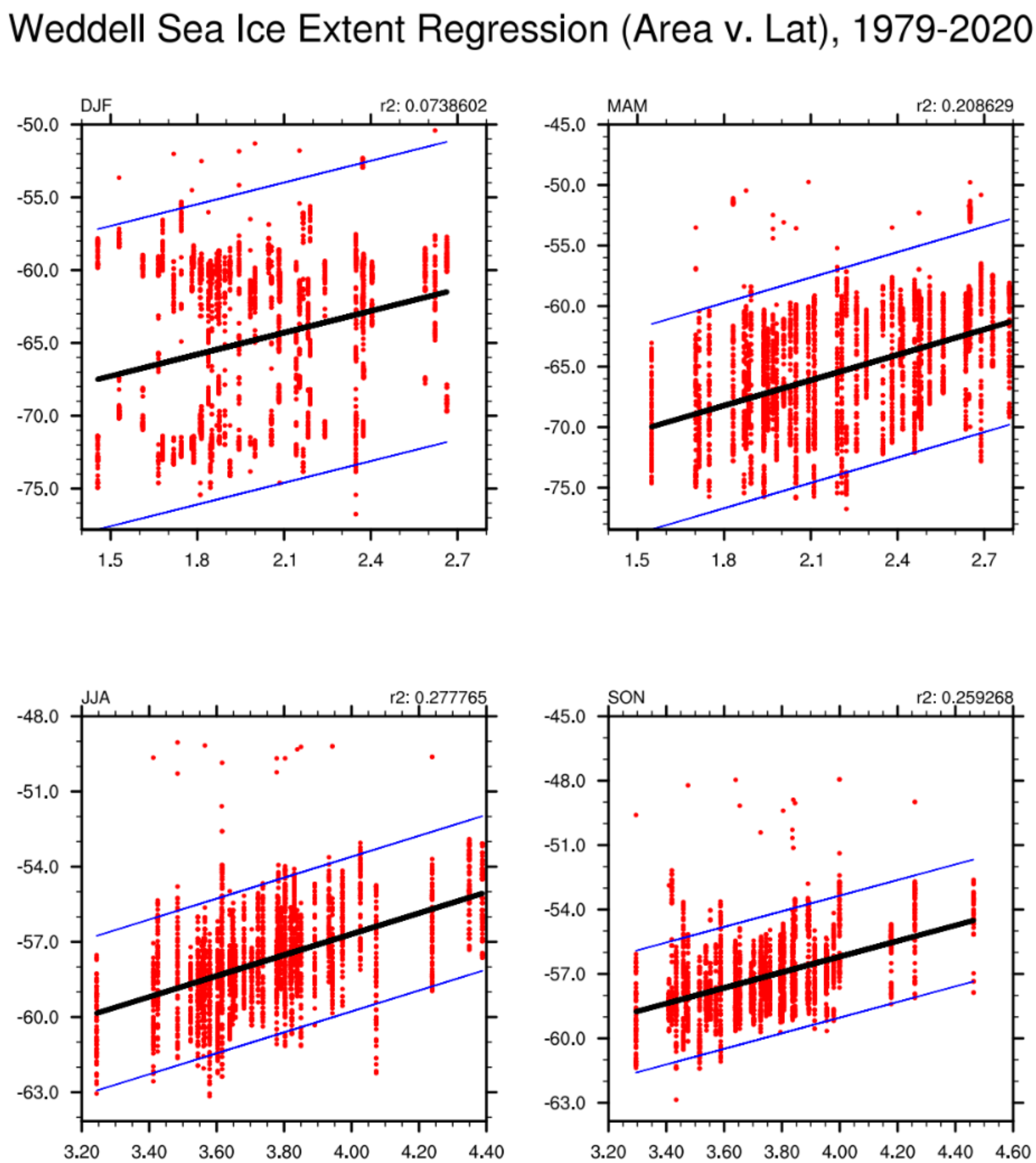
Relationship Between the Daily Maximum (equatorward) Latitudes of the 15% SIC with the Seasonal Means for Ross-Amundsen Seas



Note: Seasonal mean SIE along the x-axis is expressed as an area, 10^6 km^2.

Figure 4.6

Relationship Between the Daily Maximum (equatorward) Latitudes of the 15% SIC with the Seasonal Means for Weddell Sea



Note: Seasonal mean SIE along the x-axis is expressed as an area, 10^6 km^2.

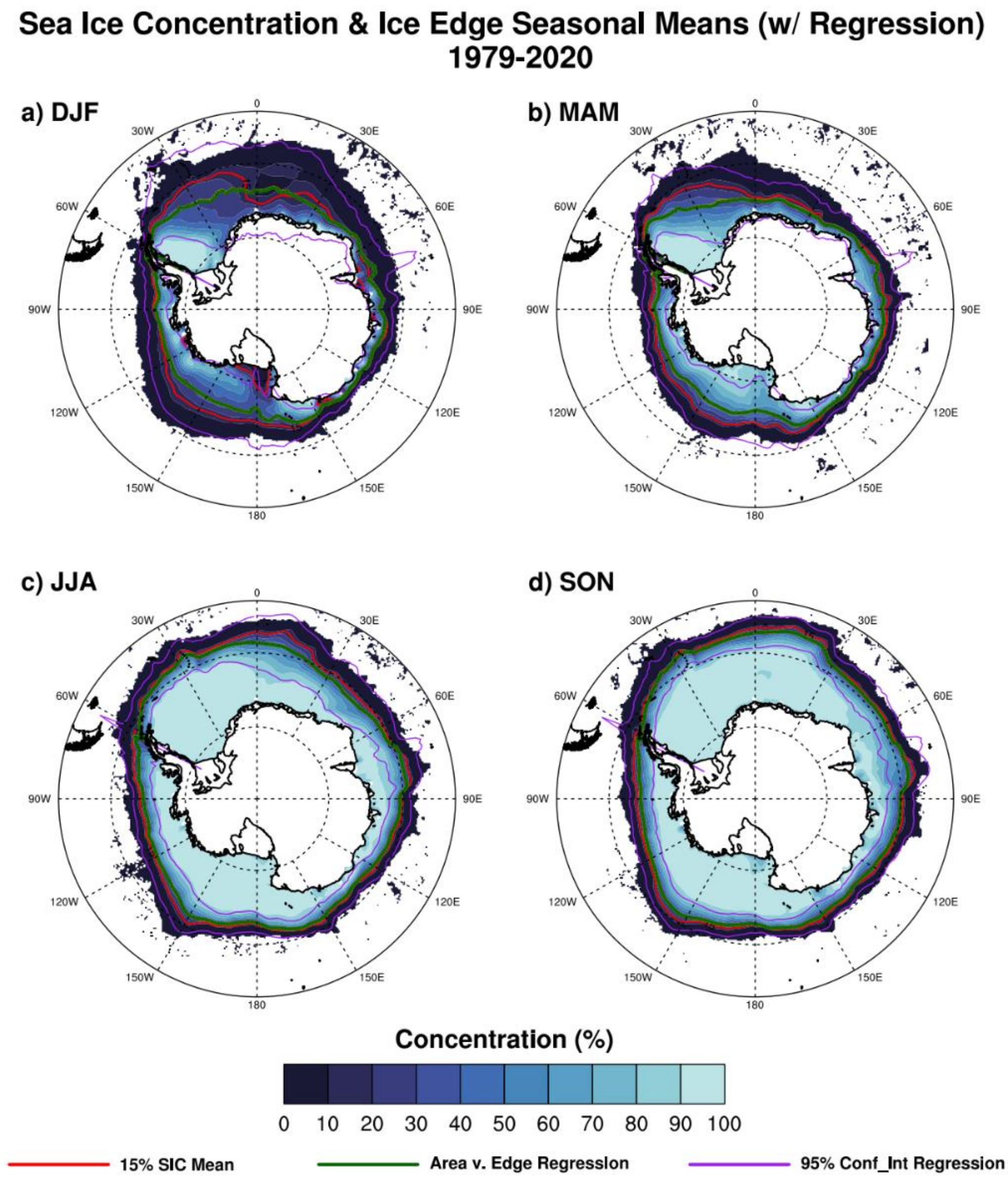
While Figs. 4.2-4.6 provide some detail on the challenges of making logbook to reconstruction comparisons of ice edge, these only represent certain longitudes and do not help to see variations around the entire Antarctic continent. Visualizing the regression relationships spatially was done by using a program that plotted the 15% seasonal concentration with the regression estimate. This program averaged the seasonal mean sea ice concentration during 1979 to 2020, and plotted the 15% SIC isoline on the time-averaged field as a red line (Fig. 4.7). To approximate the ice edge based on area, it used the seasonal mean sector area and the regression coefficient and y-intercept, to provide a best estimate of where the 15% SIC isoline would be, based on the seasonal mean, sector mean regression relationships (Figs. 4.2-4.6 as examples). This plot visually displays the errors/skill associated with linking seasonal mean, sector mean SIE to a possible daily value of SIE at a specific longitude. From Fig. 4.7, the observed average 15% SIC mean is plotted in red, while the estimated 15% line based on the area verses edge regression is plotted in green (the average of data points, as shown as a black line in Figs. 4.2-4.6), and the 95% confidence interval about the regression is plotted in purple. Within Fig. 4.7, the 95% of the daily latitudes of the ice edge observations based on seasonal, sectoral mean should fall within the purple lines, and if the regression is reliable, the red (actual average 15% isoline) and green (estimated average 15% isoline) should match closely. The blue shaded region within each season represents the long-term mean SIC. If the SIC is great (full cover), the area will appear more white. If the SIC is less, then the area will be a darker blue (more open ocean).

As seen in Fig. 4.7, the SIC is clearly much less in DJF (summer) than in JJA (winter), but it is also much more dense in winter than in the summer. As mentioned earlier, the lower density in summer aids in the variability of the edge of the daily sea ice edge. As such, the relationship between seasonal values of the ice extent and daily measurements of its edge at a given longitude are much weaker in DJF. In Fig. 4.7, this reduced relationship in DJF is apparent by the wide range of variability in the sea ice through the 95% confidence interval (purple lines). The range between the outer purple line and inner purple line is great, essentially stretching from the coast of Antarctica outward to the 0% ice edge – nearly all of the sea ice zone. This reduced relationship in Fig. 4.7 reflects what is seen within Figs. 4.2-4.6. Also, within DJF the estimate 15% isoline (green line) does not match up well with the actual average 15% isoline (red line). However, when looking at JJA and SON, the range of the data is much less (as shown in Figs. 4.2-4.6). In turn, the regression estimate improves (Figs. 4.2-4.6), and the estimate of the ice edge follows the actual ice edge very closely, indicating the estimates during these seasons are much better.

Apart from the seasonal changes in the regression relationship skill, another important aspect to consider in making these comparisons is the fact that observations coming from the ship logbooks will be off from the actual ice edge. When the logbooks are recording the ice edge, they are likely not giving the 15% isoline (what the regression estimates in Figs. 4.2-4.7); rather they are likely reporting the most equatorward edge of the ice, approximately the 0% ice line. The ships will record the sea ice in the logbooks at a latitude that is north of the actual ice edge because the ship is not going to sail into the ice. If the ship sailed into the ice, this would likely impede their mission or they would risk crashing the ship. Because of the ships reporting the ice edge further north than what it actually was, especially north of the 15% concentration, many of the observations will likely not be within the 95% confidence interval (Figs. 4.2-4.7). Further, as will be discussed in the following section, when the ships sailed, they mostly sailed during the summer season (DJF) because the conditions during this time of year in Antarctica are calmer. Also during this time, there is less sea ice for the ships to get stuck in, and the whales and seals hunted by many commercial ships are more active (and easier seen in prolonged daylight).

Figure 4.7

Sea Ice Concentration and Ice Edge Seasonal Means from 1979-2020



Note: Red line represents the 15% SIC isoline, the green line represents the area versus the edge regression estimate of the latitude of the 15% line, and the purple lines represent the 95% confidence intervals.

4.3 Comparison of Ship Logbook Data to Reconstruction Seasonal, Sectoral Means

4.3.1 Logbook Sea Ice Data Counts

Using the procedure listed above, the available data from ship logbooks used in this research was analyzed to help determine the skill of seasonal Antarctic SIE reconstructions from Fogt et al. (2022) and how its skill varies spatially and temporally prior to 1979. Since the year, month, day, latitude, and longitude were already documented in an Excel file during the key entering of the ship logbook data, the season and sector corresponding to the individual daily logbook records were identified. This information made it useful to determine the regression coefficient (as a function of longitude and season), the y-intercept, and the regression standard error to be used for the specific date listed. The corresponding reconstructions from Fogt et al. (2022) were also put into the Excel file to help determine the reconstruction best guess for the sector mean, seasonal mean SIE (as an area).

Table 4.4 summarizes the logbook by data counts per decade, sector, and season. From the various sources in Table 4.3, the total amount of ice latitude estimates that were keyed and available for this research from the logbooks is 1,586 points. Through time, the 1930s had the most data points (702 points), with the 1960s and 1990s having zero data points within the data used for this research. Spatially, Amundsen-Bellingshausen Seas and Weddell Sea had the most data points in their sectors, each having over 500 data points; these sectors are nearest the Antarctic Peninsula and many sub-Antarctic islands that were used as bases for commercial whaling and sealing businesses, so it is not too surprising to see that most of the logbooks contain data in these regions. Ross-Amundsen Seas had the least amount of data points (75 points), which is the sector farthest from major continents / landmasses in the Southern Hemisphere, so it is also not surprising to see it having less data. Seasonally, the summer (DJF) had a majority of the data points (1,072 points), while winter had the least amount of data points (6 points), again as expected given the rough conditions of Antarctic winter, including the limited daylight, that made voyages near the ice edge dangerous and often impossible. Unfortunately, this is also the season where the reconstruction vs. daily ice edge regression estimates perform the lowest (Figs. 4.2-4.7, section 4.2), which will likely hinder some of the comparisons.

Table 4.4

Data Counts from the Ship Logbooks

Total	**1,586**
Decade	
1900s	41
1910s	110
1920s	167
1930s	702
1940s	332
1950s	144
1960s	0
1970s	55
1980s	35
1990s	0
Sector	
Amundsen-Bellingshausen	575
King Hakon	196
East Antarctica	192
Ross-Amundsen	75
Weddell	548

Table 4.4 continued:

Season	
DJF	1,072
MAM	239
JJA	6
SON	269

4.3.2 Overall Comparison Between Logbook and Reconstruction Data

Table 4.5 shows the overall reconstruction skill, using the statistical measures outlined in chapter 3, for all of the sectors and decades, each of the sectors for all decades, and each sector for each decade. This bias, mean root square error (MAE), root mean squared error (RMSE), and correlation were all calculated.

The bias in Table 4.5 have values ranging from positive to negative, in degrees latitude. When calculating the bias, the difference (reconstruction – logbook) was calculated. When the number is negative this indicates the logbook latitude is further north than the reconstruction. As mentioned before, this is consistent with the ship logbooks reporting the ice north of where it actually is, so it is not surprising to see a negative bias. Overall, this bias of the data provides a favorable comparison between the reconstruction and logbook data, especially given the large errors in the regression statistics in DJF. A majority of the sectors and decades are showing the reconstructions to be within a few degrees latitude of the observations from ship logbooks (most of the data is 5° or less). In fact, overall of the sectors and decades combined show a 3° difference between the reconstruction latitudes and the logbook latitudes.

When calculating MAE, the closer the value is to zero means the observation is closer to the reconstruction, regardless of the sign of the difference. For this research, when MAE values are close to the value for the bias, this would indicate consistent signed differences between the logbook and reconstruction data, which better helps in understanding reconstruction skill than when the differences frequently change sign (i.e., causality of persistent signed differences is easier to attribute than fluctuating signs in the difference). Table 4.5 indicates based on the evaluation made here that the MAE values range from 1 to 8. Spatially, the MAE for King Hakon is the lowest of all the sectors, implying the actual (logbook) and predicted values (reconstruction and regression values) for the latitudes give similar estimates of the ice edge, overall, within about 1° latitude. There are a few notable decades where the bias and MAE are close such as the 1950s where the bias was -3.788° latitude and the MAE was 3.891° latitude. When the bias and MAE are close like this, then the observations from the ship logbooks is within that number of degrees latitude on either side of the ice edge (for example, the 1950s is within 3° latitude of the reconstruction).

The RMSE follows the pattern of the MAE values, if the MAE value was higher, the RMSE value is also higher; however, due to the squared differences in the RMSE, this statistical measure is more sensitive to outliers than the MAE. In comparisons with little outliers, the values for RMSE should not be too far off (larger than) the MAE values. When the values are significantly different from the MAE, then this indicates there is an outlier that is affecting the data. Once again, when this data lines up with the bias and MAE values, the logbook latitudes are that amount of degrees latitude with the reconstruction. Looking at the 1950s again, the RMSE value is 4.805, which is close to the bias and MAE value. Overall, the RMSE shows a relatively favorable comparison of seasonal, sector mean reconstruction estimates with the daily values from the logbooks.

The final calculation in Table 4.5 is the correlations. The correlations are looking the latitudes from the reconstructions and the latitudes from the logbook observations, and how these two values vary together through time. Given the uncertainties in the regression relationships and even the ship logbook entries, it is not surprising that many of the correlation values are near zero, or even negative, suggesting a poor relationship between the logbook values and reconstruction values. However, a few strong correlations do exist, namely within the 1930s and 1950s and the Weddell and Hakon sectors. These higher correlations occur in regions of high data points (Table 4.4), leading to an overall correlation of $r=0.311$. This is a considerable correlation (statistically significant at $p<0.01$), given that there are over 1,500 data points in the sample, even if it implies that only ~10% of the daily ice values are explained by the seasonal sectoral mean reconstructions. Nonetheless, this r^2 value is higher than the majority of the r^2values from the regression estimates (Figs. 4.2-4.6), providing another positive evaluation of the overall reconstruction skill.

Table 4.5

Statistics calculated for the latitude from the ship logbooks and sea ice reconstructions

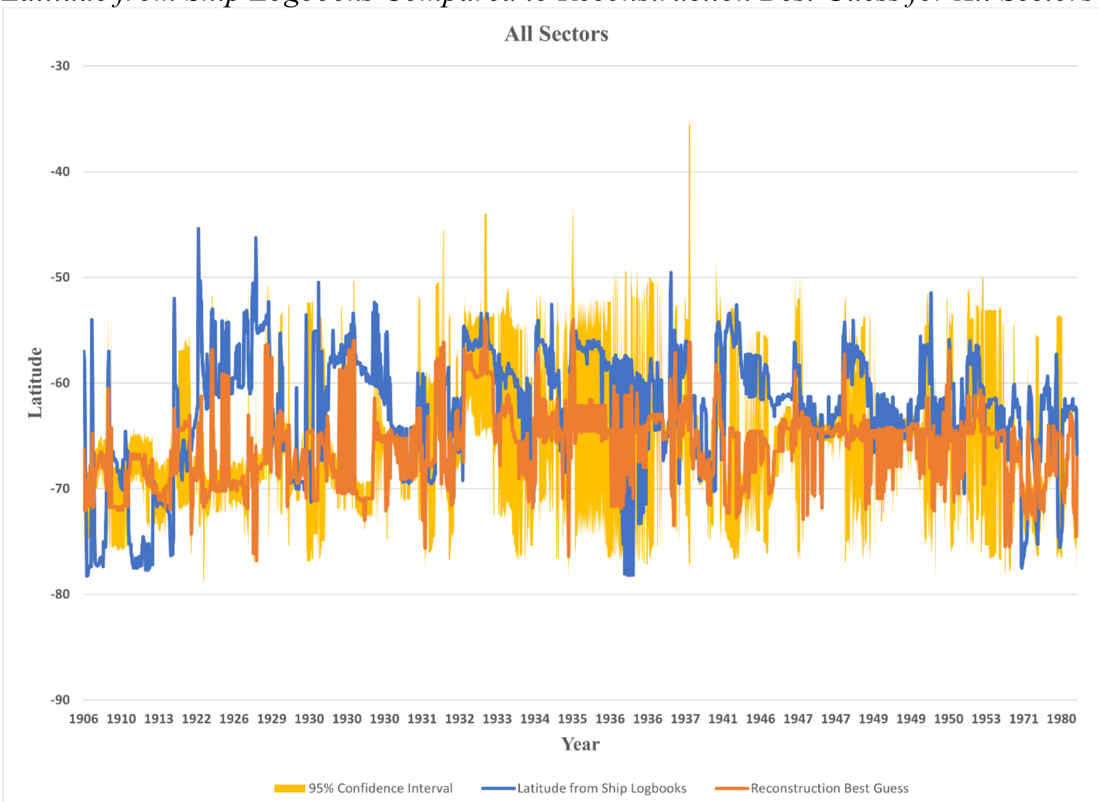
	Bias (reconstruction-logbook) (° latitude)	MAE (° latitude)	RMSE (° latitude)	Correlations
Overall	-3.596	5.388	7.205	0.311
Sector				
Amundsen-Bellingshausen	-7.286	7.507	8.957	0.076
East Antarctica	2.674	4.964	6.030	-0.195
King Hakon	-0.214	1.169	2.072	0.654
Ross-Amundsen	-1.365	3.411	5.831	0.272
Weddell	-3.674	4.826	5.456	0.500
Decade				
1900s	3.617	7.656	8.323	0.070
1910s	2.111	5.442	6.457	-0.255
1920s	-8.211	8.951	10.565	-0.010
1930s	-3.594	4.637	6.012	0.455
1940s	-4.748	4.910	6.500	-0.150
1950s	-3.788	3.891	4.805	0.528
1970s	-2.137	5.836	6.655	0.091
1980s	-2.376	6.597	7.449	-0.315

To provide a more visual depiction of the reconstructions and logbook data, Fig. 4.8 shows the latitude from the ship logbooks compared to the reconstruction best guess estimate calculated for all sectors and seasons from 1906-1980. The yellow-shaded region in Fig. 4.8 shows the 95% confidence interval of the reconstructions. The reconstruction 95% confidence interval is calculated from the upper 95% confidence interval and the lower 95% confidence interval (as represented by the purple lines in Fig. 4.7, or the blue lines in Figs. 4.2-4.6). The orange line is the latitudes estimated from the reconstructions (regression relationship) and the blue line is the latitudes from the logbooks. Ideally, the logbook latitudes should fall within the 95% confidence interval region (yellow-shaded area). An even better result is when the blue lines match with the orange lines, as this indicates the latitudes for both sea ice estimates are very close to each other. If the latitude from the logbooks (blue line) falls outside the 95% confidence interval (yellow-shaded region) then there may be something within the data causing it to fall outside these limits, such as the reconstruction (or regression relationship) providing an erroneous estimate or there is an error within the ship logbook entry.

Looking at Fig. 4.8, the data from the logbook overall frequently falls within the 95% confidence interval. This is a good indication that the reconstruction estimates of the ice edge are close to the logbook observations of the ice. One notable region in Fig. 4.8 where the logbook falls outside of the 95% confidence interval reconstruction estimates is in the 1920s. The logbook latitudes show a northward (equatorward) deviation in the latitude of the sea ice edge that are not within the reconstruction estimates. This deviation is seen in Table 4.5 for the 1920s, as the observation values are within MAE 8° latitude of the reconstruction latitudes (the bias is near -8° as well). This decade needs more research to figure out why logbook observations are outside the 95% confidence interval of the reconstruction estimates to help determine if the issue is within the logbook entry or if the issue is within the reconstruction estimate for that decade, especially if the error is large enough to be outside much of the daily variation of the latitude of the ice edge (Fig. 4.7).

Figure 4.8

Latitude from Ship Logbooks Compared to Reconstruction Best Guess for All Sectors



Note: The latitude from the ship logbooks is in blue, the reconstruction best guess is in orange, and the 95% confidence interval about the reconstruction estimate is in yellow.

4.3.3 Temporal and Spatial Changes in Logbook and Reconstruction Ice Edge Estimates

The ship logbook data was sorted into the 5 different sectors (Amundsen-Bellingshausen, East Antarctica, King Hakon, Ross-Amundsen, and Weddell) to be able to look at the differences between the logbook latitudes for the visual sighting of the ice edge and the reconstruction latitude of the ice edge to see if the logbook latitudes fell within the 95% confidence interval of the reconstruction estimates. Figure 4.9 displays the latitude from the logbook in blue, the latitude from the reconstruction in orange, and the 95% confidence about the reconstruction estimate in yellow. The same reasoning applies from Fig. 4.8; a skillful reconstruction and comparison would have the ship logbook latitudes (blue lines) to fall within the 95% confidence interval (yellow-shaded region) of the reconstruction estimates. This would imply that the logbook observations and the reconstruction estimates are close to each other (within the upper and lower 95% confidence).

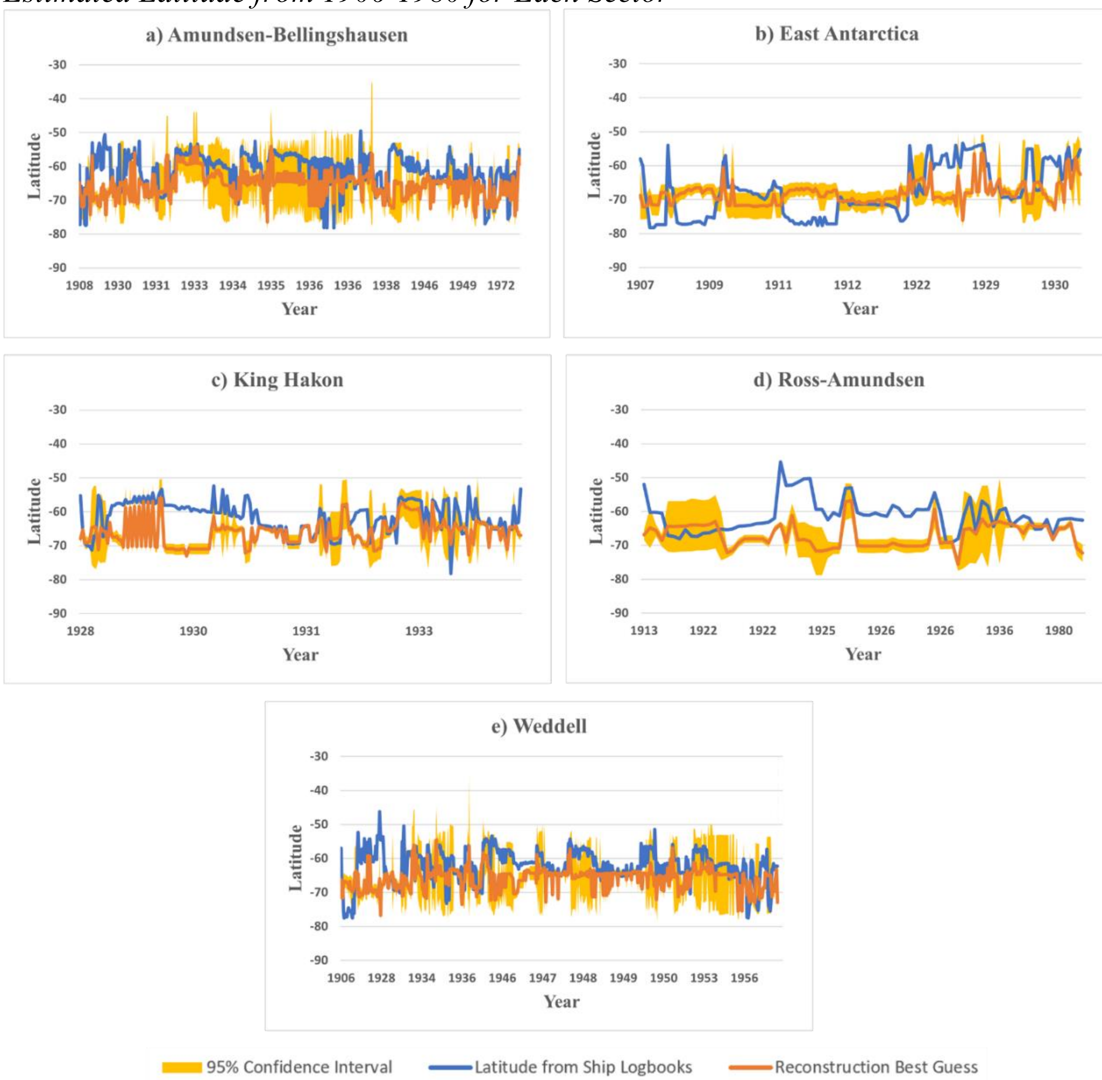
The Ross Amundsen graph in Fig. 4.9d shows that during the 1920s the logbook latitudes (blue line) does not fall within the 95% confidence interval (yellow-shaded region) reconstruction estimate. This pattern within the Ross Amundsen sector in the 1920s is consistent with the bias and MAE values in Table 4.5. This indicates that the latitudes from the logbooks are off (farther north) of the reconstruction estimates falling outside of the 95% confidence interval. It is also important to note for this sector that it is the farthest Antarctic sector from major continents and landmasses, which could lead to potential errors within the ship logbook entries as there is less data points for this sector compared to most others (Table 4.4).

The Amundsen Bellingshausen Seas (Fig. 4.9a) and Weddell Seas (Fig. 4.9c) show a better relationship between the latitudes from the ship logbooks and the reconstruction estimates. In fact, as seen in Fig. 4.9, the logbook latitudes fall within the 95% confidence interval a majority of the time in these sectors. This is consistent with the bias, MAE, and correlation values in Table 4.5. These values (in Table 4.5) for both Amundsen Bellingshausen and Weddell show a close relationship between the degrees latitude with the logbook and the reconstructions. The Weddell Sea falls within 4° latitude of the reconstruction estimate, with a correlation value of 0.500. This indicates the logbook observations and the reconstruction estimates are in general good agreement for the latitude of the sea ice edge in this sector. While the Amundsen Bellingshausen Seas falls within 7° latitude of the reconstruction estimate, the MAE value is very close to the bias value of -7.373, suggesting that the ship logbook values are consistently north of the ice edge. Given the large regression error and broad 95% confidence interval, a ~7° error is still a fairly robust indication of good reconstruction skill in this sector as well (limited by the ability to make easy direct comparisons). Furthermore, the Amundsen-Bellingshausen Seas and Weddell Seas (from Fig. 4.9) can be compared to Fig. 4.2 and Fig. 4.6 to tell the story of challenge in understanding the relationship between the daily ship logbook observations and the seasonal sectoral mean values from reconstructions. In Fig. 4.2, the range and slope during DJF for the Amundsen-Bellingshausen Seas weaken the ability to make direct comparisons (as discussed earlier) but the relationship between the daily sea ice observations and the area is much better in MAM, JJA, and SON. Despite these challenges, this fact that the observations are falling mostly in line with the 95% confidence interval from the reconstruction estimates, Fig. 4.9 is quite remarkable. The same can be said for Fig. 4.6 (Weddell). This sector follows the same pattern as in Fig. 4.2 with DJF have a much greater range and variability in the sea ice, but during MAM, JJA, and SON the relationship is much stronger. This is also seen in Fig. 4.9 as the observations from the logbook are falling within the 95% confidence interval. Since both the Amundsen-Bellingshausen Seas and Weddell Seas are nearest the Antarctic Peninsula, more ships sailed to this region to use as bases so the observations of the sea ice are more likely closer to where the sea ice actually was compared to locations like Ross-Amundsen sector where very little ships sailed as there were very little, if any, bases to set up (Table 4.4).

Overall, Fig. 4.9 solidifies the findings seen in Table 4.5 as it visually shows a majority of the sectors have the logbook latitudes falling within the 95% confidence interval from the reconstruction estimates. This tells us that the skill of the logbooks to be used to estimate the ice edge is good as it is following the reconstruction estimates for most sectors. Indeed, in both Fig. 4.9 and Table 4.5 as the values of the observations and reconstructions are falling mostly within 5° latitude of each other, with a few exceptions (as mentioned above), much smaller than the daily ranges of sea ice edges in DJF for any given seasons (Figs. 4.2-4.6).

Figure 4.9

Difference Between the Logbook Sighting of Sea Ice Latitude and the Reconstruction Estimated Latitude from 1906-1980 for Each Sector

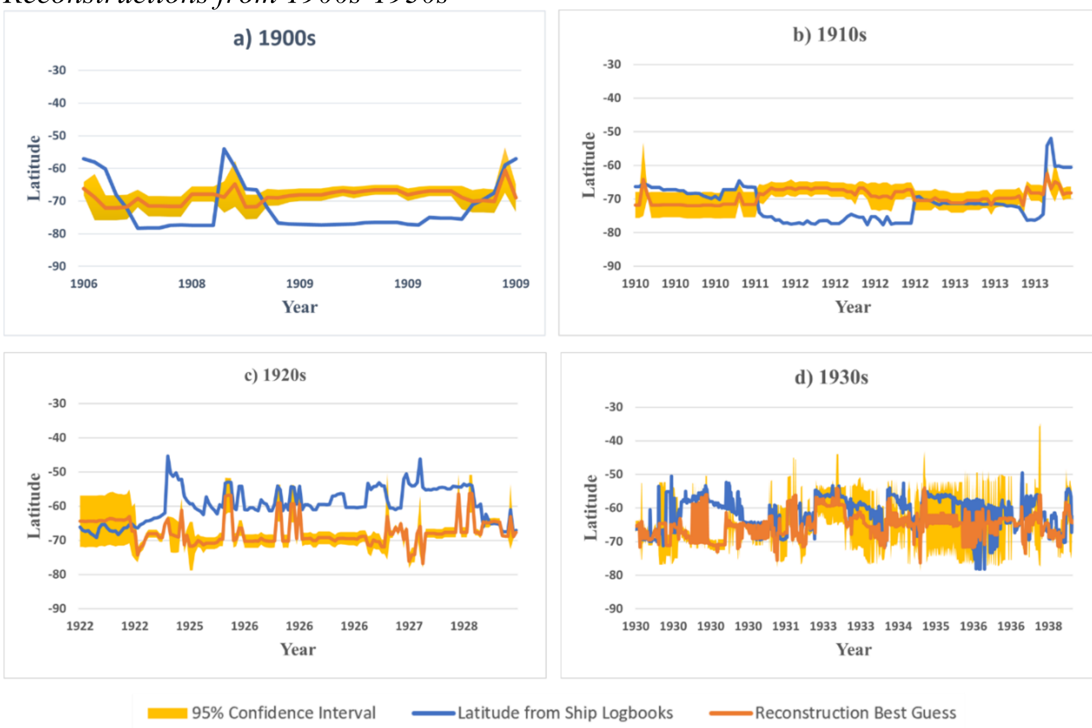


Note: The ship logbook sighting latitude is in blue, the reconstruction latitude is in orange, and the 95% confidence interval about the reconstruction estimate is in yellow.

The ship logbook records of the sea ice were not only divded by sector, but also by decade (1900s-1990s). With the data available and keyed for this research (Table 4.3 and 4.4), the decades of 1960s and 1990s did not have any data points. A majority of the data points in this research lie within the 1930s, as indicated in Table 4.4. By grouping the data into the different decades, it is easier to visually see the temporal variations and which decades the latitudes from the ship logbooks (blue line) fall within the 95% confidence interval (yellow-shaded region) from reconstruction estimates. The comparisons for each decade can be seen in Figs. 4.10 and 4.11.

Figure 4.10

Difference Between the Latitude from Ship Logbook Records and the Latitude from the Reconstructions from 1900s-1930s



Note: The ship logbook sighting latitude is in blue, the reconstruction latitude is in orange, and the 95% confidence interval about the reconstruction estimate is in yellow.

Figure 4.11

Differences Between the Latitudes from the Ship Logbook Records and the Latitudes from the Reconstructions from 1940s-1980s

Note: The ship logbook sighting latitude is in blue, the reconstruction latitude is in orange, and the 95% confidence interval about the reconstruction estimate is in yellow.

In Fig. 4.10c, the 1920s, once again, shows the trend of the logbooks being consistently outside and north of the 95% confidence interval from the reconstruction estimates. With this trend in the 1920s reoccurring throughout this research, more research is needed to figure out whether the error is a result of the logbook or if the error is in the reconstruction estimate.

The 1900s in Fig. 4.10a is showing the logbook latitude to fall outside the 95% confidence interval more times than not. When comparing this graph to Table 4.5, the pattern between the two values can be more precisely identified. While the bias is showing the data falling within 3° latitude of the reconstructions (since the number, 3.617, is positive this indicates the logbook latitudes are further south than the reconstructions), the MAE is much higher showing a 7° latitude difference. This indicates the logbook latitudes and the reconstruction estimates are not in agreeance and there may be an error in the logbook or the reconstruction during this decade may be poor. A similar story of northward reconstruction estimates of the ice edge are also observed in the 1910s (Fig 4.10b), but not as persistently as in the 1900s.

Another decade of importance is the 1980s (Fig. 4.11d). The first part of the graph shows the logbook latitudes falling within the 95% confidence interval; however, the logbook latitudes are outside of this range for the second part of these data. This is especially important to note because the reconstructions during the 1980s were able to use satellite observations (as satellite observations started in 1979) to reconstruct the SIE during this decade confidently. With this information, it is likely that there is an error within the logbook data and/or the regression methods to relate the reconstruction data to the daily logbook data. The latitude from the logbooks is consistently showing the sea ice to be farther north than what the reconstructions are showing.

A pattern seen in Figs. 4.8-4.11 is the latitudes from the ship logbooks tend to be more equatorward while the reconstruction latitudes are more poleward. This is somewhat expected, as the ships likely recorded the sea ice at a greater distance than what it actually was, visually sighting the 0% rather than the 15% line of sea ice concentration from miles away. It is promising in this research, as seen throughout the figures and tables, a majority of the time in each sector, decade, and overall that the logbook observations are falling within the 95% confidence intervals from the reconstruction estimates. While the variability of the ice edge is much greater in DJF (Fig. 4.7), the estimates of the ice edge are falling within the 15% isoline a majority of the time. This is consistently and repetitively reflected within the bias, MAE, RMSE, and correlation values along with the graphs in Figs. 4.9-4.11. While the average error of 4°-5° latitude or about 400-500km seems extreme, the daily variations in sea ice edge within a given season, and even the distance between the 0% and 15% sea ice concentration are often much farther than this distance, as represented in Fig. 4.7. Thus, these preliminary comparisons overall show a remarkable agreement between daily ship logbook estimates of the sea ice edge and the reconstruction estimates of the same, based on a seasonal sector mean reconstructed value. As such, the logbooks are helpful to validating the skill of the early (pre-satellite) reconstruction depictions of the Antarctic sea ice cover.

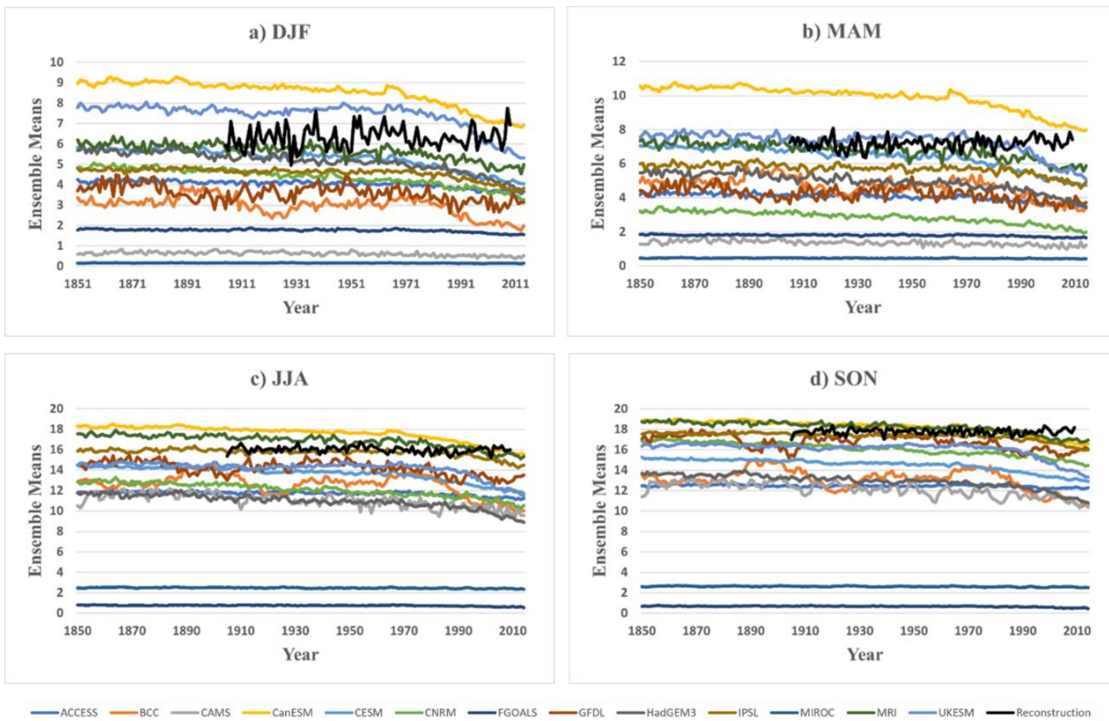
4.4 Comparing the Ranges of Variability of Antarctic SIE in CMIP6 to the Seasonal Sea Ice Reconstructions

Coupled climate models are a useful tool to use to look at the variability of sea ice. In this research, CMIP6 climate models were used to look at the SIE from the years 1850-2014. Specifically, the historical simulations were used. The historical simulations from CMIP6 were used to compare against the seasonal sea ice reconstructions from Fogt et al. (2022). Within the 13 models (from Table 3.1), each of them had multiple ensemble members. By having multiple ensemble members in the models, an average across all members, called an ensemble mean, can be created, which allows for a more detailed investigation of the similarities between the ensemble members, often called the forced response. However, because climate models do not assimilate observations, only prescribe forcing, looking at each ensemble member helps to illustrate possibilities that are more similar to the range of physically possible changes observed on Earth's variable climate. Fig. 4.12 is a time series plot of the ensemble model means and the reconstructions for each season for all sectors from 1850-2014. In all of the seasons, the means for MIROC and FGOALS models are all consistently lower than the rest of the models and the reconstructions. This indicates these models are showing lower sea ice means. The CanESM model is consistently showing the highest mean out of all the models and reconstructions, indicating this model is showing more sea ice, much greater than in the reconstructions (and by extension, observations). The reconstructions in each of the seasons show more sea ice than a majority of the models. The time series of the models do show the pattern of less sea ice in the summer months and more sea ice in the winter months. It is also interesting to see within these time series that the models are showing a decrease in the sea ice in the later periods, while the reconstruction is showing the positive trend that is shown within the observations; this is the known flaw in the CMIP6 models detailed in the literature review and discussed in Roach et al. (2020) – the models appear to be too sensitive to the greenhouse gas forcing and reduce Antarctic sea ice in response.

Figure 4.12

Time Series of the Model Ensemble Means and Reconstructions from 1850-2014 for Total Sea Ice Extent

Total Sea Ice Extent



Note: Each various color represents a different model. The black line represents the reconstructions.

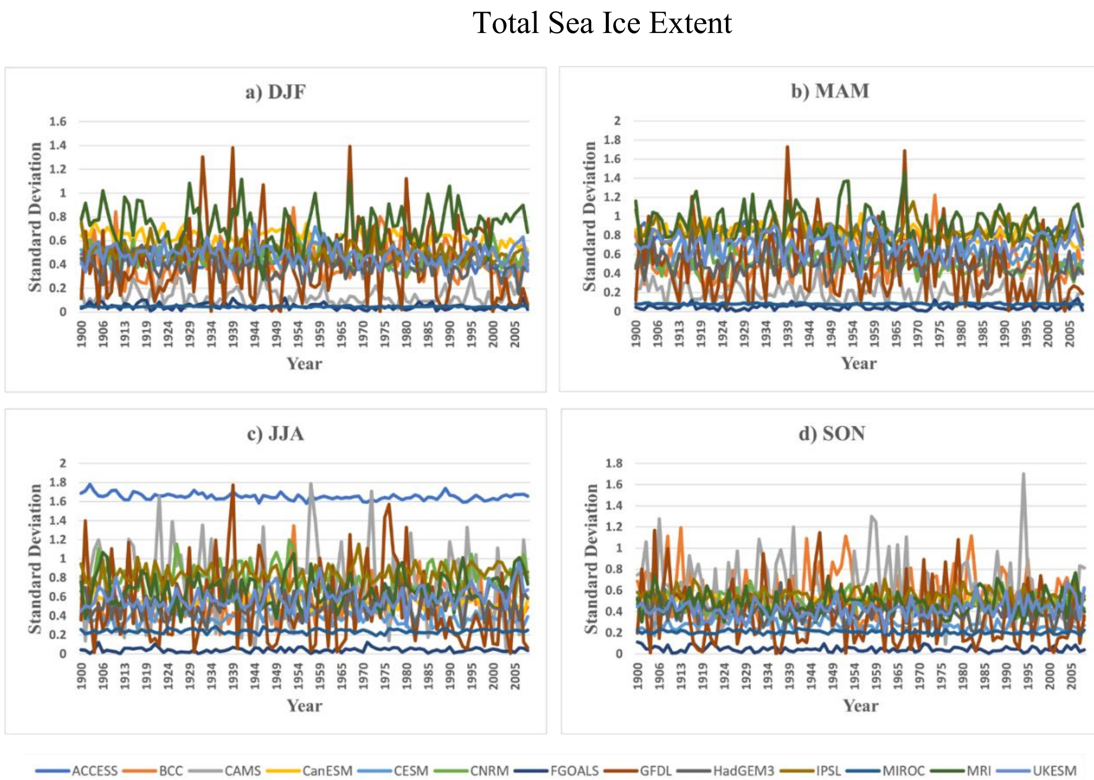
4.4.1 Interannual Variability

While the means (and trends after 1979) are markedly different between the reconstruction and the models, they may still produce realistic variability that will allow for more investigation into the reconstruction skill, specifically if the reconstruction produces physically possible interannual variability in the reconstruction period (pre-1979). The interannual variability between each ensemble member within a given year was calculated from 1900-2009, and is shown in Fig. 4.13 for the total extent. The x-axis represents the different years, and the y-axis is the standard deviation across the ensemble members for each year. The various colors in the graphs represent the 13 models used in this research.

In Fig. 4.13, while the standard deviation between the ensemble members for each model is different, it is quite prevalent how variable the sea ice is in each model. The large variation between each ensemble members tells that the sea ice edge can change in each realization substantially from natural and internal variability, even without strong direct forcing. While most of the models show a large variation in each season, there are a few models that stand out where there is little variation. In each season, there is a pattern where at least 2 of the models have a persistently lower standard deviation across the ensemble members than the rest of the models, suggesting that each ensemble member in these models is depicting similar values each year (at the same time). For all seasons, the models FGOALS and MIROC have this flat, lower standard deviation appearance. Notable, since the lines are more persistently low in DJF and MAM, near the times of sea ice minimum (Figs 4.13a-b), these models are likely melting all or nearly all the Antarctic sea ice each year, which would result in less variability. The flat line is also an indication that the sea ice minimum in these models is consistent from year to year, providing a further possible explanation that these models are melting the sea ice out every year to values of near zero (0) total sea ice extent. In JJA (Fig. 4.13c), the ACCESS model stands out because the standard deviation is much higher for this model compared to the rest of the models. This indicates the sea ice values for this model during times of growth approaching its maximum extent are spread out over a wider range than the other models. Also, with the standard deviation of ACCESS being much higher in JJA, it is also showing change through time – indicating that there is a consistent high degree of spread between the various ensemble members each year. In contrast, the majority of other models and in other seasons in Fig. 4.13 show a narrow range, but with marked change from year to year, of the standard deviation across models, suggesting that at times the ensemble members are more than others, again highlighting the complex sea ice variability in the models.

Figure 4.13

CMIP6 Interannual Variations from 1900-2009 for Total Sea Ice Extent



Note: Interannual variations of the CMIP6 models. The x-axis is year, and the y-axis is the standard deviation values. Each various color represents the different models.

4.4.2 Correlations Across Ensemble Members

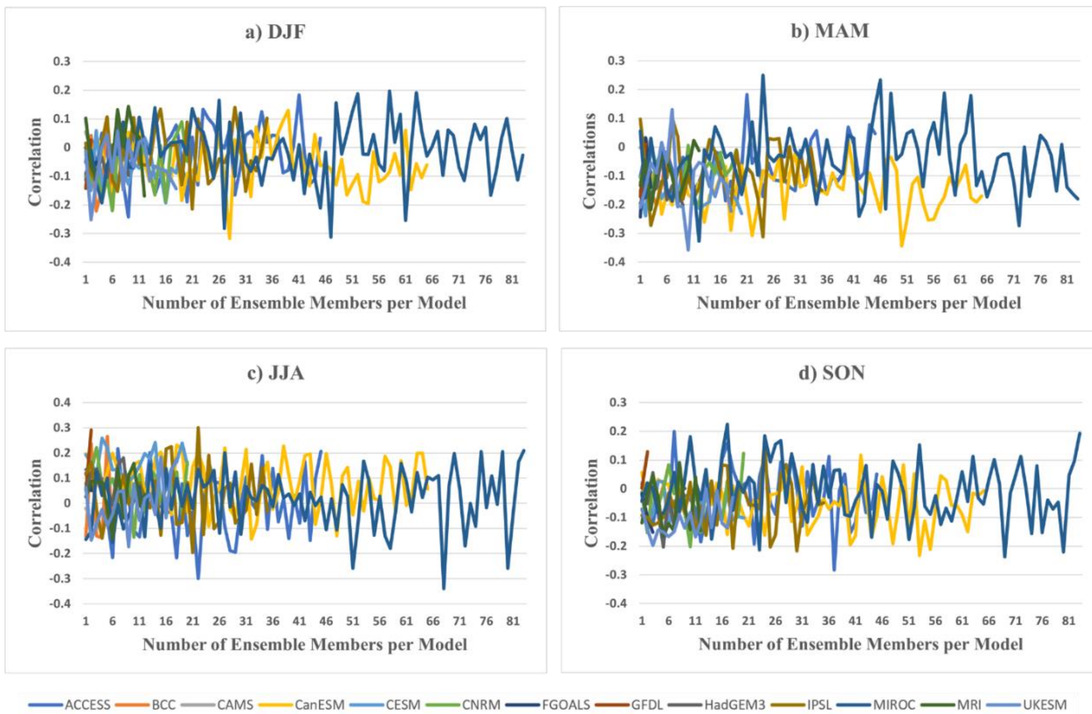
Each ensemble member is a physically possible representation of the real world, although as noted in Roach et al. (2020), each model has considerable error. However, the models do not assimilate direct observations (they only apply consistent forcing), so despite this physically realistic variability, there is no reason a priori that any of the model's variability coincides with the real-world variability, approximated by the reconstructions. To demonstrate this, the different ensemble members and the seasonal sea ice reconstructions are correlated. Calculating the correlation (equation 3.2) makes it easier to see if there are similar patterns within the models compared to what is seen within the reconstructions. When calculating the correlations, the seasonal means for each year (1850-2014) for each ensemble member were used. Each ensemble member was then correlated with the seasonal sea ice reconstructions. Figure 4.14 shows the correlations between the ensemble members from each model and the reconstructions in all of the sectors in Antarctica. The correlations look at how the sea ice is changing across each ensemble member and how this agrees with the sea ice reconstruction. Within Fig. 4.14, the x-axis will represent the ensemble member numbers within each model. Before these correlations are discussed, it is important to stress again that it is not expected for the models to have a strong correlation with the reconstructions, for several reasons. First, the models are flawed in terms of their depiction of Antarctic sea ice trends (Fig. 4.13), which likely also influences their ability to simulate interannual variability accurately (Roach et al. 2020). Although this would weaken the correlations, the main reason why no strong correlation is expected with the sea ice reconstruction is that these ensemble members do not assimilate direct observations, and so their connection to the timing and intensity of events in the real-world (represented by the reconstructions) is not likely to align, giving them different patterns of interannual variability from the real-world, and thus, weak correlation. Rather, this check on the correlation between the models and reconstruction is to further highlight the high variability in Antarctic sea ice within and across models, and to what extent any of this variability may (arbitrarily) align with the sea ice reconstruction – it is not exactly a test on the reconstruction's skill.

A quick glance at Fig. 4.14 again shows the variability in each of the models, as the correlations are quite different between various ensemble members within a model, and between the models as well. If all the models said the same thing about the sea ice, then the 13 colors (indicating the 13 different models) would all look the same. This was seen in Fig. 4.13 and the pattern continues in the correlations. The variability is especially seen within the CanESM (yellow line) and MIROC (blue line) models. Even with the variability being quite different within the models and the ensemble members, there are some similarities. The models all range from having a positive correlation to a negative correlation, generally within ± 0.02.

Figure 4.14

Correlations from 1850-2014 Between Ensemble Members and Reconstructions for Total Sea Ice Extent

Total Sea Ice Extent



Note: Correlations between the ensemble members from the 13 models and the reconstructions. The x-axis is the number of ensemble members per model, and the y-axis is the range of the correlation values. Each various color represents the model.

Seasonal variations in Antarctic sea ice variability can also be seen through the correlations. In the summer months (DJF) the correlation is the weakest overall (Fig. 4.14a). The variability in the different ensemble members is also greater in DJF (Fig. 4.14a), most readily apparent within CanESM (yellow line) and MIROC (blue line) models. Even the correlation for the fall months (MAM, Fig. 4.14d) is weaker than the correlation in the winter (JJA) and spring (SON). This pattern makes sense as the sea ice in summer is highly variable, as noted in section 4.2, due to the daily sea ice easily changing from one day to the next. In MAM, the sea ice is starting to advance for the winter months. In the winter, the SIC is much greater (Fig. 4.7) which aids in the sea ice being less variable day-to-day compared to summer sea ice (this relationship is also seen in Fig. 4.7 as the range of the sea ice edge is less in JJA). The weak correlation between the ensemble members and reconstruction in DJF in Fig. 4.14a matches the variability of the ice edges in DJF seen in Fig. 4.7, and the stronger correlation in JJA in Fig. 4.14 matches the variability in JJA in Fig. 4.7

As mentioned previously, with any given color line, there is much change seen from each of the ensemble members. This indicates a large natural variability within each of the models. The models change depending on what ensemble member is being examined, and its relationship with the reconstruction is quite different. With there being a change in each of the ensemble members, the representation of the sea ice within them is quite different. Figure 4.14 displays, in each season, all of the models have both positive and negative values for various ensemble members, even with a single model (colored line). Out of all the models, CanESM (yellow line) and MIROC (blue line) stand out the most in each season as varying most dramatically across ensemble member from positive to negative correlations. These two models tend to have stronger correlations (in an absolute sense) between the ensemble members and the reconstructions, which suggest that very few of their ensemble members are even positively correlated. Given that the models variability will be different from the real world, so negative correlations within the models can be expected.

One interesting pattern that is seen in Fig. 4.14 is in MAM (Fig. 4.14b), where most of the ensemble members appear to have a negative correlation. This opposite relationship in the variability of these models compared to the reconstructions need further research for MAM to determine why this particular season is seeing more negative signs than the other seasons. The other seasons (Figs. 4.14a, c, and d) tend to fall around 0, with a slight prominence for negative correlations. There is similar variability in the models and the reconstructions. The correlations, whether positive or negative, are not extremely strong as the correlations range from 0.4 to -0.4, with a majority of the models being close to 0; thus as expected the models have very little consistent (arbitrary) similarities with the reconstructions.

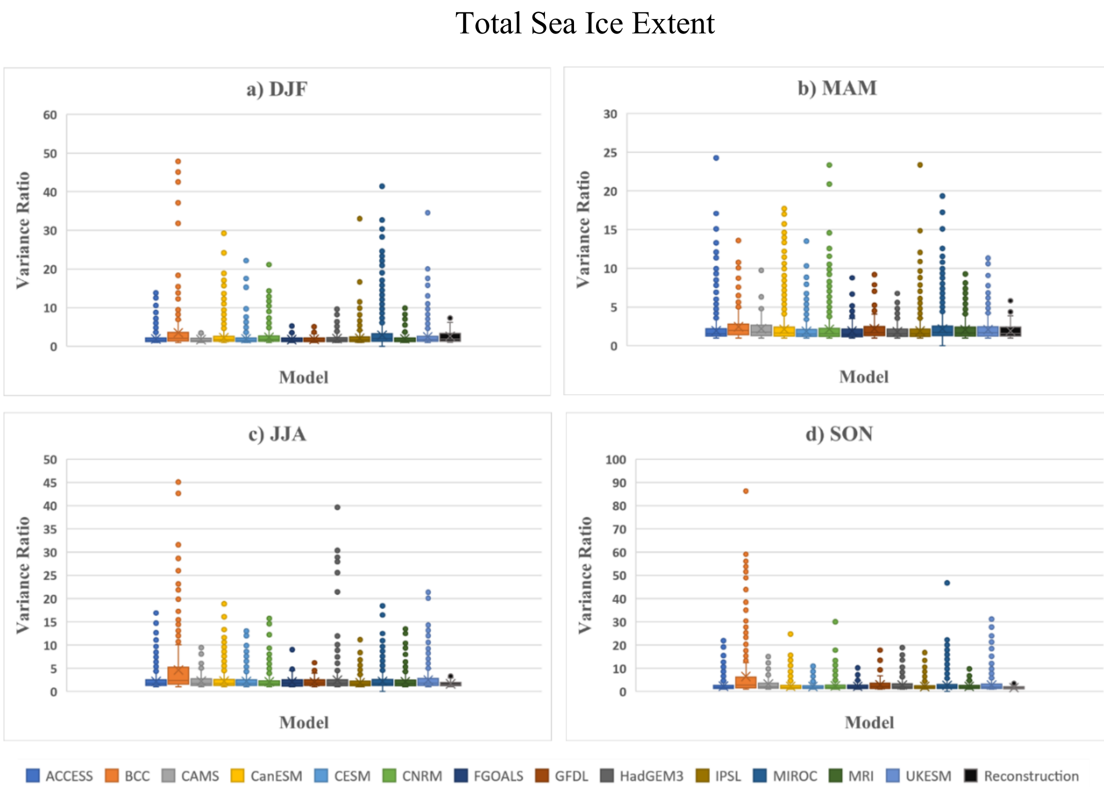
4.4.3 Boxplots of Variance Ratios Between the Models and Reconstructions

Despite the models being flawed and having very different interannual variability than the reconstruction (Fig 4.14), they can still be used to understand physically possible changes in sea ice cover through time. Realistic changes in variability in the reconstructions and models were determined by conducting an F-test looking at changes in variability between all the possible 10-year periods during 1900-2014 within each reconstruction member, for each sector and season. For one ensemble member, there are 11 decades within 1900-2014, but 55 unique possible combinations of decades to compare (possible combinations = {11 decades * 11 decades) – 11 pairs of same decades] / 2 to get only unique pairs = 55). This is then multiplied by the number of ensemble members to get the total possible combinations for one model. This large sample allows us to physically see the possible decadal changes in variability, and examine both how the models agree on these possible changes, and where the reconstruction changes (essentially 1 ensemble member) fall within these distributions. To visualize these changes, boxplots of variance of ratios for all possible 10-year periods were created to compare the models to the reconstructions. Fig. 4.15 shows the boxplots of the variance ratios from 1900-2014 for the total sea ice extent. The multiple colors represent the 13 different models (and all the possible ensemble members within that model, using the same colors as in Figs. 4.13-4.14), and the black boxplot is the reconstruction.

For total sea ice extent (Fig. 4.15), all of the models have a similar change in variability for 10-year periods, and this change is consistent with the reconstructions. This consistency could be a measure of the reconstruction skill, suggesting its depiction of decadal scale variability changes are consistent with a physically possible solution seen in a climate model. This distinction is important: the reconstructions were statistically generated (not constrained directly to obey physical laws), yet they are producing the same kind of variability that is present in the model, with the model being constrained by physical governing laws. While the reconstructions, in each season, overall see the same variability as the models for total sea ice extent (Fig. 4.15) some of the models have strong outliers in the variability that are inconsistent with the reconstruction. For example, the BCC model has strong outliers in all of the seasons compared to the constructions in Fig. 4.15 for total sea ice extent. MIROC is also showing strong outliers in a majority of the seasons that are not consistent with the reconstructions. Also, within Fig. 4.15, the ratios change seasonally, which the reconstructions also show, again suggesting the reconstructions are providing an accurate depiction of total sea ice extent variability throughout time, and how this changes seasonally. There is more variability between 10-year periods in MAM (Fig. 4.15b) and JJA (Fig. 4.15c) and less variability in DJF (Fig. 4.15a). In contrast to the correlations, which look at year-to-year changes in sea ice extent, the weaker variability in DJF is to be expected as most of the ice is consistently gone in DJF (Fig. 4.7), and the values do not fluctuate much from summer to summer as the ice is melted away in the same way as the previous year (the exact values of the ice extent do change substantially, though, giving the weak correlations in Fig. 4.14). Another explanation for the models showing less variability in DJF is that a few of the models get rid of some of the ice versus some of the models getting rid of all of the ice. During MAM, the sea ice is starting to advance and is changing. Overall, Fig. 4.15 does a decent job of showing the broad picture of the variability as it is encompassing all the sectors, and the reconstructions mimic the behavior seen in the models very well.

Figure 4.15

Boxplots of 10-year Variance Ratios in Models Compared to the Reconstructions for Total Sea Ice Extent



Note: The various colors represent the different models. The reconstruction is in black on the far right boxplot in each season.

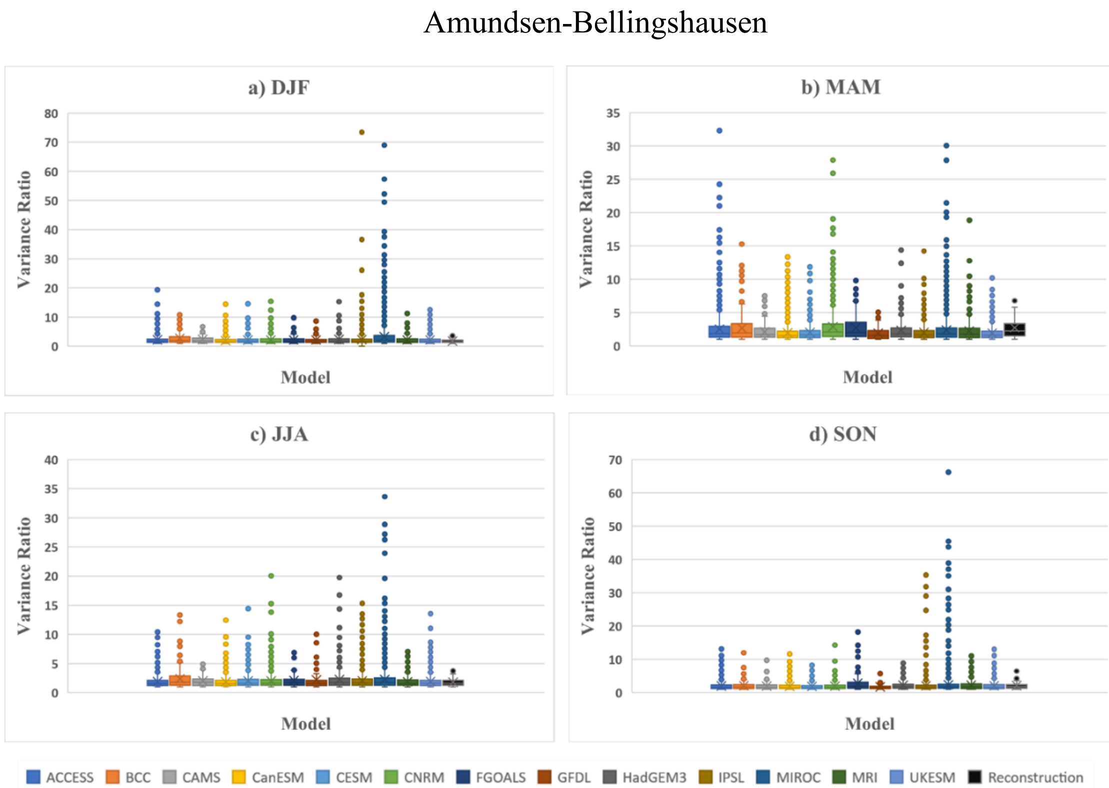
Figures 4.16 (Amundsen-Bellingshausen Seas) and 4.17 (East Antarctica) show the regional variability in the models compared to the reconstructions. In Fig. 4.16, there is very little variability at the maximum in SON (Fig. 4.16d), in both the reconstructions and models. However, in MAM (Fig 4.16b) and JJA (Fig. 4.16c) the variance ratios are much larger, indicating a considerable difference in sea ice variability from year to year in these seasons for this sector. This is especially true during the advancing growth time of the sea ice in MAM (Fig. 4.16c). There are also a lot of strong outliers in multiple models in MAM for Amundsen Bellingshausen Seas which is inconsistent with the reconstruction for that season and sector. In particular, the MIROC model in all of the seasons in Fig. 4.16 shows strong outliers compared to the other models. Nonetheless, each model is consistent with the variability in the reconstructions in each of the seasons in Amundsen Bellingshausen Seas. The minimum values in all seasons are close with each other and with the reconstructions, suggesting that the variability in the reconstruction aligns well with physically plausible variability in the models.

For East Antarctica, Fig. 4.17 shows more variability overall compared to all the other sectors (not shown for brevity). The models, and even the reconstructions themselves, are a lot more variable than in Amundsen Bellingshausen Seas (Fig. 4.16). Despite the increased variability in this sector, with each season, the reconstructions still match very well with the model variability, further validating the reconstruction variability in this sector is physically plausible. It is also apparent in Fig. 4.17 that the minimums are very consistent with one another and match up very well. Within these boxplots, along with Figs. 4.15 and 4.16, 50% of the variability is within the boxes (the interquartile range), and once again, this specific range matches really well between the models and reconstructions. The outliers in the MIROC models, once again, are strong in this sector (as in Fig. 4.16). Overall from Figs. 4.15-4.17, the boxplots here are showing that the reconstructions have similar patterns of what is seen in the models.

Since Figs. 4.15-4.17 display ratios of variance between 10-year periods, it is possible to determine the statistical significance of these variance ratios using an F-test. When calculating the F-test probabilities, the reconstructions have a higher probability (i.e., that there are not significant differences in variance across the 10-year periods, not shown). Even though there is a lot of variability within the models, a majority is not significant based on the F-distribution, including many of the outliers. The same thing can be said for the reconstructions. While there is high variability in the models and reconstructions in certain seasons (such as MAM in Fig. 4.16b), most of these changes in variances is not significant based on the f-test.

Figure 4.16

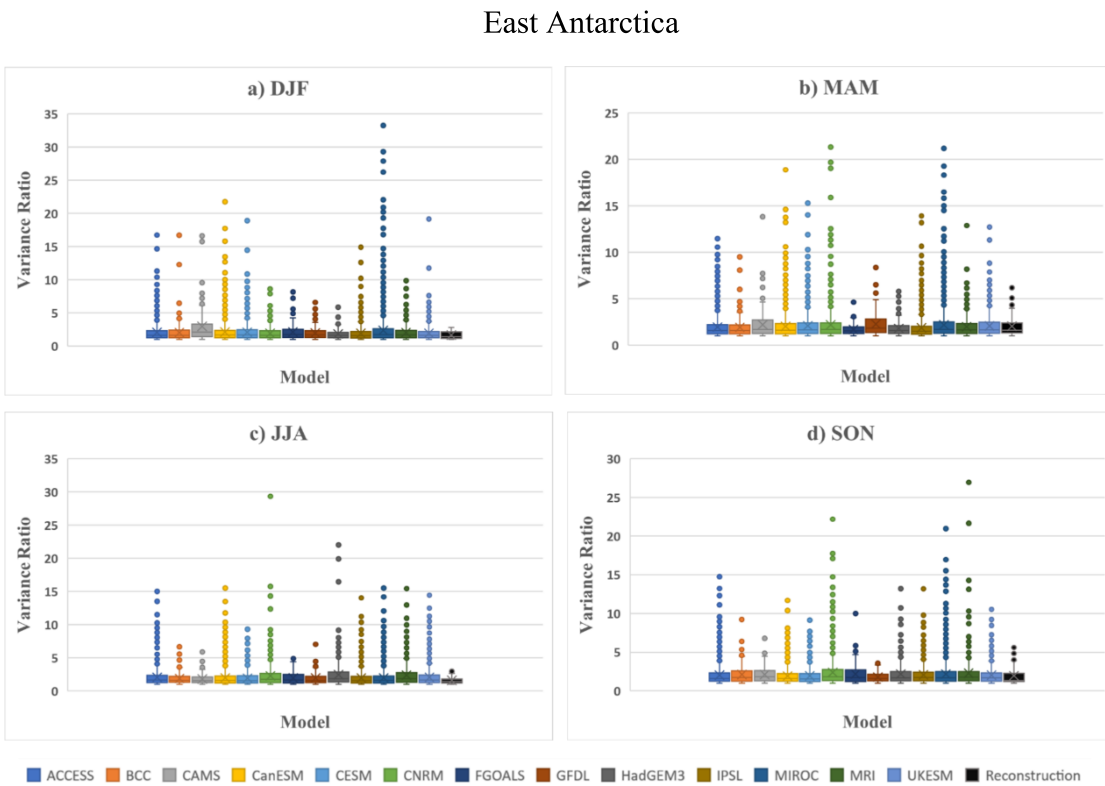
Boxplots of 10-year Variance Ratios in Models Compared to the Reconstructions for Amundsen-Bellingshausen



Note: The various colors represent the different models. The reconstruction is in black on the far right boxplot in each season.

Figure 4.17

Boxplots of 10-year Variance Ratios in Models Compared to the Reconstructions for East Antarctica



Note: The various colors represent the different models. The reconstruction is in black on the far right boxplot in each season.

4.4.4 Boxplot of the Differences Between the Models and Reconstructions

To understand how physically plausible changes in mean sea ice extent are occurring in the models, and how these agree with the models, a similar procedure as above was employed on comparing 10-year means from within the individual ensemble members, along with the reconstructions. To understand if these means are statistically different, a T-test was performed using Eqn. 3.11 to analyze the decadal-scale average sea ice extent changes. The T-test looks at the distribution (the boxplots) of differences between every possible pair of 10-year averages and how the ice acts in this period. Notably, because there are large negative trends in the models through time (Fig. 4.12), the difference for the time periods was calculated by subtracting the earlier period from the later period in 10-year comparisons. Essentially, the later period is being compared to something that has happened earlier. When the difference is negative, this indicates that the sea ice is decreasing through time.

Figure 4.18 is a boxplot of the differences between the models and reconstructions for total sea ice extent. The reconstruction, in all of the seasons, shows a much smaller in total change compared to what the models would suggest. This is more than likely a problem within the models rather than a problem with the reconstructions, due to their negative trends (Fig. 4.12). The possible changes in the total sea ice extent are larger in models because the models tend to have more uniform changes spatially than the reconstructions; the regional variations in the seasonal sea ice reconstructions often offset and oppose each other, giving weaker changes in the real-world (Fogt et al. 2022). These regional patterns could be where some areas gain ice and some areas lose ice, as discussed in the literature review, and so the changes in the reconstruction are smaller than what is seen in the models where the overall signal in the models is in the loss in sea ice. In contrast, the uniform changes within the models, reducing everywhere in response to greenhouse gas forcing (Roach et al. 2020), contribute a large portion of difference shown in Fig. 4.18.

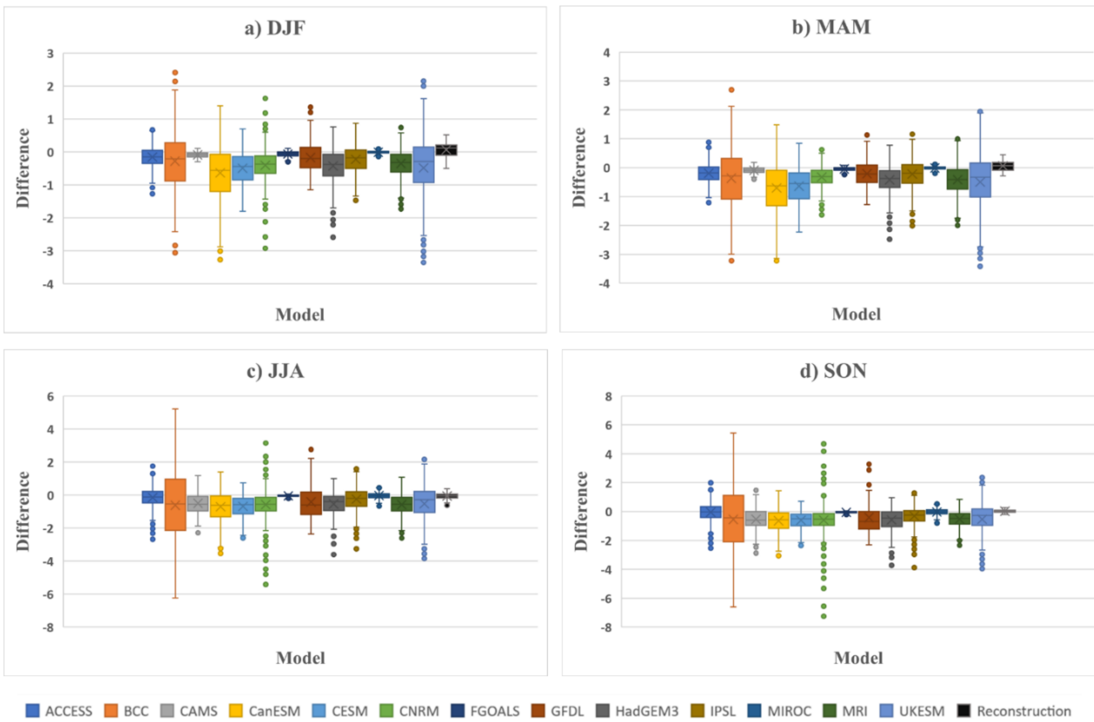
To examine some of these spatial patterns more closely, Fig. 4.19 displays the boxplots/distributions for 10-year mean sea ice changes for East Antarctica. Compared to Fig. 4.18, Fig. 4.19 shows the difference seen in the model are negative a majority of the time and not showing much of an increase. The models tend to lose ice in an unrealistic way, too sensitive to greenhouse gas forcing (Turner et al. 2019). The 10-year differences in the seasons for nearly all of the models show a largely negative difference through time. The models are showing a lot more decrease in the sea ice than what is being observed in the real world, which is a problem. Nonetheless, the absolute magnitude of the difference is consistent with the reconstructions. In comparison, there is not such a large change over the 10 years reconstructions, and the reconstructions seem to fall within what a lot of the models suggest. Even though the models are decreasing, the important idea to look at is whether or not the models have a similar change in a 10-year period as the reconstruction, which it does, suggesting that the reconstructions are producing physically plausible changes in decadal mean sea ice extent in East Antarctica, even if the sign of these differences isn't always the same.

In Fig. 4.20, the boxplot is showing the 10-year difference for Ross Amundsen Seas between the models and the reconstructions. In this sector, the reconstructions are a little different in their sign compared to the models, which warrant further discussion. For example, in DJF (Fig. 4.20a), the reconstruction is showing a positive difference (an increase for sea ice) and the models are showing a more negative difference (a decrease for sea ice); this is due to increases in sea ice in the real-world in this sector since 1979 (Parkinson 2019), which is in stark contrast to the models continual decrease even in this region (Fig. 4.12). However, even with the reconstructions signs being slightly off, the magnitude of the differences in the reconstructions is similar to what the models are showing. In MAM, JJA, and SON (Figs. 4.20b-d) the reconstructions and models both have similar, and slightly positive, changes in the 10-year period. This is a good test for both the reconstructions and models to have this consistency, and suggests the reconstructions are producing physically plausible decadal changes in sea ice overall in this sector as well.

Figure 4.18

Boxplots of 10-year Differences Between the Models and Reconstruction for Total Sea Ice Extent

Total Sea Ice Extent

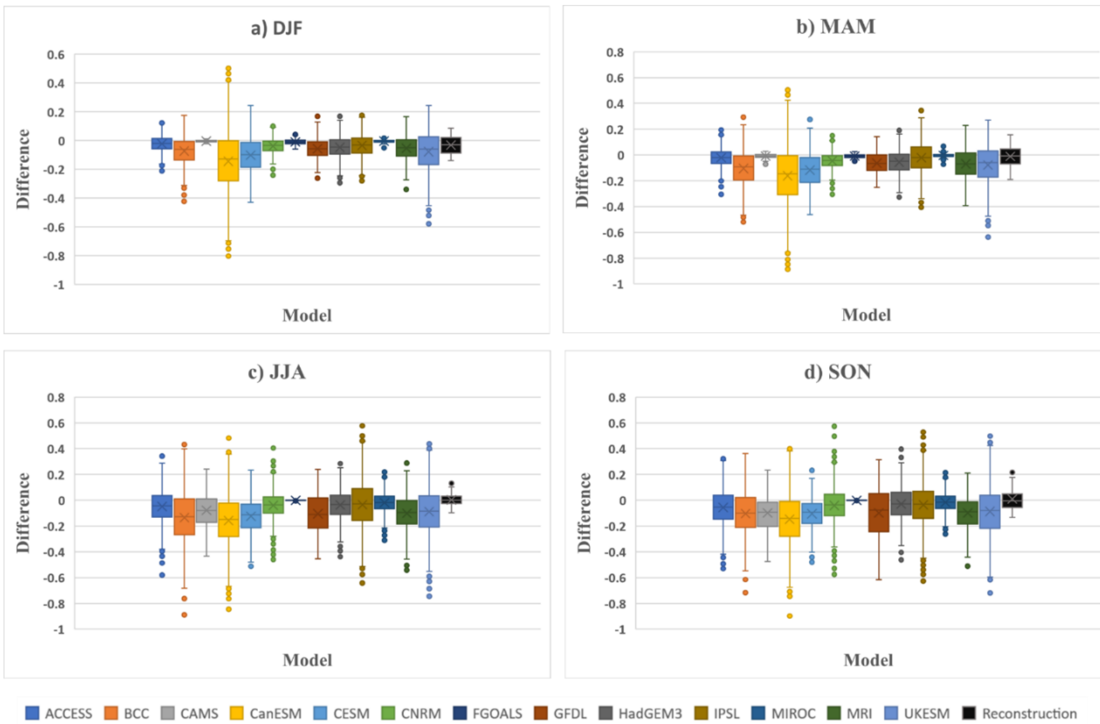


Note: The various colors represent the different models. The reconstruction is in black on the far right boxplot in each season.

Figure 4.19

Boxplots of 10-year Differences Between the Models and Reconstruction for East Antarctica

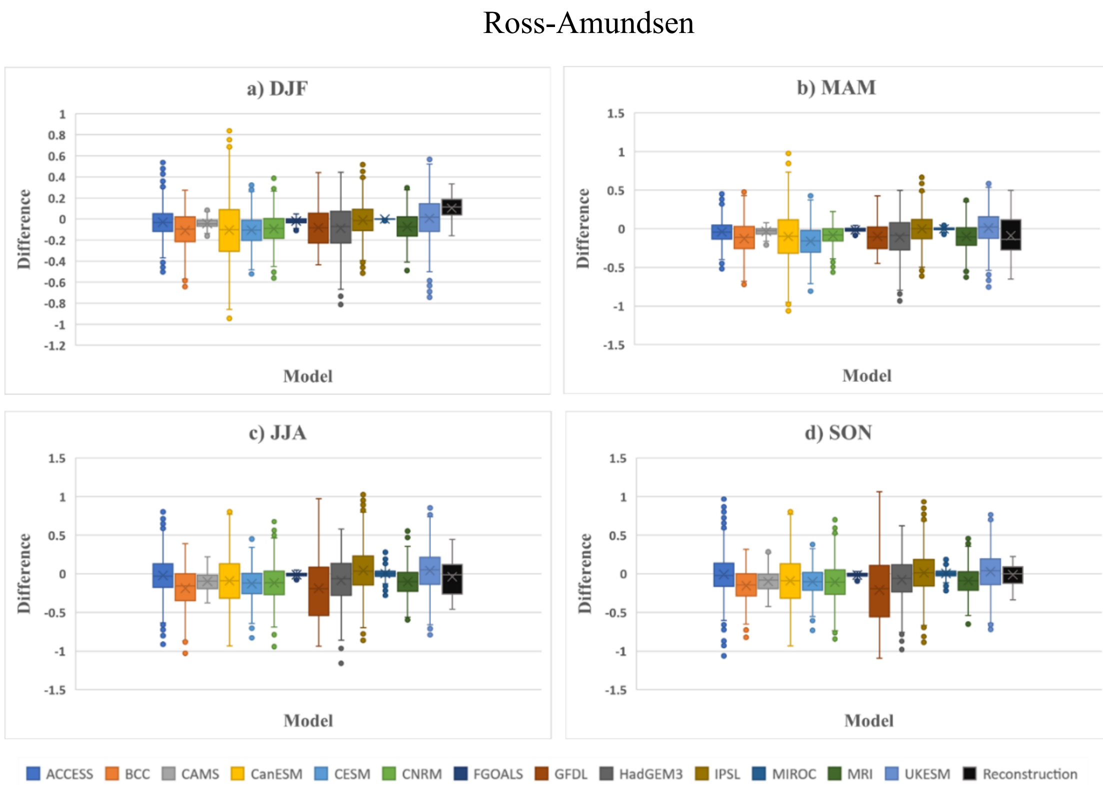
East Antarctica



Note: The various colors represent the different models. The reconstruction is in black on the far right boxplot in each season.

Figure 4.20

Boxplots of 10-year Differences Between the Models and Reconstruction for Ross-Amundsen



Note: The various colors represent the different models. The reconstruction is in black on the far right boxplot in each season.

4.4.5 30-Year Trends

Given that the trends between the models and reconstructions are likely different, it is ideal to more explicitly examine this, especially through time as forcing mechanisms evolve. To help identify the direction of the trends within the models, a 30-year trend was calculated. These trends allow for a comparison to be made between the reconstructions and the models over 30-years to see if the reconstructions are physically valid and consistent with the changes in sea ice depicted in the models, and more importantly, when these two datasets diverge (and the quality of the model trends are questionable as they differ from observations). Fig. 4.21 shows the 30-year trend between the models and the reconstruction for total Antarctic sea ice extent. The trends in Fig. 4.21 are based on the model ensemble means, so the averages for them are going to be smaller, and more reflective of the forced response. The reconstructions will appear larger because the reconstruction is not taking an average, it is just one real-world simulation and therefore contains larger amplitude internal and natural variability. As expected from previous analyses in this thesis and through the literature (see literature review), the trends in the majority of the models in Fig. 4.21 are largely negative. However, there are some similarities between the timing of the positive and negative trends within the models and the reconstructions. Since the model representations used for the calculations in Fig. 4.21 are based on the ensemble means, the similarity is likely pointing to a forced, external response in Antarctic sea ice extent that is captured in the reconstructions; these periods are interesting, and deserve further research in the future (beyond the scope of this thesis). However, as noted in studies looking at the models in comparison to observations after 1979, as seen in Fig. 4.12, the models in Fig. 4.21 show the same decreasing trend in the later periods, while the reconstructions generally are showing a positive trend.

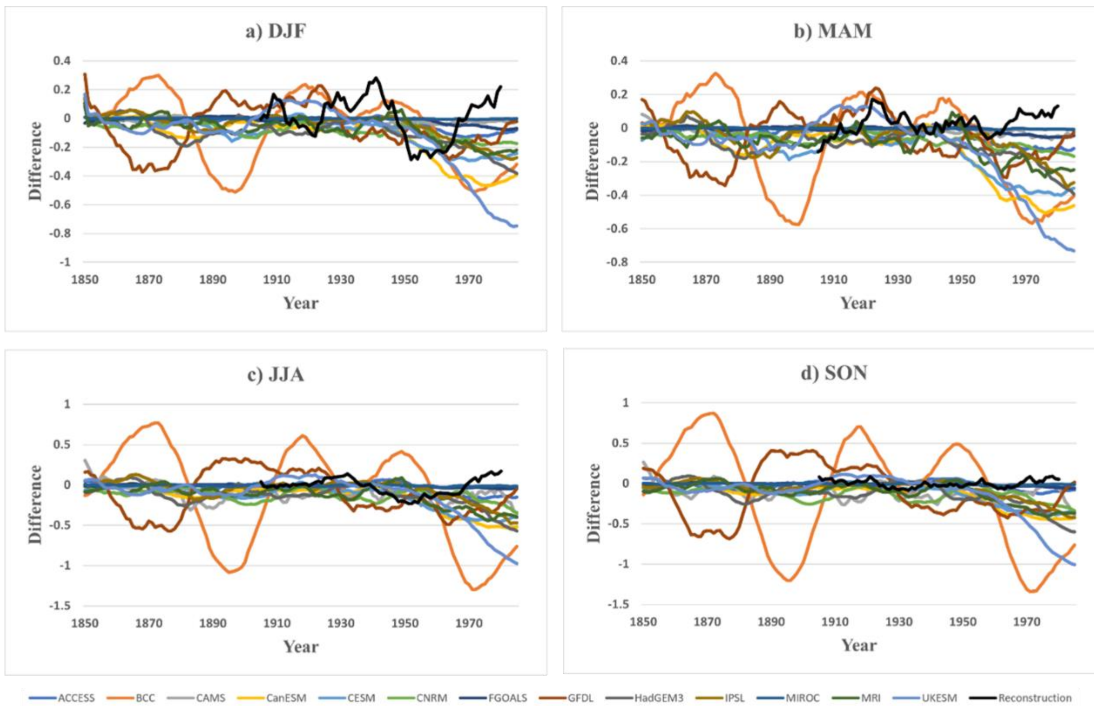
To investigate the trends regionally, Fig. 4.22 shows the trends for the King Hakon sector. Again, the majority of the models are showing a negative trend. In MAM (Fig. 4.22b) during the early 1900s and the 1930s, the reconstruction is opposite of what the majority of the models are showing. In the 1900s the reconstructions are showing a negative trend when the models show a positive trend in the sea ice. The opposite is occurring in the 1930s when the reconstruction is showing a positive trend in the sea ice while a majority of the models are showing a decrease during this particular decade.

The trends in the Weddell sector are shown in Fig. 4.23 as with all other sectors (not shown) the differences the models and reconstructions in the observation period are quite remarkable, where in the 1970s the reconstruction had a positive trend while all of the models are showing a decreasing trend. This agreement throughout much of the 20th century in this sector is much better in JJA, where the trends are much weaker (but still negative, Fig. 4.23c). One other interesting note from the model trends: in Figs. 4.21-4.23, there is an oscillatory trend occurring with the BCC model. This oscillatory trend can be seen in the time series plot (Fig. 4.12), although it is not as prominent in the time series plot as it is in Figs. 4.21-4.23. For reasons unknown, this model's trend changes dramatically and unlike any other model through time, flipping from positive to negative multiple times throughout the 20th century. Nonetheless, despite the frequent disagreement between the models' and reconstructions' trends, overall, there are periods of positive trends in the models where they seem to be showing some of the variability that has been seen throughout this research in the reconstructions, even though the dominant signal in the models is negative in the 30-year trends.

Figure 4.21

30-year Trends of the Models and Reconstructions for the Total Sea Ice Extent

Total Sea Ice Extent

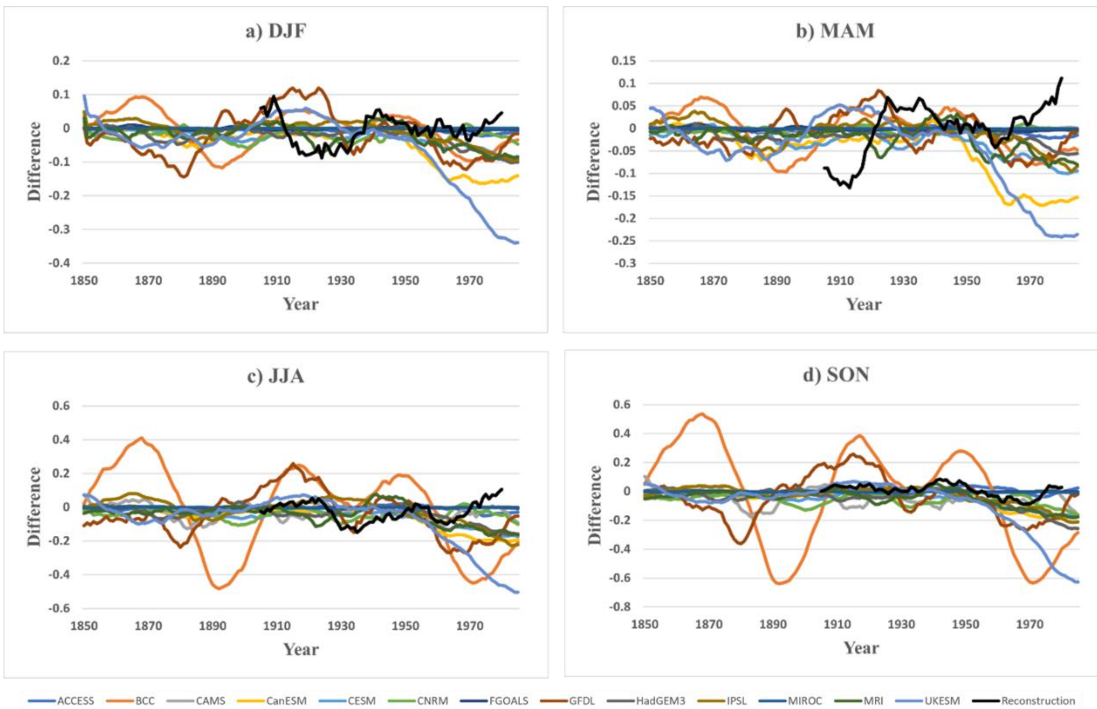


Note: Each color represents a different model. The black line represents the reconstructions.

Figure 4.22

30-Year Trends of the Models and Reconstructions for King Hakon

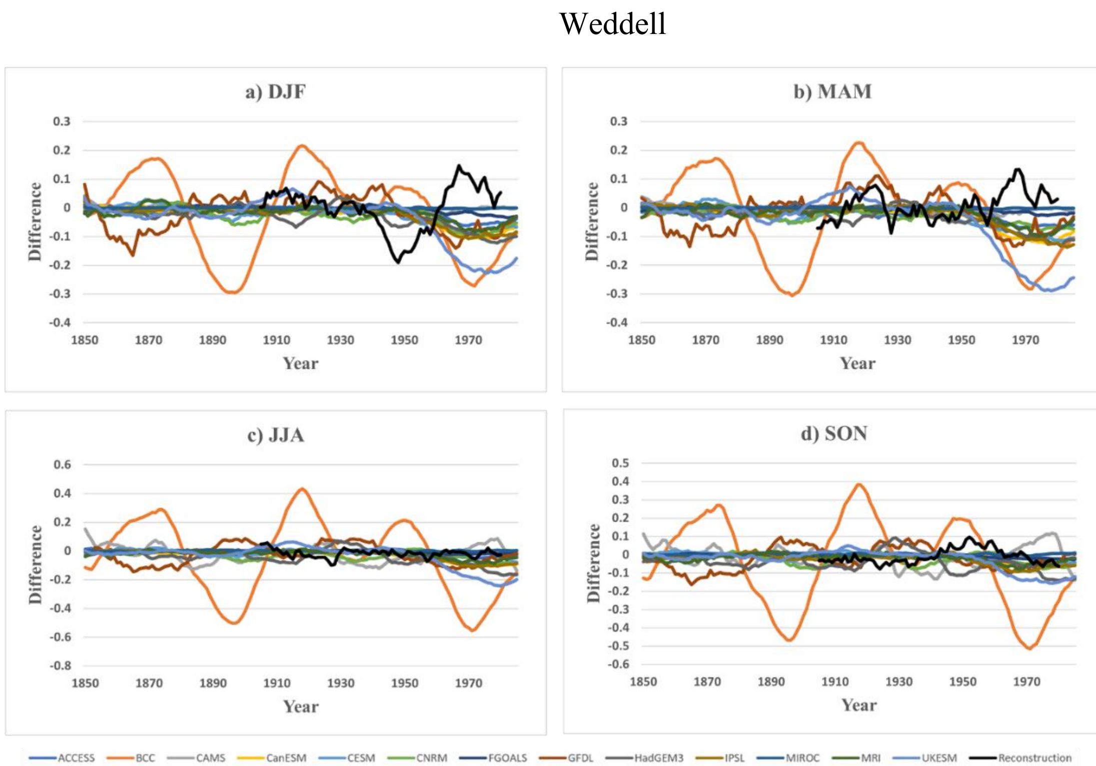
King Hakon



Note: Each color represents a different model. The black line represents the reconstructions.

Figure 4.23

30-year Trends of the Models and Reconstructions for Weddell



Note: Each color represents a different model. The black line represents the reconstructions.

4.5 Chapter Summary

Overall, the results from this chapter conclude that: one, the ship logbooks provided to Ohio University are reliable enough to be used to evaluate Antarctic sea ice extent reconstructions and, by using the ship logbooks, the skill of the seasonal Antarctic sea ice extent reconstructions appears consistent with the errors during the calibration period. Two, variability within estimates of early 20th century sea ice extent from control and historical simulations of Antarctic SIE in couple climate models compare well with the sea ice reconstructions, despite different trends in the models from both observations and reconstructions.

Ohio University was able to gather numerous ship logbook archives, where undergraduate student researchers are working to digitize the archives. Not only does the university have archives the student researchers are working to digitize, but the project was also able to use logbooks from other published sources. Due to the tedious and time-consuming process it takes to digitize the logbooks for the student researchers, only a handful of the total available ship logbooks were used for this research. The ship logbooks can provide valuable information about sea ice, as when sea ice was sighted, the ship would often log this observation in the remarks section of the ship logbook with the approximate latitude and longitude (often times the latitude was more equatorward than the actual position due to the ship not wanting the sea ice to impact the expedition). A lot of the ship logbooks sailed during DJF (summer) because there was not as much sea ice (to impede their expedition or cause harm to the vessel) and the conditions during JJA (winter) are harsh (such as limited daylight making it dangerous for the ships to sail during this season).

The ship logbook observations were daily sightings of sea ice at a single point, so in order to compare them to the seasonal sectoral means of the reconstructions the logbook observations needed to be adjusted by converting them to an area. A procedure was created to achieve converting the daily logbook observations to an area and Figs. 4.6-4.7 to depict the relationships between the daily ice edge from logbooks and the seasonal sectoral means of sea ice extent as an area to understand this relationship. The variability in the latitude of the sea ice edge is greatest in DJF (as the SIC is lower in this season and it is easier for the ice to change from day-to-day) and is lowest in JJA (where the SIC is higher and the sea ice edge is less likely to be influence by forces such as storms); most ship logbook data were from DJF, however, as mentioned earlier.

The ship logbook observations of the sea ice edge were used to help determine the skill of the reconstructions and how its skill varies spatially and temporally prior to 1979. When the bias for the latitude from the ship logbooks was calculated, each sector and decade (1900s-1980s) had a combined approximately 3° latitude difference between the estimated reconstruction latitudes and the logbook latitudes. This is remarkable as the bias provides a favorable comparison between the latitudes from the ship logbooks and the latitudes from the reconstructions. To add to the favorable comparison of the bias, the RMSE also shows a favorable comparison of the reconstruction estimates with the daily values of the sea ice from the ship logbooks. The values of RMSE were fairly close with MAE (which were fairly consistent with the bias values) values, further validating the degree difference between the latitude in the logbooks and reconstructions. The other statistical measure calculated was the correlation. The correlation, even though it was only $r=0.311$, is a considerable correlation as it is statistically significant at $p<0.01$. This r^2 value is higher than a majority of the r^2 value from Figs. 4.2-4.6 which gives another positive evaluation for the reconstruction skill. Further, Figs. 4.8-4.11 visually show the logbook latitude frequently falling within the reconstruction estimate 95% confidence interval overall.

Finally, the last sections in this chapter compared 13 different CMIP6 models to the seasonal sea ice reconstructions in terms of range of variability and linear trends. The models all show the complex sea ice variability. Even though the means between the reconstructions and models are different, realistic (and physically constrained) variability may still be produced within the models. The correlation between the model and reconstructions highlights the high variability in the Antarctic sea ice seen in the models and how this variability aligns with the sea ice reconstructions. While there are differences in the variability within the models and ensemble members, similarities do arise such as seasonal aspects of variability change in various sectors around Antarctica. Overall, the reconstructions of the sea ice edge mimic the behaviors seen in the models very well and the reconstructions are producing physically plausible decadal changes in sea ice.

Chapter 5: Conclusion

This thesis gathered archives of ship logbooks that were digitized to look through for information about sea ice sightings to help see what is available to estimate the sea ice extent prior to the modern satellite era. Statistical measures were used to determine if the logbooks are reliable to be used to evaluate Antarctic sea ice extent reconstructions. Coupled climate models were also used in this research to compare to seasonal sea ice reconstructions. Statistical relationships were used between the ship logbook data reconstructions, as well as for CMIP6 models and reconstructions to evaluate the skill of the reconstructions prior to the modern satellite era (before 1979). The role Antarctic sea ice has on the global climate is important, and having a better understanding of what the sea ice extent was like prior to the modern satellite era using historical data is very valuable. In order for the ship logbooks to be a valuable tool for estimates of early sea ice extent, it is essential that they are reliable. Once it is determined the logbooks are reliable, research can continue further gathering sea ice data from the observations in the logbooks to estimate the sea ice extent and compare them with sea ice extent reconstructions. Understanding the coupled climate models and how they vary in terms of ranges of variability and linear trends can be compared to those of the seasonal sea ice reconstructions to provide information on how well both the models and reconstructions are for Antarctic sea ice.

Various ship logbook data were gathered along with the seasonal, sector mean reconstructions from Fogt et al. (2022) and CMIP6 historical simulations to determine and evaluate the Antarctic sea ice extent reconstructions. In this research, statistical analyses were used to provide evidence of reliable, skillful results in both the logbook data and the CMIP6 models compared to the reconstructions from Fogt et al. (2022). The purpose of this thesis is to investigate the comparisons of Antarctic sea ice with ship logbook observations of sea ice, as well as sea ice from coupled climate models, with seasonal and sector mean reconstructions through the following research questions:

1. What are the available early estimates of sea ice extent prior to the modern satellite era from the logbooks provided to Ohio University, and are those reliable enough to be used to evaluate Antarctic sea ice extent reconstructions?
2. Using ship logbook data, how does the skill of seasonal Antarctic sea ice extent reconstructions vary spatially and temporally prior to 1979?
3. How do estimates of early twentieth century sea ice extent from historical simulations of Antarctic sea ice extent in coupled climate models compare to the seasonal sea ice reconstructions in terms of range of variability and linear trends?

Digital scans from multiple ship logbooks, ranging from various expeditions to commercial logbooks, were gathered by Ohio University, which helped to determine the sources of available early estimates of sea ice extent (question 1). Ship logbooks are valuable as they help to put together a greater picture of what the sea ice extent was like prior to the satellite era by synthesizing pieces from day-to-day explorations and commercial ventures to tell where the ice edge is. When the ships encountered ice, this was typically written in the logbook in the remarks section described as ‘pack ice’ or ‘ice pack’.

There were around 33 archives that were digitized that Ohio University was able to collect. Within these archives, there are multiple vessels that each had logbooks of meteorological records, including times when sea ice was visually spotted. There are still an unknown additional number of voyages and information that has not yet been recovered, and additional work is needed in order to recover these historical data to help in other research projects like this one. The additional voyages will help to better understand climate variations where direct observations are very little or nonexistent. The process to key enter the logbook data into Excel took a considerable amount of time, so instead of keying in all of the meteorological data in the logbooks, only mentions of sea ice were logged with the associated latitude and longitude (and date). A majority of the expeditions occurred in the months between November and March due to the harsh conditions in Antarctica in the winter months. Outside of the logbooks keyed by Ohio University undergraduate student researchers were archives keyed by other researchers.

Since there was an overwhelming number of ship logbooks to be digitized and the process is tedious to do manually as it requires scanning each logbook page individually, opening up to 100 files to just read every possible entry, only certain logbooks were used for this research. Out of the available logbooks, only 9 of the archives were used for this research (Table 4.3 in section 4.1.3). The information gathered from these archives included the date, latitude, and longitude of when the vessel saw the sea ice. In most cases, the sighting of the sea ice was at a distance north (equatorqard) of the actual ice edge as the ship did not want to get too close to the sea ice to damage the vessel or prolong their expedition. The observations within the logbooks were daily observations of the ice edge.

A challenge with using the observations of sea ice from the logbooks is to compare the daily ice edge to seasonal, sectoral means of the ice edge. A procedure was created to be able to make the comparison and determine if the observations of the daily sea ice edge are reliable enough to be used to evaluate the sea ice extent reconstruction (question 1). Using the date, latitude, and longitude of the logbook sea ice sighting allowed for the daily sea ice sighting to be converted to an area that is able to be compared to the seasonal, sectoral means (procedure is outlined in chapter 4, section 4.2). Even with the challenge of relating a single point observation of the ice edge latitude to the seasonal, sectoral mean sea ice extent from the reconstruction, the reconstructions and daily sea ice observations are in agreeance with the pattern of the variability of the ice edge, especially in DJF (summer). The range of the ice edge in DJF (summer) is high and lessens in JJA (winter). In the East Antarctica and King Hakon sectors, the relationship between the reconstruction of the seasonal mean area of the sea ice and the latitude of the daily observations is low, and most of the relationships in each of the seasons in these sectors fall below 10% explained variance. The regression line in East Antarctica and King Hakon barely has a slope indicating the predictions of the daily latitude of the ice edge from a seasonal sector mean reconstruction value is not great. However, East Antarctica and King Hakon experience the lowest amount of summer Antarctic sea ice, and what ice remains moves around considerably each day giving a wide range of daily values for any given location in these sectors. Other sectors, such as Ross-Amundsen and Weddell sectors, have a much greater slope in the regression line, which marks a greater relationship between the daily observations and the reconstructions. When there is a greater relationship between the daily ice edge observations and the reconstructions, the reconstructions are better for predicting the daily SIE as given in ship logbooks. Overall, it was determined that the reliability of the ship logbooks is good enough to be used to evaluate the Antarctic sea ice extent reconstructions, although there are definitely challenges in making these comparisons.

The ship logbook data were also used to help evaluate the skill of the seasonal Antarctic sea ice extent reconstructions both spatially and temporally prior to the modern satellite era (question 2). The total number of data points from the ship logbooks used in this research is 1,586. Each of these data points had the date, latitude, and longitude listed for each that were able to be calculated into an area. A majority of the data points are from DJF, which is not surprising given the rough conditions of Antarctic climate in the winter. Unfortunately, having a majority of the ship logbook data in DJF is the season where the reconstruction versus the daily ice edge regression estimates is the lowest (question 1).

Bias was calculated to help determine the overall reconstruction skill. A majority of the values in the bias were negative, which indicates the logbook latitude is further north than the reconstruction values. This is not surprising, however, as the ship logbooks are reporting sea ice north of where it actually is, especially north of the 15% concentration that is used in the reconstructions. The bias of the data, overall, provides a favorable comparison between the logbook data and the reconstruction, even with the errors in the regression statistics for DJF. Most of the sectors and decades, and even overall, are showing the reconstructions to be within a few degrees latitude of the observations from the ship logbooks. MAE (mean absolute error) was also calculated to help determine the skill of the reconstructions. The values of the MAE are similar to the bias values, which is a good indication the skill of the reconstructions compared to the ship logbook data (the difference if often the same sign throughout time). The RMSE (root mean squared error) values followed the same pattern at the MAE values in that when an MAE value was higher, the RMSE value was also higher. Outliers do play a bigger role in this statistical measure as when the values are significantly different than the MAE value then there is an outlier that is affecting the data. Just like with the MAE and bias, the RMSE shows a favorable comparison of seasonal, sectoral mean reconstructions estimates with the logbook daily values. The correlation for this research looked at how the latitudes from the reconstructions and the logbooks observations vary together through time. It is not surprising that the correlation values were near zero, or negative because of the uncertainties in the regression relationships and the ship logbook entries (question 1). These correlation values suggest a poor relationship between the logbook values and the reconstruction values. However, there are some sectors and decades that have few strong correlations. The higher correlations occur in regions where there are more data points. The overall correlation for the data is r=0.311. This r^2 value is higher than the majority of the r^2values from the regression estimates which is a positive in the skill of the reconstruction.

Overall, the latitudes from the logbook observations frequently fell within the 95% confidence interval of the seasonal, sector mean reconstructions. This is promising as it is an indication the reconstruction estimates of the ice edge are close to the sea ice observations in the logbook. There was one decade (the 1920s) where it is noticeable that the latitudes from the logbook were not within the 95% confidence interval, and this pattern is seen within the bias and MAE statistics as these also had their highest values, indicating the observations are within ~7° latitude of the reconstruction latitudes. Further research is needed on this particular decade to figure out why the logbook observations are outside of the 95% confidence interval to see whether the issue is with the logbook entry, the reconstruction values, or the reconstruction estimate of the daily ice edge for that decade. For a majority of the decades, sectors, and overall the latitudes from the ship logbooks do fall within the 95% confidence interval from the reconstruction estimates. There is an agreement between the daily logbook estimates of the sea ice edge and the reconstruction estimates, so the logbooks are a useful tool to help validate the skill of the pre-satellite reconstructions of the Antarctic sea ice cover (question 2).

The seasonal sea ice reconstructions were also compared to estimates of early twentieth century sea ice extent from historical simulations of Antarctic sea ice in CMIP6 (question 3) – there was an abundance of historical simulations and ensemble members that it was deemed unnecessary to further investigate various control simulations. In this research, 13 different CMIP6 models were used, and each model had numerous ensemble members. The ensemble means of the sea ice showed each model as being highly variable. Compared to the reconstructions, a majority of the models are showing less sea ice than the reconstructions. Towards the end of the time series, the models are showing a decrease trend in the sea ice (inconsistent with observations), while the reconstructions are showing a positive trend in the sea ice, (which is consistent with the observations).

When looking at the interannual variability, most of the models have a large variability. There were a few models that had a persistently lower standard deviation compared to the rest of the models. This is likely from the ensemble members in these models having similar values each year, at the same time. The lines are persistently low in DJF and MAM (sea ice minimums) for 2 of the models, and it is likely these models are melting all or nearly all of the Antarctic sea ice each year, resulting in less variability. With that being said, the interannual variability does show the sea ice as being highly variable in the models overall, as it is in the reconstructions. This variability is also seen within the correlations between the different ensemble members and the seasonal sea ice reconstructions. While the ensemble members are a physically possible representation of the real world, the models do not take in direct observations, so even though they are showing physically realistic variability, there is no reason that any of the model's variability coincides with the real-world variability estimated by the reconstructions. It is expected for the models to not have a strong correlation with the reconstructions as the models are flawed with how they depict Antarctic sea ice trends, and the ensemble members do not take in the direct observations of the sea ice (the connection to the timing and intensity of real-world events, shown in the reconstructions, is not likely to align , weakening the correlation). These correlations look at how the sea ice is changing across the ensemble members and how this agrees with the sea ice reconstructions. The correlation in DJF (summer) is the weakest and the correlation is higher in JJA (winter). This pattern makes sense due to the sea ice in the summer being highly variable due to the daily sea ice easily changing daily. In MAM (spring), most of the ensemble members have a negative correlation, and this opposite relationship in the variability between the models and reconstructions need further research to determine why this season in particular is more negative compared to the other seasons. The rest of the models have a correlation of around 0, as described above, is expected since the models have very little consistent similarities with the reconstructions.

The realistic changes in variability between the reconstructions and models were determined by F-tests. The F-tests were of variance of ratios for all possible 10-year periods and were compared to the reconstructions. It is shown through the variance of ratios that, for the total sea ice extent, all of the models have a similar change in variability that is consistent with the reconstruction (suggesting the decadal scale variability changes in the reconstructions are consistent with physically possible solutions seen in a climate model). The models tend to have stronger outliers than the reconstructions. The ratios also change seasonally in both the models and the reconstructions, suggesting the reconstructions are providing an accurate picture of total sea ice extent variability throughout time. There is more variability in MAM and JJA and less variability in DJF when most of the ice is gone. The variability in MAM will be higher due to the advancement of the sea ice going into JJA. A t-test was also used to help analyze the decadal-scale changes in the sea ice between the models and the reconstructions and understand if the means are statistically different. The t-test looked at the distribution of every possible pair of 10-year averages and how the sea ice acts within these periods. Anytime there was a negative number when calculating the differences for the t-test, this indicated that the sea ice was decreasing through time. The reconstructions show a smaller total change in all the seasons compared to the models. It is likely that this is a problem within the models and not the reconstructions, due to their negative trends. Nonetheless, the models have a similar (absolute) change in a 10-year period as the reconstructions (even though the models show a negative decrease) which suggests the reconstructions are producing physically plausible changes in decadal mean sea ice extent in East Antarctica, even if the sign of the difference is not always the same. Overall, this this a good test for both the reconstructions and models to have the same consistency as it suggests the reconstructions are producing physically plausible decadal changes in sea ice.

Comparisons between the reconstructions and the model linear trends over 30-years are able to show if the reconstructions are physically valid and consistent with the changes in the sea ice. Once again, the models show a decreasing trend, but there are similarities between the timing of the positive and negative trends between the models and the reconstructions. The similarity is likely pointing to a forced, external response in Antarctic sea ice extent that is seen in the reconstructions (this needs further research in the future and is beyond the scope of this research). There is one model in particular, BCC, where there is an oscillatory trend and for reasons unknown; BCC trends change dramatically unlike any of the other models, going from positive to negative multiple times in the 20th century. Even with the disagreement between the models and reconstructions trends, overall, the CMIP6 Antarctic sea ice extent compares very well with the seasonal sea ice reconstructions in terms of range of variability and linear trends. While there are differences, the overall pattern between the two are similar, as seen in the F-tests, T-tests, correlations, interannual variability, and 30-year trends, suggesting that the reconstructions are similar in terms of variability and decadal changes as in the models (question 3).

The available early estimates of Antarctic sea ice through ship logbooks and the reliability of them to evaluate the sea ice extent in the reconstructions, the skill of seasonal Antarctic sea ice extent reconstructions using ship logbook data, and CMIP6 early estimates of sea ice compared to the seasonal sea ice reconstructions were evaluated all in this research. Further research is needed for all of the available ship logbooks, as this research did not warrant all the time needed to fully go through all of it. The results from this thesis show the skill of seasonal sea ice reconstructions prior to 1979 and how they do compare to the models in terms of range of variability and linear trends. The research completed here opens up further research to be explored into the historical early estimates of sea ice extent prior to the modern satellite era.

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